Bt505 biosensor - Ppts and handout by Ahmad saeed final term ppts and handout

Biosensor (Virtual University of Pakistan)



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Bt505

Biosensor (Merged Final Term Handout)

Dr. Madiha Chaudhry

Merged by Ahmad Saeed

Lesson 1 : Characteristics of Biosensor

Summary/Handouts

Introduction

Some characteristics of biosensors must be considered before making them. These characteristics are:

- Sensitivity
- Selectivity
- Stability
- Reproducibility
- Response
- Biocompatibility
- Low background

Sensitivity is the response per unit. It is the response of the bio receptor element to the Analyte. Selectivity is the response of receptors elements to the analyte. It changes with a change in the concentration of the analyte. Stability is the degree of susceptibility to ambient disturbances. There must be internal and external stability. Reproducibility is the identical response. It provides high reliability that is characterized by precision and sensitivity. The biosensor must be user friendly and should give accurate results. The biosensor must have low noise. Accuracy and precision are very important for low noise. Another important property is LOD (Limit of detection). This is the minimum amount that a biosensor can detect.

In this topic, students will learn about:

- Characteristics of biosensors
- Important features of biosensors
- Limit of detection

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- Internal and external stability

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Lesson 2 : Analyte Background

Summary/Handouts

Introduction

Biosensors are designed to detect targeted molecules at the analytical stage. These targeted molecules or elements are known as Analyte. Analyte could be a toxin, molecule and pathogen or other. Analyte binds with its bio receptor molecule and generated a recognizable signal. For example, a glucose molecule is an Analyte in Glucometer that generates a signal after binding with its bio receptor element 'glucose oxidase'. The biosensors available in the market can detect glucose, lactic acid, uric acid, cortisol and bacteria.

Chemistry and biochemistry play important roles in designing biosensors. The characteristics of analyte, bio receptor elements, and support material must be completely understood to design a biosensor. Clinical chemistry is an important branch of biochemistry and chemistry. It provided many important detection methods for biological and chemical molecules. For example, a rapid urine test for glucose detection by dipstick method. The dipstick consists of a plastic strip on which the chemical is fixed in dry form. The application of the sample detects glucose in urine.

Physical components are basic units of bio sensors. These are:

- Detector
- Signal converter
- Amplifier
- Measuring of signals
- Display unit

In this topic students will be able to learn:

- Basic principles involved in designing biosensors
- Chemistry of biosensors and analyte
- Physical components of biosensors
- Role of basic sciences in designing of biosensors

Lesson 3 : Components of a Biosensor

Summary/Handouts

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Introduction

Biosensors consist of two types of components: Biological components or Biological Recognition Element (BRE) and Physical components consisting of input and output devices. Proteins are used as Biological Recognition Element (BRE). Functional proteins are most important. Two types of functional proteins are used in biosensors at a large scale. These are enzymes and antibodies. Proteins are liable to denaturation that can lose binding capacity at high temperatures and pH. The enzymes (proteins) are biological catalysts produced by living cells. These proteins bind to specific receptors known as substrates at specific receptor sites. The other functional proteins are antibodies. Antibodies (blood proteins) are produced by the immune system and can bind with counteracting antigens. These are Y-shaped. Besides this, microbial cells, tissues, and viruses can be used as BRE. These are sources of enzymes. They can regenerate after immobilization of cells. Deoxyribonucleic acid (DNA) is also used as BRE in biosensors.

Physical components of biosensor include:

- Detector
- Transducer
- Amplifier
- Display Unit

The signals generated in biosensors are usually weak that are amplified by the amplifier. The detector detects the signals produced by the biorecognition process. Detector sent signals towards the transducer and the transducer converts signals from physical form to electronic signals and vice versa. The display unit of the biosensor consists of input and output devices. These signals are processed after amplification and displayed through LCDs or small printers in a user-friendly manner.

In this topic, students will learn about:

- Components of biosensors
- Types of biological and physical components
- How biological components work
- The functionality of physical components

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Lesson 4 : Working of Biosensor

Summary/Handouts

Introduction

There are five min steps in working of biosensors

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1. Bio recognition
2. Signal generation
3. Signal recognition
4. Signal conversion
5. Display of information

Analyte (target molecule) binds with its bio-recognition element (BRE) in biosensors. The binding generates a signal that is picked up by the transducer and detectors. Different bio-recognition elements produce different types of biological signals. So, transducers are used according to the nature of biological signals. The transducer could be optical, electrochemical, colorimetric, potentiometric, etc. the transducer converts the biological signal into electric current. The amount of this generate electric current is in proportion with the analyte amount that is displayed in a display unit. For example, in a glucometer blood sample is used. The analyte is a glucose molecule and BRE is an enzyme called glucose oxidase. An electrode is used as a detector while physical components of glucometer are software and hardware with a powerful processor.

A renowned example of biosensor is Glucometer. Glucometer has two platforms: Wet platform and Dry platform. The disposable test strip is a wet platform. The blood sample is uploaded on it through a microfluidic channel. The glucose oxidase (BRE) is immobilized on this test strip and electrodes are fixed. The dry platform of the glucometer consists of software and hardware. It has a powerful processing unit with a voltage converter, amplifier, filter and display unit.

In this topic, students will learn about:

- General working system of a biosensor
- Components of glucometer
- Detection of glucose by glucometer

Lesson 5 : Characteristics of Biosensor

Summary/Handouts

Introduction

The parameters used to detect the performance of biosensor are:

1. Sensitivity
2. Range
3. Linearity
4. Selectivity
5. Reproducibility
6. Response time
7. Stability

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Sensitivity is the first main parameter of a biosensor. It is the minimum input of an Analyte that can create a detectable output change. Linearity of an analytical method is its ability to deduce test results for Analyte in a sample, within a given range. The range of the sensor is the maximum and minimum values of applied parameters that can be measured. For example, a glucometer can measure 20-400mg/dl and the normal physiological range of blood glucose level is 70-140 mg/dl. Selectivity is the ability of a sensor to respond only to the target molecule. For example, the glucometer only detects and measures glucose molecules. The selectivity of biosensors depends on the selection of Biological Recognition Element (BRE). Reproducibility is the ability of a biosensor to obtain identical responses for duplicate experimental setups. Reproducible signals provide high reliability. Response time is the time that biosensor takes to produce a result. For example, a glucometer takes 3 to 60 seconds to display the results for different models.

In this topic, students will learn about:

- Characteristics of biosensors
- Parameters of biosensors

Lesson 6 : Economics and Impact

Summary/Handouts

Introduction

Biosensors are commercial products that are known for rapid results and use. 90% of the biosensor market is mainly a healthcare market and 10% relates with other fields. The healthcare market aims to benefit the consumers that are current or future patients. The objective of biosensors is to improve the population's health by encouraging the value of a person's health. Half of the biosensor market is based on glucose monitors. The first glucose meter was introduced in 1987 and now it is currently experiencing a boost with new technology.

The biosensor market is categorized as a growth market. The major growth areas are:

- Point-of-care (POC)
- Diagnostic

They make real-time remote health monitoring. POC testing or bedside testing is undertaken close to the patient and patients do not have to go to the laboratory. The specimen is obtained, the test performed and the result reviewed where care is provided. This is currently the largest and most profitable area for the development and commercialization of biosensors.

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In this topic, students will learn about:

- Objectives of biosensors
- The market value of biosensors
- Point-of-care (POC) testing

Lesson 7 : Applications of Biosensor

Summary/Handouts

Introduction

Biosensors have applications in multiple fields of life including: Health, Drug designing, Environment, Process technology, Food industry and agriculture. Biosensors have high demand in medical sciences as they provide fast and accurate test results. Biosensors have many applications in the environment because the increasing number of potentially harmful pollutants (heavy metals, phenols, and nitrogen compounds) in the environment demands the use of fast and cost effective sensors. The environmental biosensors provide the following advantages over conventional analytical techniques:

- Portability
- Work on-site
- Can measure pollutants in complex matrices
- The minimal amount of samples is needed
- No pre-treatment

In agriculture, biosensors are widely used for testing the pesticides, crop deterioration, gases detection, ripening of fruits and to assess the pH of soil. The small size device is hooked up to a smartphone app that can allow scanning a piece of fruit or a vegetable to determine the pesticide levels. Similarly, In the food industry, biosensors have high demand. They are used to examine the food quality and to detect the pathogens.

Enzyme-based biosensors are used for security surveillance. Recently a biosensor has been introduced that rapidly detects both explosives and nerve agents. It detects hydrogen peroxide. The technique could replace two time-consuming tests that are currently used to assess these threats.

In this topic, students will learn about

- Application of biosensors in multiple fields
- Enzyme based biosensors

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Lesson 8 : Receptors in Living cell

Summary/Handouts

Introduction:

Receptors are specialized cells. The idea of the biosensor is taken from receptors. There are five general senses. These receptors are located on sense organs. Receptors are specific to stimuli. A sensory receptor is a specialized cell. The receptors provide information to CNS by ionic movement. Olfaction or sense of smell is provided by the paired olfactory organs on either side of the nasal septum. The nasal area contains 10-20 million olfactory receptors with cilia. These cilia are responsible for olfaction. A complex of odorant receptors leads to activation of an enzymatic mechanism involving ATP, for the opening of sodium-calcium gate channels leading to Na+ influx. Sense of taste or gestation provides information about the taste of foods and liquids. The gustatory receptors or taste buds are about 3000 that are distributed over the tongue, superior surface, pharynx, and larynx. The movement of calcium, potassium, and sodium ions initiates the gustation process. Vision is a sense through which a person or animal becomes aware of light color, etc. using the eye. The eye receptors (Rods and Cones) transfer light energy into nerve impulses. They transmit the information to the cortex of the brain's occipital lobe. The flow of sodium ions plays important role in vision. The hearing receptors are present in-ear and are topped by hair bundles. The hair bundles move in response to the sound waves. The ear acts as a transducer. The potassium and calcium ion channels play important role in the hearing mechanism. During the hearing mechanism, the polarizing and depolarizing potential changes have the same frequency as the original sound.

In this topic, the student will learn about:

- Receptor's characteristics and working
- Five basic senses
- Working mechanism of olfaction, and gestation
- How eye and ear receptors work

Lesson 9 : Enzymes as Bioreceptor

Summary/Handouts

Introduction

Enzymes are biological catalysts that are used as biological recognition elements in biosensors. Enzymes can be divided into three different types, depending upon the detection of the molecule as a signal. Enzymes bind with a substrate molecule and generate a biorecognition signal. A diverse form of sensors is available to detect that specific signal. Various biosensors are designed using different classes of enzymes. The first step in the process of enzyme biosensors is the

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immobilization of enzymes. During this process, the solid support must not interfere with enzyme activity. The solid support can be:

- High charge density or hydrophobic: surfaces are possible causes of enzyme conformational changes and inactivation.
- Enzyme co-immobilized with hydrophilic: polymers or tethered can reduce unfolding and inactivating support-enzyme interactions.
- Incorrect orientation and multilayer formation: cause reduction of specific activity

In this topic, the student will learn about:

- Different enzymes used in biosensors
- Working of enzyme biosensors

Lesson 10 : Mechanism of Enzyme Action

Summary/Handouts

Introduction

Enzymes increase the rate of biochemical reactions by reducing the activation energy. Enzymes have catalytic and substrate binding sites. Enzymes activity is inhibited by the inhibitors. The reaction rate depends on the physical and chemical structures of the enzymes and the substrates. Designing of the biosensor is dependent on the mechanism of enzyme action.

Biosensors for the hydrolase enzyme 'acetylcholine esterase' are based on the inhibition of enzymes for the detection of organophosphorus chemical pesticides. The second mechanism is the involvement of the transfer of electrons in oxidation-reduction reactions. Application in biosensors involves the release of electrons by glucose oxidase that is detected by biosensors.

Some other enzymes based on functionality are

1. Transfer of radicals: Transferase enzymes catalyze the transfer of groups from one reactant to other.
2. Addition or removal of carbon bonds: lyases catalyze reactions.
3. Change of geometry within a molecule: The isomerase enzyme catalyzes reactions that transfer the functional groups within the molecule so that isomeric forms are produced.
4. Another important function of an enzyme is the joining of two macromolecules. The ligase enzyme joins the two molecules through hydrolysis of pyrophosphate in ATP or other triphosphates. For example, T4 DNA ligase.

In this topic, students will learn about:

- Mechanism of enzyme action

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- Different types of enzyme action mechanisms
- Functioning of ligase enzyme

Lesson 11 : Antibody as Bioreceptor

Summary/Handouts

Introduction

An antibody is also very popular by receptor when a foreign molecule invades the body, the immune system recognizes and memorizes the molecules shape by creating an antibody molecule that is a perfectly engineered to be invaders. In is Y shape molecules and called immunoglobulin. It consists of two heavy chain (H-chain) and light chain. The antibody has a constant and a variable part, the variable part sometimes acts like a key which specifically fits into a lock. Human body can produce the estimated 100 million different antibodies, recognizing all kinds of foreign substances. It does this by shuffling different sections of part of the genes which produce the variable reagent (Like building different models from basic set of shapes). An antibody can also recognize and viral particle or a bacterial by recognizing the proteins on the surface. Therefore, antibodies can be used as excellent bio-receptor for wide variety of protein molecules as well as viruses and bacteria.

For the immobilization of antibody on biosensor, orientation of antibodies on the surface is important for sensitivity. Covalent cross linking or oxidation of carbohydrate groups can be used to control the interaction between specific reactive groups on the surface and on the antibody.

In this topic, students will learn about

- Importance of antibody to be utilized as bio-receptor
- Immobilization of antibody on sensor surface

Lesson 12 : Nucleic acids as Bioreceptor

Summary/Handouts

Introduction

DNA and RNA are the biorecognition elements in nucleic acid biosensors. The detection methods are based on recognition of complementary nucleic acid sequence. DNA biosensors are evolved to be, rapid, simple and economical. For the immobilization of nucleic acids, the DNA fragments have to be immobilized in a way that retains,

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2) Reactivity

3) Accessibility

4) to target analyte for hybridization.

5) Adsorption

DNA probes are immobilized via electrostatic adsorption between a negatively charged phosphate group of DNAs on the positively charged surface. The sensing methods includes the Label based method in which specific organic dyes, metal complexes and enzymes are used for hybridization detection. For example, Methylene blue binds at minor groove of DNA.

In this topic, students will learn about

- How nucleic acid can act as bioreceptors
- Immobilization of DNA as bioreceptor
- DNA sensing methods

Lesson 13 : Cells tissue, Tissue Virus Bioreceptors

Summary/Handouts

Introduction

Cell-based biosensors use cells as biological receptor element like prokaryotic cells (bacteria), eukaryotic cells, yeast, invertebrates and vertebrates. Certain cells that possess strong affinity for a specific target can also be used as a bioreceptor. Cell-based biosensors use cells as bioreceptors. They apply the principles of cell-based assays by employing living cells for detection of different analytes from environment, food, clinical or other sources. But the major thing is to learn that how these cells can be immobilized.

Entrapment is the ideal method to immobilize the cells or tissues on support material. Different materials can be used for their immobilization. For example,

a) Cellulose based polymer

b) Chitosan

c) Polyacrylamide

d) Porous silica

e) Polyvinyl chloride based polymers

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These cells or tissues as bioreceptors can offer multiple applications including the detection of bio-toxins. Cells report the cellular responses upon exposures to toxins under Physiological conditions. It reveals functional information about mode of action, toxic potency, bioavailability and target tissue or organ.

In this topic, students will learn about

- Cell-based biosensors
- Immobilization of cells as bioreceptors
- Applications of cells or tissues as bioreceptors

Lesson 14 : An introduction to Miniaturization

Summary/Handouts

Introduction

Miniaturization is the process of making something very small using modern technology. All the objects have size dependent properties. These physical properties change when the size of an object changes. Examples of size dependent properties are Length, Width, Height, Volume, Mass. The advantage of miniaturization is that smaller objects perform faster simply because they have, less mass which leads to less inertia (tendency of a body to resist acceleration). It leads to products that perform quicker. Smaller objects need less energy to function. They are economical.

For miniaturization, scaling which is the study involve in the operation of sizing, is very important. Different scaling factors are involved in downsizing of the objects including gravitational force, strength, speed, heat capacity, resistance and surface tension. Generally, smaller things are less effected by volume dependent phenomena such as mass and inertia, and are more effected by surface area dependent phenomena such as contact forces or heat transfer. Size change in living bodies has numerous effects and profound biological significance. With this biological significance of miniaturization, different biosensors with high sensitivity can be fabricated that will detect the analyte rapidly and with more selectivity.

In this topic, students will learn about

- Principles of Miniaturization
- Scaling factors and laws
- Effects of Miniaturization on Biosensors

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Lesson 15 : Microsystems

Summary/Handouts

Introduction

Micro electro mechanical System (MEMS) is the making and combining of miniaturized mechanical and electrical components on a common silicon substrate through micro fabrication technology. MEMS consist of mechanical microstructures, microsensors and microelectronics, all integrated onto the same silicon chip. Microsensors detect changes in the system's environment by measuring mechanical, thermal, magnetic, chemical or electromagnetic information or phenomena. Firstly, the interdisciplinary nature of MEMS technology and its micromachining techniques, as well as its diversity of applications has resulted in an unprecedented range of devices. Secondly, MEMS with its batch fabrication techniques enables components and devices to be manufactured with increased performance and reliability, Thirdly, MEMS provides the basis for the manufacture of products that cannot be made by other methods.

The basic techniques for the fabrication of MEMS are Deposition, Patterning and Etching. Different materials are used for the fabrication of MEMS including Silicon, Crystalline silicon and some other materials like metals and ceramics. Silicates, Polysilicon and Silicon dioxide can also be used in manufacturing of MEMS. Basic unit of Micro electro mechanical System (MEMS) are

1. Silicon Wafers
2. Integrated Circuit

In this topic, students will learn about

- Principles of Miniaturization
- Scaling factors and laws
- Effects of Miniaturization on Biosensors

Lesson 16 : Sensing Techniques

Summary/Handouts

Introduction:

The main feature of a biosensor is to detect a target molecule by using a sensing technique. Sensing techniques are used to detect any change in the physical quantity of a system. The electrochemical technique detects electric signals, while the optical technique uses light as a probe. The acoustic technique involves the detection of a wave or vibration in a system. The electrochemical technique is most commonly used in biosensors that are further divided into three types: Potentiometric, Amperometric and Impedimetric. The Potentiometric technique

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detects variation in potential in volts; Amperometric detects current and Ampedimetric measures the impedance of a system. Basically, Sensing techniques rely on two types of detection methods that are Label-based method and Label-free method.

The Label-based method is a very rapid process that involves the attachment of biotin to a molecule through covalent bonds. The other technique is the fluorescent probe. Fluorescent is a chemical compound that can reemit light upon light excitation—for example, Rhodamine dye. The third and most important labeling technique is the enzyme-label-based sensing technique. The enzymes used in this technique are alkaline phosphatase, glucose oxidase, beta-galactosidase, horseradish peroxidase, etc. The Label-free method depends upon the Physico-chemical parameters of a system and allows direct detection of the targeted molecule.

In this topic, students will learn about:

- Sensing techniques and types
- Types of Electrochemical technique
- Detection methods of sensing techniques

Lesson 17 : Optical Techniques

Summary/Handouts

Introduction:

Optical sensing techniques are widely used in biosensors to detect target molecules through light rays. These are economical and reliable. They detect the physical quantity of light and convert it into electrical signals. Later, the electronic signals are converted into a readable form. The optical sensors use monochromatic light because this is unidirectional, intense, and has one color. The other light source used in the optical technique is laser-light. LASER stands for Light Amplification by Stimulated Emission of Radiation. The artificially produced Laser beams generate a very narrow light beam with a similar wavelength. These are coherent, monochromatic, unidirectional, and intense—for example, a laser pointer. The biosensors use different optical techniques, including Surface Plasma resonance (SPR), Fiber-optics sensing techniques, Mass-spectrometry Techniques and Reflectometric interference spectroscopy techniques.

Mass Spectrometry detects the mass-to-charge ratio (m/z) for one or more molecules in a sample. It includes ionization, acceleration or deflection, and detection of the molecule. Only positively charged ions are selected for acceleration. The Fiber-optics sensors are thin fibers made up of glass or polymers, about the thickness of human hair. Optical fiber has three parts: Core, Cladding, and Coating or buffer. The Surface plasma Resonance (SPR) detects the interaction between different molecules in which one is mobile (target), and the other is fixed (bioreceptor

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element) on a thin film. Plasmon is density waves of electrons created when light hits a metal surface under precise circumstances. Reflectometric interference spectroscopy techniques are a physical method that uses white light. In this, white light hits the surface onto a multiple-layer system of silicon oxide or metals, and interference in its reflection detects the targeted molecule.

In this topic, students will learn about:

- Working of Optical Sensing techniques
- Laser-light
- Types of Optical Sensing techniques

Lesson 18 : Transistors

Summaries/Handouts

Introduction:

Transistors are microelectronic compounds that are the building blocks of many electronic devices. A transistor is a semiconductor device that transfers a weak signal from a low-resistance circuit to a high resistance circuit. It also controls the flow of current by controlling voltage. For example, a hearing aid that uses transistors. One hearing aid may contain at least four transistors. It is a tiny microphone that picks up sound from outside and turns it into fluctuating electric current, which is then fed into a transistor. The signals are amplified and sent to a tiny loudspeaker.

Dopping is the addition of any impurity (dopant) into a pure semiconductor to improve its electrical properties. The dopant from the 5th and 3rd groups of the periodic table give N-type and P-type extrinsic semiconductors, respectively. Each transistor has three parts called as Emitter, Collector and Base. The collector collects current, and the emitter emits current, while the base is the region between collector and emitter.

In this topic, students will learn about:

- Transistors
- Working of hearing aid
- N-type and P-type extrinsic semiconductors
- Dopping
- Parts of transistors

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Lesson 19 : Electrodes

Introduction:

These electrical conductors are an important part of biosensors. These are made up of two types of materials: reactant and non-reactive materials. These are produced in different sizes and shapes to monitor and connect different systems. The electrodes are divided into two types based on charge: Cathode and Anode. The cathode is an electron-donor carrying a negative charge, while the electron-charged electrode is an electron acceptor. Electrodes are an important part of biosensors. These are extensively used in the medical and health fields. Different types of electrodes are designed for different purposes. For example, a CO2-Sensitive electrode is used to determine the amount of carbon dioxide. It is immersed in HCO3 buffer and 0.1M KCl solution. A gas-permeable membrane is used to keep electrode solutions separate from biological samples. The glass electrode measures the resulting pH of the bicarbonate solution to indicate CO2.

ECG electrodes or electrocardiogram electrodes are used to detect any coronary artery disease. They contain silver and silver chloride in a water-soluble electrode gel. The disposable electrodes are stuck to the patient's skin and easily removable. The signals are converted into electrical voltage by electrodes. Another type of electrode is the Microelectrode used in electrophysiology. Its needle is very thin (o.5 um). It comprises glass, optical fiber, metal or alloy wires, and carbon fiber. It is usually filled with a 3M KCl solution as an electrical conductor. It is also used as a hair-removing device.

In this topic, students will learn about:

- Working of electrodes
- Types of electrodes
- ECG electrodes

Lesson 20 : Enzyme Sensitive Electrodes

Introduction:

Enzyme Sensitive Electrodes or Enzyme sensors are miniature chemical devices that combine an electrochemical procedure with immobilized enzyme activity. The first enzyme-sensitive electrode was reported in 1967 that uses the enzyme glucose oxidase. It was an Amperometric electrode based on the physiological electron acceptance by oxygen. A negative potential is applied to the platinum cathode for a reductive detection of oxygen consumption. It was subjected to errors from fluctuation in the oxygen tension. Different approaches are introduced to

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overcome these errors. These are enzyme electrode with a mediator, three electrode system and screen-printed electrodes.

Two enzymes were used in the Enzyme electrode with a mediator: glucose oxidase and Horse reddish peroxidase. The three Electrode system is used in biosensors. Three electrodes are used in this system: a working electrode that contains a sample, a reference electrode that sets the potential, and a counter electrode as a current source. These Enzyme-sensitive electrodes are used in screen printing for the fabrication of electrodes. Screen-printed Electrodes are carbon, gold, platinum, and silver ink. Carbon is widely used as an inner electrode due to its neutrality feature. Enzyme-printed electrodes are used in Biosensors. The enzyme layer is placed between the carbon electrode and the mediator layer. Other enzymes like urease are also used. It catalyzes the hydrolysis of urea to form ammonia and carbon dioxide. A urease biosensor is used to determine the human serum urea level. A liquid urease layer is trapped in a cellophane layer over a cation electrode. It reacts with urease and produces ammonia that increases the pH measured by a pH meter.

In this topic, students will learn about:

- Enzyme Sensitive Electrodes
- Approaches used in Enzyme Sensitive Electrodes
- Amperometric electrode

Lesson 21 : Field Effect Transistors

Introduction:

Field Effect Transistor (FET) or unipolar transistor is a type of transistor that uses only one type of pole: N-type or P-type. It has three types of terminals: source, gate, and drain. FETs control the flow of current by applying a voltage to the gate. The biosensor containing a FET is called Bio-FET. IT combines a FET and biological recognization element (BRE). BRE could be a protein, enzyme, or DNA molecule. The Bio-FETs provide an effective, label-free, and real-time biosensing platform for detecting the target molecule. In addition to BRE and FET, a chemically and electrically insulating layer is used as a biofunctionalized surface. Generally, the membrane is fabricated by triethoxysilane (3-aminopropyl). Bio-FETs are classified based on BRE used for detection.

The working of Bio-FET includes binding of the analyte with BRE on a bio-functionalized surface. It introduces changes in the electrostatic potential of the surface. The measurable changes in current between electrodes are recorded as detecting signals. The other type of Bio-FET is DNA-FET using field effect due to the partial change of DNA molecule. In DNA-FET, a layer of immobilized ssDNA acts as surface receptors. after analyte hybridization, the charge distribution changes, and signals are generated. The cell-transistors hybrid uses biological cells

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to receive, process, and transmit information. They provide information about biological phenomena. An interesting example of BioFET is the beetle Antenna to detect crop damage. The antenna of a beetle is coupled via an electrolyte to bioelectronic field-effect sensors. It is highly sensitive to different odors. Damaged plants are identified due to the presence of odors.

In this topic, students will learn about:

- Working and terminals of Field Effect Transistor
- structure of Bio-FET
- Classifiaction of Bio-FET
- Bio-FET with beetle Antenna

Lesson 22 : Ion Selective Electrodes

Introduction:

The basic structure of the ion-selective electrode requires a membrane that excludes all ions except for the ion of interest. It is composed of three parts.

The membrane creates impermeability for different ions, the voltmeter determines the potential difference, and the electrode is the important part of the Ion Selective Electrode. For example, a glass electrode comprises amorphous silicate glass. Glass comprises an irregular network of silicate tetrahedrons. It contains cations of metals like Na, K, Ca, and Li that coordinate with oxygen atoms. The most important part of the Ion Selective electrode is its Glass. The glass membrane selects ions present in the electrode.

The glass electrode bulb is made up of a very thin glass membrane. A K-sensitive electrode is a potassium-sensitive electrode. It is made up of an ionophore (Valinomycin). Ionophore is a substance that facilitates the transport of particular ions across the membrane. Valinomycin is a naturally occurring peptide used in the transport of potassium ions. The circular-shaped peptide is a natural antibiotic produced by streptomyces and is a decapeptide. It contains lactic acid and DNL amino acids. The other type of Ion-Sensitive Electrode is Ion-exchange resins. These are special organic polymer membranes containing a specific ion-exchange substance. Ion-exchange resins are widespread electrodes with anionic selectivity mostly used for water purification. These are less stable electrodes.

In this topic, students will learn about:

- Structure and types of Ion-selective electrode
- Glass membrane

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- Structure of Valinomycin
- Ion-exchange resins

Lesson 23 : Latest Trends in Miniaturization

Introduction:

The latest trends in the miniaturization of devices are noted in the health sector. Many small devices are introduced for diagnostic and wearable biosensors—for example, smartwatches and Tattoos. A microbial detection array is a small chip that identifies approximately 2000 viruses and 900 bacterial microbes in a sample device. The chip can identify thousands of pathogens within 24 hours. HIV detection chip is a microfluidics-based detection chip that functions to detect HIV. It also determines viral load in patients. It has the capacity to test 100 samples at a time. The chip can determine the viral load of the patient within 24 hours.

Tattoo Ink is a mixture of chemicals that can sense pH, blood sugar, and sodium levels. For example, the Tatoo Ink color changes from blue to brown when blood sugar goes up. Similarly, the sodium sensing ink becomes a more vibrant green under the UV light when the salt level increases.

Wearable Tattoos have cosmetic values and are based on gold. The first step in the production of the Tattoo is the sketching of the circuit. Then it is fabricated on gold leaf on top with spray adhesives. The next step is the mounting of electronics. These electronics are transistors. These DuoSkin devices are planted on the user's skin like a regular temporary tattoo.

In this topic, students will learn about:

- Advancement in miniaturization
- Microbial detection array
- Wearable Tattoos
- HIV detection chip
- Tattoo Ink

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Lesson 24 : Bio-signals Definition

Introduction:

Biosignals are inductors that provide information about any physiological phenomena. Biosignal is the key to determining the function of a living organism. These biosignals record biological events occurring in a time frame within the human body, such as heart rate, body temperature, and hormone level. There are at least eleven organ systems in humans. Biosignals are the only source that gives information about the functioning of the human body in health and disease conditions. Biosignal is a measurable body variable. Any change in biosignal indicates any physiochemical and pathological condition.

Animal also contain Biosignals. For example, the octopus has richly colored ink stores used for chemical deterrence, sensory distraction, and invasion when attacked. Similarly, cuttlefish change their appearance to camouflage themselves from predators. The plant generates signals for survival. These signals are electrical current or volatile compounds. For example, wounded plants produce volatile odor compounds. Some carnivorous plants generate electrical signals to trap their prey.

In this topic, students will learn about:

- Working of biosignals
- Biosignals in humans, plants, and animals

Lesson 25 : Bio-signals Classification

Introduction:

Biosignals are heterogeneous. It isn't easy to classify them separately. There are at least four ways on th basis of which classification is possible. These are Permanent biosignals, Induced biosignals. Nature of biosignals and Origin of biosignals.

Permanent biosignals exist without artificial impact, triggers, or excitation from outside the body. Their source is already present inside the body. These are available at any time. The signals that need any artificial source are called induced signals. These are short and used in different diagnostic and cosmetic processes. For example, cosmetic electrotherapy. The classification according to the system of origin of biosignals is broad. In this system, biosignals are classified based on the endocrine system, nervous system, cardiovascular system, respiratory system, auditory system, etc. The next type of classification is based on biosignals' physical and chemical nature. The nature of biosignals is the most convenient classification system. These signals could be:

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- Electrical
- Magnetic
- Mechanical
- Optic
- Acoustic
- Chemical
- Thermal

In this topic, students will learn about:

- Classification of biosignals
- Nature of biosignals

Lesson 26 : Optical Bio-signal

Introduction:

Optic Biosignal is an electromagnetic signal produced by a living cell. It has electric and magnetic fields that are orthogonal to each other. For example, generation of light by a Firefly. Firefly produces biosignals called Bioluminance. It is an enzyme-based reaction occurring through an enzyme Luciferase and substrate luciferin in the presence of magnesium ions and ATP. A proton is released that generates light. Several marine animals also use the process of Bioluminance by using the enzyme Luciferase. Jellyfish shows its defense system by bioluminescence. Jellyfish, for example, comb jellies produce bright flashes to shock a predator. The siphonophores produce a twinkling red light to attract their prey. Cyanobacteria also use a Bioluminance system. These bacteria involve oxidation-reduction reactions to produce blue-green light by oxidation of an aliphatic aldehyde by a reduced flavin mononucleotide in the presence of oxygen.

Bioluminance is widely applied in Biotechnology. The luciferase reporter assay is commonly used to study gene expression at the transcriptional level. The gene is used as a marker in molecular biology. The DNA sequence of the luciferase enzyme is inserted into a plasmid, which provides expression on the positive colon as a light. It is convenient, inexpensive, and gives quantitative measurements instantly.

In this topic, students will learn about:

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- Mechanism of Optic Biosignals
- Process of Bioluminance in living organisms
- Applications of Bioluminancein biotechnology

Lesson 27 : Electrical Bio-signals

Introduction:

Electrical bio-signals, or bioelectrical time signals, usually refers to the change in electric current produced by the sum of an electrical potential difference across a specialized tissue, organ or cell system like the nervous system. Electric field is generated in cells (nerve and muscle) and organs because of intra- and extracellular ionic currents. Typical values of membrane potential are in the range –40 mV to –70 mV. Bioelectrical signals are generated from the complex self- regulatory system in human. These signals are very low in frequency and can be measured. Differences in the concentrations of ions on opposite sides of a cellular membrane lead to a voltage called the membrane potential. Our body generates electrical signals and control the body functions. Different cells produce different types of signals for example, the signals generated by Neural cells can be measured by the techniques like Electroneurogram ENG, Electroencephalogram EEG, and Electroretinogram ERG

Signals generated by muscle cells can be measured by Electrocardiogram ECG and Electromyogram EMG. With the development of methods for remote measurement of electric fields using new sensor technology, electric bio-signals such as EEG and ECG can be measured without electric contact with the skin. This can be applied for example for remote monitoring of brain waves and heartbeat of patients who must not be touched, in particular patients with serious burns.

In this topic, students will learn about:

- Bioelectric signals
- Process of Bio signals in living organisms
- ECG Electrocardiogram

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Lesson 28 : Magnetic Bio-signals

Introduction:

Magnetic biosignals Weak magnetic fields are generated by different organs and cells. For example, neural cells signals can be tested by Magnetoneurogram (MNG) and Magnetoencephalogram (MEG). The signals of Muscle cells can be tested by Magnetocardiogram (MCG) and Magnetomyogram (MMG).

MEG is based on measuring the magnetic field outside the head using an array of very sensitive magnetic field detectors (magnetometers). Because the magnetic field in in the human body is generated under the influence of electric field, the magnetic biosignals are also considered under electric biosignals. Magnetocardiography (MCG) is a technique to measure the magnetic fields produced by electrical currents in the heart. It uses extremely sensitive devices such as the superconducting quantum interference device (SQUID).

Birds use Earth's magnetic field as sort of a heads-up display to help them navigate the globe. The Earth's magnetic field is a result of the movement or convection of liquid iron in the outer core. Iron-rich structures have been found in the beak of homing pigeons, chickens and several species of migratory birds and interpreted as magnetoreceptors. As the liquid metal in the outer core moves, it generates electric currents, which lead to a magnetic field. Cryptochrome protein helps birds navigate via magnetic field. Researchers have found one possible explanation for some birds' ability to sense the earth's magnetic field and use it to orient themselves: a magnetically sensitive protein called cryptochrome that mediates circadian rhythms in plants and animals.

In this topic, students will learn about:

-
- Magnetic bio-signals
- How birds use magnetic bio-signals

Lesson 29 : Mechanical Bio-signals

Mechanical biosignals are related to displacement and change of position or vibration using a mechanical force by the organisms. These signals could affect by expansion, as in the case of respiration. In the honey bees, these mechanical biosignals lead to specific displacement.

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The respiratory system is an excellent example of mechanical biosignals produced by inhalation and exhalation. Lungs are mainly composed of an elastic functional protein, 'Elastin.' The system facilitates the exchange of gases. During inspiration, the diaphragm contracts and pulls downward. The muscles between the ribs contract and pull upward. During expiration, the diaphragm relaxes; lungs contract, and the air is forced out. It all happens because of elastin protein. Honey bee workers perform a series of movements called 'waggle or round dance.' It tells the location of the food source more than 150 meters from the hive. The pigeons flap their feathers as alarm signals. Male peafowl performs a complicated dance in which they tilt, shake their feathers, and makes dazzling moves.

Plants also exhibit mechanical biosignals. For example, sunflower sense change in light and temperature as day progresses and change their side. Night-blooming flowers use their fragrance as a signal to attract pollinators at night.

In this topic, students will learn about:

- Mechanical biosignals
- Different types of Mechanical biosignals in other organisms

Lesson 30 : Acoustic Bio-signals

Acoustic biosignals are related to sound. Different biological sounds such as lung sounds, heart sounds, bowel sounds, and joint sounds are emitted in the human body. These sounds provide an easy way of indicating existing pathology. In healthy adults, there are two normal heart sounds, often described as lub and dub occurring in a sequence with each other. These are produced by the closing of atrioventricular and semilunar valves, respectively.

Snoring is the vibration of respiratory structures during sleep that results in sound due to obstructed air movement during breathing.

Normal breath sounds are tracheal, bronchial, broncho vesicular, and vesicular sounds. These breath sounds are described by duration, intensity, and pitch timing.

A burb is a belch or the loud release of trapped air loudly through the mouth. A reflex is initiated, leading to the relaxation of the lower esophageal sphincter and upward air movement through the esophagus. It passes through the upper esophageal sphincter that produces an audible belch.

Hiccups are a sudden and involuntary contraction of diaphragm muscles. As the muscles contract, the opening between the vocal cords snaps shut to check the air inflow and make the

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hiccup sound. Also, irritation of nerves that extend from the neck to the chest can cause a hiccup.

These acoustic biosignals have led to many techniques and instruments for diagnostic purposes.

In this topic, students will learn about:

- Acoustic biosignals
- Advantages of Acoustic biosignals
- Hiccups and burb sound

Lesson 31 : Chemical Bio-signals

Summary/Handouts

Chemical biosignals are released in and outside the body in response to specific stimuli. For example, insulin is released when blood sugar level rises, and glycogen is released when blood sugar level lowers. These chemicals are produced inside the body. Some chemicals are produced outside the cell in response to external stimuli. These are released in the environment and are produced by all types of organisms.

Plants produce biosignals under stress conditions. For example, jasmonic acid is produced in the environment during a parasitic attack, but this is dangerous to herbivorous animals.

Animals also produce biosignals under stress conditions. For example, cephalopods produce Ink like compounds against predators. The blue-ringed octopus has a deadly toxin named tetrodotoxin in its Ink. Similarly, sea hares produce colorful sticky Ink, which is a mixture of amino acid and different salts that destroys the sensory receptors of the predator. Humans produce a pheromone, which acts outside the body on other individuals. Androstenes is the most common one. These are similar to hormones, but they work outside the body. Hence different animals produce different chemicals in response to external stimuli. These include pheromones, Ink, and volatile organic compounds.

In this topic, students will learn about:

- Types of chemical biosignals
- Examples of biosignals

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Lesson 32 : Thermal Bio-signals

Thermal biosignals are related to heat release or heat absorption leading to a change in the standard temperature of the body. This phenomenon is the indication of a physiological change or a pathological condition. The average body temperature of the human body is 370C to 37.80C. Thermoregulation is a process that allows our body to maintain its core internal temperature. All thermoregulation mechanisms are designed to return the human body to homeostasis. The factors that can raise body temperature are fever and heat stroke. Onion contains sulfur compounds that can raise body temperature by touching the body. The factors that can lower the body temperature are metabolic conditions, including under-functioning of the thyroid gland, cold sponging, and hypothermia. Hypothermia occurs as body temperature falls below 35C. Thermal biosignal tells about the death time of a body. After death body begins to cool at a set rate: about 1.50F falls in an hour. The average temperature of the skin is about 330C. It varies slightly throughout the body. The inflamed part of the body has a higher temperature indicated by thermal biosignals.

In this topic, students will learn about:

- Application of thermal biosignals
- Hypothermia
- Factors lowering and raising body temperature

Lesson 33 : Latest Research and Current Trends in Bio-signals

Summary/Handouts

The knowledge of biosignals is utilized for diagnostic purposes. Different equipment and instruments are designed to use normal and pathological biosignals' values. These biosignals are used in different medical devices. Some of them are:

- Electrocardiogram
- Electroencephalogram
- Electromyogram
- Mechanomayogram
- Galvanic Skin Respons

Electrocardiogram (ECG) is a technique that relates the electrical activity of heart muscle in a time frame. It records heart activity in graphical representation. It tells about two ways: P (Atrial depolarization) and ventricular QRS. Electroencephalogram (EEG) determines the electrical

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waves of the brain. It is used to study the nervous system, monitors sleep stages, and diagnose diseases like epilepsy. Its analysis is based on external frequency. In epilepsy, a comparison is made between normal and pathological signals for diagnosis. Electromyogram (EMG) detects mechanical signals. The contraction and relaxation of muscles produce electrical signals. This technique detects and records any change occurring during contraction time. EMG technique can be used for clinical and biomedical applications. Mechanomayogram (MMG) is a low-frequency mechanical signal technique that detects automated signals when a muscle is contracted. The gross changes in muscle shape during contraction cause a prominent peak to appear in MMG. Galvanic Skin Respons (GSR) is a technique in which skin's electrical activity is recorded. A galvanometer is used that measure the autonomic nervous system activity of the system.

In this topic, students will learn about:

- Latest trends in biosignals
- Biosignal based medical devices

Lesson 34 : An Introduction to Support Material

Summary/Handouts

Support material is an essential component of a biosensor. The support material immobilizes the enzyme for a short period or permanently. It also functions in transporting generated signals to transducers. The ideal support material is cheap, inert, physically strong, and stable. It must discourage microbial growth, and there should be no specific absorption. Three physical characteristics must be present in the support material. These are mean particle size, hydrophilic character, and mechanical strength. Mean particle size is a parameter that decides the porosity of a carrier. The hydrophilic nature determines the activity level of an immobilized enzyme, and mechanically robust material is required as a support material for immobilization. Generally, support materials are classified as organic and inorganic. The organic support materials are polymers that may be of biological origin or synthetic origin. The inorganic support materials are further divided into natural and processed materials. Natural support materials include zeolite, calcite, and silica, while processed materials include porous glass, metal oxides, activated charcoal, carbon-based Nanomaterials, ceramic, etc.

The natural support materials are the organic support materials taken from animal and plant sources. They have a proteins and carbohydrate nature. Examples are alginate, cellulose, starch, gelatin, collagen, chitosan, and chitin. Synthetic polymers consist of different types of monomers. For example, polyacrylamide, polyaniline, DEAE Cellulose, polyvinyl chloride, polyethylene glycol, sepharose, TEOS derived silicates, etc. The synthetic polymers are stable and can be designed according to need. Carbon can form long carbon chains by polymerizing at the atomic level. It can link through single, double, or triple covalent bonds. Amorphous carbon has long been used in biosensors because of its porosity.

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In this topic, students will learn about:

- Support matrix in a biosensor
- Characteristics of support material
- Classification of support material

Lesson 35 : Natural Materials

Summary/Handouts

There are at least four types of materials that are being used as a platform for biosensors. These are:

- Plant-derived materials
- Animal-derived materials
- Carbon-based materials
- Silicon-based materials

Cellulose is a plant-based polymer. It is composed of glucose subunits joined through beta linkage. It is an essential structural component of the primary cell wall of green plants that is the most abundant organic polymer on earth. Humans cannot digest cellulose due to the absence of cellulose enzymes.

The other important material is Nano cellulose. It is stiff, lightweight, electrically conductive, highly absorbent, and non-toxic. Its crystalline form is transparent and gas permeable. It has a very high tensile strength, nearly eight times higher than steel.

Alginate is another material that is a naturally occurring anionic gelatinous polysaccharide. It is present in the cell walls of algae and seaweed. It has two basic units: D-mannuronic acid and L-guluronic acid.

Diatomaceous earth is a naturally occurring soft, siliceous sedimentary rock that forms a fine white to off-white powder. It has a particle size of 10-200 μm.

Chitosan is a linear polysaccharide. It is composed of randomly distributed β-linked D-glucosamine and N-acetyl-D-glucosamine. It is made by treating the chitin shells of shrimp and other crustaceans with an alkaline solution.

Collagen is the main structural protein in the extracellular spaces in the various connective tissues in the body. It consists of amino acids bound together to form triple-helices of elongated fibrils. It is primarily present in fibrous tissues such as tendons, ligaments, and skin. It is capable of holding water.

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In this topic, students will learn about:

- Different types of support materials
- Inorganic and processed materials

Lesson 36 : Synthetic Material

Summary/Handouts

A variety of synthetic materials are used as a platform for immobilizing the Biorecognition element. These are:

- Ion exchange resins
- Aniline based polymers
- Carbon-based materials
- Silicon-based materials

Ion exchange resins are tiny beads that are composed of organic polymer chains. These beads are covered with cationic or anionic resins. The cationic resin has a negative functional group and attracts positively charged ions. The anionic resin has a positive active group and attracts negatively charged ions.

Polyaniline (PANI) is common and economical material. Its basic unit is aniline. It forms long threads with an average diameter of 30 nm-100 nm. PANI is semi-flexible and shows different colors in oxidized and reduced forms. It has three forms: oxidized, reduced, and intermediate forms. It is a poor conductor in the oxidized and reduced state.

Carbon-based materials are successfully used in biosensors as a platform. It requires a simple construction procedure with low cost. It has a low background current and has the ability for surface regeneration. But they have a short lifetime. Glassy carbon is a smart form of carbon that combines glassy and ceramic properties with graphite. It is mechanically stable, chemically resistant, highly porous, and impermeable to gases and fluids. It produces a low background current with low sensitivity to peroxide and other mediators.

In this topic, students will learn about:

- Different synthetic materials
- Ion exchange resins
- Use and characteristics of Polyaniline
- Use and characteristics of glossy carbon

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Lesson 37 : Hydrogels

Summary/Handouts

Hydrogels are three-dimensional networks of natural or synthetic polymers. They can absorb a large amount of water or biological fluids and do not dribble water due to the presence of a novel exclusive zone (about 100 μm). Different types of hydrogels are made from natural and synthetic polymers. The natural polymers could be carbohydrates or proteinases in nature. Proteins used are collagen and gelatin, while polysaccharides are starch, alginate, and agarose. Synthetic polymers are prepared using chemical polymerization methods.

Gelatin is a natural material used for the preparation of hydrogels. It is a heterogeneous mixture of single or multi-stranded polypeptides that form gels. When gelatin is heated, the chains arrange them regularly to cool and absorb water. Each strand is composed of 50-1000 amino acids. It is a food additive and is used in chemicals.

An example of hydrogel uses the application of hydrogels in the eye lens. Modern contact lenses are made of hydrogel and silicone hydrogel polymer.

Hydrogels are also used in diapers. Sodium polyacrylate is used as a primary material in preparing hydrogels to absorb water. The Sodium polyacrylate swells on exposure to water and absorbs up to 1000 times more water than its weight. The diaper soaks up urine and keeps moisture away from the baby's skin.

Hydrogels can be regarded as valuable biomaterials for human beings. They are used in:

- Tissue engineering
- Dressing of wounds
- Cosmetics
- Hair gels
- Biosensing

In this topic, students will learn about:

- Characteristics of hydrogels
- Production of hydrogels
- Applications of hydrogels in medical and other sciences

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Lesson 38 : Immobilization

Immobilization is the binding of biomolecules onto solid support with functional activity. It gives a stable environment to the biomolecules. In biosensors, the proximity of the biomolecules with the transducer is required. There are three components of immobilization:

- Biological recognition element
- Support matrix
- Mode of attachment

There are two methods for immobilization: chemical and physical methods. Chemical methods include covalent binding and copolymerization. Physical methods include adsorption, entrapment, and encapsulation, and weak forces are used.

Adsorption is a straightforward process that involves weak forces such as hydrogen bonding and Vandar Wall forces. The matrix particle size must be small for adsorption. It is a straightforward process with limited loss of enzyme activity. Its disadvantages include desorption and Non-specific adsorption.

Another method is covalent coupling, chemical binding between functional groups of enzyme and support. The method is stable, and there is no diffusion barrier. The disadvantages include high enzyme activity loss and matrixes being non-regeneratable.

Entrapment is the incorporation of the enzyme in gel or polymer. Several types of enzymes can be immobilized within the same polymer diffusion barrier. The disadvantages include enzyme leakage and the need for a high concentration of enzyme and monomer for electro-polymerization.

Cross-linking biological recognition elements with the support material involves bonding between the amino group of the matrix and the amino group of an enzyme—for example, Glutaraldehyde. The advantages include minor loss of enzymes, high enzyme stability, and low cost.

In this topic, students will learn about:

- Immobilization
- Components of Immobilization
- Methods of Immobilization

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Lesson 39 : Reversible Immobilization

Reversible Immobilization is the method in which biological recognition element (BRE) can be removed from the matrix. It is temporarily fixed on the support material. Generally, no extreme conditions are required for the process, and it allows multiple times of use of BRE. Adsorption is the method used for reversible Immobilization. Only weak forces like hydrogen bonding are required. There are four methods or strategies used in adsorption: static, dynamic, reactor loading, and electro-deposition. The static method involves the Immobilization of enzyme and carrier molecules without agitation. Mixing enzyme with the carrier under constant agitation is called the dynamic process. Reactor loading involves transferring both enzyme and carrier in the reactor with agitation. The enzyme gets deposited on the surface in electro-deposition due to the electric current.

Another method used in reversible Immobilization is encapsulation. This BRE is enclosed in a vessel. It can be done by encapsulation as a reaction vessel, encapsulation by hollow fibre membrane, microencapsulation by chemical polymerization by using 1-6-diamino hexane and encapsulation by liposomes. Entrapment is another method of reversible Immobilization that further has various types:

- Inclusion in gel
- Inclusion in fibre
- Inclusion in microcapsules

In this topic, students will learn about:

- Reversible Immobilization
- Methods of adsorption
- Methods of entrapment

Lesson 40 : Irreversible Immobilization

Irreversible immobilization methods are permanent methods that involve the formation of a covalent bond between biological recognition elements and support materials. The complex formed through this bonding is very stable with no enzyme loss. Covalent bonding occurs between the active parts, such as the functional group of an enzyme and carrier molecules. Lysine is a critical amino acid that could bind covalent bonding sites. It has a wide surface and high reactivity, especially slightly alkaline solutions. Epsilon amino group lysine takes part in the reaction.

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Different support materials are available for immobilization, including cellulose, gelatin, collagen, porous glass, silica, polyacrylamide, and CNBr-Agarose.

Usually, three types of methods are used for irreversible immobilization. A Diazotization reaction occurs between a primary aromatic amino acid and nitrous acid to give a diazo compound. Peptide bond formation occurs between a carboxyl and amino group of carrier and protein. The third method is forming a bond between the amino group of carrier and support.

The reagents used in the diazotization are sodium nitrite, hydrochloric acid and cyanogen bromide. It activates the sepharose or agarose hydroxyl group that reacts with the Biological recognition element. Another reagent is glutaraldehyde, which reacts with the primary amino group. It requires low ionic strength and acts as a link between support material and protein.

In this topic, students will learn about:

- Chemistry of irreversible immobilization
- Support material for immobilization
- Methods of irreversible immobilization
- Reagents used in immobilization

Lesson 41 : Immobilization Application

Summary/Handouts

Immobilization is characterized as the method of fixing the cells, organelles or compounds/different proteins onto a strong supportive network, into a strong support lattice or held by a film, to keep up the stability and reproducibility. Immobilization is an important step used in the fabrication of biosensors. Covalent binding, Adsorption, Cross linking, Encapsulation and Entrapment are some of the common methods of immobilization used to make the biosensors more accurate and efficient. Likewise, different support materials are used in the fabrication of biosensors such as porous carbon fiber whose size can be tailored and is very useful in immobilization of thermostable enzymes. Similarly, Polyaniline (PANI) is also used as a support material in fabrication of biosensors. PANI is an insulator and is very useful in redox reaction and have high conductivity. Moreover, nanoporous gold is also used as a support material due its high surface area, more than one protein can be immobilized.

In conclusion, immobilization is a modern technique and applied in biosensors to increase the efficiency, thermal stability, sensitivity, storage stability, and reproducibility. Above mentioned, gold nanoparticles are successfully used in the immobilization due its high surface area, more binding capacity and less toxic.

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- Application of Immobilization in Biosensor
- Material for immobilization

Lesson 42 : Immobilization Latest Trends

Summary/Handouts

Various trends are utilized in the fabrication of biosensors in order to increase the stability, thermal efficiency, sensitivity, selectivity and reproducibility. One of the techniques is Photonic Immobilization technique used to immobilize antibody on the surface of biosensor, utilizing gold nanoparticles and the Triade of amino acids (tryptophan, cysteine and cysteine). This triade is present on the Fc region of antibodies. In this technique, UV signals are used that react with tryptophan and as a result tryptophan absorb the proton and transferred it to neighbouring cysteine and as a result energy is produced due to which thiol channels are opened and attached to gold nanoparticles. Another novel technique is immobilization on reduced graphene oxide and gold nanoparticles in which single stranded DNA is used that is immobilized on the surface of gold nanoparticles used in the detection of TB. Bioanode is also used for highly efficient enzyme reaction on the surface of electrode.

In conclusion, surface made up of gold nanoparticles is very effective in the immobilization of DNA, protein and other biomolecules. Amino acid residues are also used for the targeted enzyme immobilization. Similarly, enzymes especially thermostable enzymes are utilized in the biosensor to increase the stability and effectivity of biosensor.

In this topic, students will learn about:

- Recent trends in Biosensor
- Novel strategies for biosensors fabrication

Lesson 43 : Nanoparticles in Biosensors

Nanoparticles are ultra-small particles with a dimension of 1-100nm and are used to fabricate biosensors. It has all three external dimensions in the nanoscale. Its longest and shortest axes do not differ significantly. Natural and artificial sources produce nanoparticles. Natural sources include combustion products like forest fires, volcanic ash, ocean spray, and the radioactive decay of radon gas. The artificial sources include metals. Metal oxides, non-metals, biological molecules, and polymers. They usually are classified as organic and inorganic nanoparticles.

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Organic nanoparticles include dendrimers, liposomes, carbon-based and polymeric nanoparticles. The inorganic nanoparticles include mesoporous silica, gold nanoparticles, iron oxide, and quantum dots.

Another particular type is known as engineered nanoparticles. These are engineered for shape, size, and surface properties. They impart particular functional catalytic behavior, electrical conductivity, improved strength, and controlled release of host molecules.

Nanoparticles have four distinct properties. They are highly mobile in a free state, have an enormous surface area, and exhibit quantum effects. They are ultrafine units with a size range of 1-100nm.

The nanoparticles are very heterogeneous in shape. Depending on the production method and manufacturing conditions, they come in different shapes, including spherical, triangular, cubical, oval, helical, pyramidal, and rod-like.

In this topic, students will learn about:

- Characteristics of nanoparticles
- Types of nanoparticles
- Shape and distinct properties of nanoparticles

Lesson 44 : Organic and Inorganic Nanoparticles

There are two types of nanoparticles: organic and inorganic. Both types of nanoparticles are used in the fabrication of biosensors. Organic nanoparticles are derived from biological materials and processed materials such as lipids and biological polymers. These are carbon nanoparticles. The organic nanoparticles are compatible and widely used in the biomedical field.

Polymeric nanoparticles are prepared from natural (gelatin) and synthetic polymers (polymeric acid) used in drug delivery. Polymeric nanoparticles are of two types: nanocapsule and nanosphere.

Liposomes are another type of nanoparticles that are known as drug delivery vehicles. They can be formulated with various natural, synthetic, and modified lipid species to deliver drugs and contrast agents. They are used in drug delivery, transfection, as a precursor for lipid coatings, and model cell membrane preparation.

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Dendrimers are repetitively branched molecules. They have three main components: a central core, interior branches, and terminal functional group. The terminal functional group is present at the outer surface of the molecule and used for in vivo and in vitro transport of DNA.

Inorganic nanoparticles are broad, including elemental metals, metal oxides, and metal salts. Compared to organic materials, these are non-toxic, hydrophilic, biocompatible, and highly stable. They are widely used in drug delivery. Metallic nanoparticles include aluminium, nickel, manganese, silver, and gold. Silicate nanoparticles exhibit intrinsic surface reactivity for chemical modifications.

In this topic, students will learn about:

- Different types of organic nanoparticles
- Inorganic nanoparticles
- Characteristics and uses of nanoparticles

Lesson 45 : Synthesis of Nanoparticles and Sol Gel

There are different methods available for the synthesis of nanoparticles. Free nanoparticles are formed either by the breakdown of large particles or by a controlled assembly process. The controlled condition provides an efficient product. For example, burning wood and other domestic activities produce nanoparticles, but these are inefficient methods. Usually, there are three types of methods used to synthesize nanoparticles. These are biological, physical, and chemical methods. Biological methods involve living organisms, and physical methods involve condensation, evaporation, etc.; chemical methods involve chemical techniques and reagents like thermal decomposition to synthesize nanoparticles.

A plasma source system is a method in which an inert gas 'argon' is used. The gas carries microscopic particles, and plasma is produced by a high-power radio frequency signal applied to the carrier gas. The ions then condense into nanoparticles.

Sol-gel is a chemical method that uses colloidal particles and chemical solutions to synthesize nanoparticles. It involves partial hydrolysis of metal alkoxides to form reactive monomers. These monomers are condensed to form colloidal-like oligomers. The addition hydrolysis is done for polymerization, followed by drying.

Nanoparticles are also synthesized by the green synthesis method. The plant leaf extracts are an excellent beginning source of metal and metal oxide to synthesize nanoparticles. It is widely used in science for the development of new chemicals.

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In this topic, students will learn about:

- How to synthesize efficient nanoparticles
- Green synthesis method
- Plasma source system
- Sol-gel method

Lesson 46 : Nanogold Particles

Nanogold or colloidal gold is 1-50nm in size. They can be easily conjugated using thiol chemistry and are highly stable. They do not have any adverse effect on BRE and provide a large surface area for BRE. Gold nanoparticles are unique in properties as they come in different shapes such as stars, rods, spheres and branches etc. Another essential property of gold nanoparticles is quantum property. They absorb light in different areas. Different ways can synthesize these. the most common method is the reduction of gold salt.

Another method for synthesis is the reduction of gold salt by sodium tetraborate, which involves seeding in the presence of sodium borate, followed by the separation of gold particles in the presence of ascorbic acid.

A green method is also available for the synthesis of nanogold particles. In this, a gold compound chloroauric acid is reduced with plant extract. The size of gold nanoparticles ranges between 10-20nm and are spherical. It is a very economical and straightforward method.

Nanogold particles are biocompatible and are used in every medical field. They are not lithal to living organisms. They are used in electronics, sensors, probes, biosensors, therapeutic agent delivery and tissue engineering.

in this topic, students will learn about:

- Properties of nanogold particles
- Method available for the synthesis of nanogold particles
- Applications of nanogold particles

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Lesson 47 : Quantum Dots

Quantum dots are a type of nanoparticle. They are nanocrystals of semiconductors with a diameter of 2-10nm. They are fluorescent and comprised of two semiconductor materials. For example, lead sulphide and lead selenide. Colour emission is an essential property of quantum dots. They can emit any colour of light from the same material simply by changing the dot size. Small quantum dots require more energy to excite them. The small dots with higher energy produce higher frequencies and shorter wavelengths. The smaller emit blue light, and the intermediate emits green light. The larger dots produce a red colour.

Several ways are available to fabricate quantum dots, including colloidal synthesis, plasma synthesis, and the green method. The quantum dots are synthesized from solutions like a chemical process in colloidal synthesis. Plasma synthesis is the gas-phase approach. The green method involves reducing plant-derived material into quantum dots.

Quantum dots can be classified into different types based on composition and structures. These are core-type quantum dots, core-shell quantum dots, and alloyed quantum dots. The core and the shell are semiconductors and have low fluorescence quantum yield.

In this topic, students will learn about:

- characteristics and properties of quantum dots
- Fabrication of quantum dots
- Types of quantum dots

Lesson 48 : Zinc Oxide nanoparticles

Zinc oxide is a metal oxide. Zinc is a brittle metal at room temperature with the atomic number 30. It is the first element in group 12 of the periodic table. Zinc oxide is used for the production of nanoparticles. Zinc oxide exists in two crystalline forms: hexagonal wurtzite crystals and cubic zinc blend crystals. The wurtzite structure is most stable at ambient temperature. These zinc oxide nanoparticles exist in the wurtzite structure. Zinc oxide has a sizeable direct bandgap of around 3.3 eV at room temperature. The advantages associated with a large bandgap include:

Higher breakdown voltage

Low electronic noise

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High power operation

High-temperature operation

Ability to sustain a large electric field

The critical property of zin oxide is its ability to give a spectrum at 355 nm. In the photoluminescence spectrum, it exhibits two emission peaks at 392nm and 520 nm. These zinc oxide nanoparticles are being used in medicine and beauty products. They are thermally resistant, chemically stable and have anti-bacterial activity. They have applications in gene therapy, immunotherapy, drug delivery, biosensors, angiogenesis, and anti-cancer agents. They are used in tissue engineering and anti-diabetic cure. They are used in medical sciences as nanogenerators, gas sensors, solar cells and photodetectors. The zinc oxide nanoparticles are synthesized by physical, chemical, and biological methods.

In this topic, students will learn about:

- Emission spectrum of zinc oxide nanoparticles
- Crystalline forms of zinc oxide nanoparticles
- Applications of zinc oxide nanoparticles

Lesson 49 : Carbon nanotubes and Graphene oxide

Summary/Handouts

Carbon Nanotube and Graphene oxide are nanomaterials used in the fabrication of biosensors. Graphene is an allotrope of carbon available in the form of a single layer of atoms. It has a two-dimensional hexagonal lattice. It is the fundamental structural element of allotropes, including graphite, charcoal, carbon nanotubes and fluorescences. In the structure, each carbon atom acts as a vertex.

Fullerene is a nanomaterial formed by the twisting and wrapping of graphene sheets. Fullerene is single and double bonds with five to seven atoms fused rings. The molecules may be hollow, sphere, cube, ellipsoid or of another shape.

The rolling of graphene forms carbon Nanotubes. They are cylindrical and elastic molecules with high tensile strength. The unique strength of carbon nanotubes is due to orbital hybridization. An example of carbon nanotubes is single-walled carbon nanotubes. They are of three types, including zigzag, chiral and armchair.

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Graphene oxide is a single monomolecular layer of graphite. It contains different functional groups such as epoxide, carbonyl, carboxyl, and hydroxyl. It has optical and electronic properties. It is used in biosensors and other electronic devices.

In this topic, students will learn about:

- Formation of Fullerene
- Characteristics of Carbon Nanotubes
- Graphene oxide

Lesson 50 : Porous Gold

Porous gold is a sponge structure made up of different gold ligaments. They have sizes of about 20-50nm. They are made up of silver (3%) and gold (97%). It has high catalytic activity and acts as a platform for multiple reactions. Its structure is based on the self-organization process. It is a three-dimensional sponge gold structure that is evolved with ligaments in the range of only a few tens of nanometers. Due to its porosity, the material can be penetrated by gases which then adsorb and interact with the surface. The nanoporous gold (NPG) is a heterogeneous catalyst with unique structural properties, including open non-porosity, high surface area, excellent electrical conductivity, nontoxic nature, easy recyclability, and tunable surface chemistry. The porous gold can catalyze CO oxidation, alcoholic oxidation, hydrogen oxidation, and reactions in the liquid phase. Reduction reactions or hydrogenation are not accessible on gold due to H-H disassociation.

Porous gold material provides support for CO oxidation. The process is very effective at low temperatures as 30 degrees in the presence of a porous gold catalyst.

Porous gold helps in alcoholic oxidation. Selective oxidation of alcohol is a critical catalytic conversion in the petroleum industry. In gold-based catalytic conversion, oxygen can be used as an oxidant under mild conditions and ambient pressure.

In this topic, students will learn about:

- Structure and composition of Nanoporous gold
- Reactions catalyzed by porous gold

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Lesson 51 : Latest Research and Current Trends

Nanoparticles are ultra-
small particles. They have a high loading capacity due to the large surface area. They can easily t ravel into the target cell due to their small size, shape and crystallinity. These nanoparticles are u sed in tissue engineering. For example, these nanoparticles are used to regenerate bones through bioactive hydrogels. These hydrogels are prepared through two types of monomers by their cross -linking.

Currently, seven types of nanoparticles are present. These are carbon-
based nanoparticles including graphene and nanotubes, silicon-
based nanoparticles including quantum dots, metal-
based nanoparticles composed of metal oxides, silver or gold, liposomes formed by lipids, polym er-based nanoparticles etc. These nanoparticles are used in drug delivery and for anti-
bacterial and anti-
fungal activities. These nanoparticles are used in biosensors, diagnostic tools, photoimaging and for purification of biological molecules. These nanoparticles are also used in the environment. Te xtile wastewater contains many carcinogenic dyes that are effectively removed by graphene dioxi de. Graphene oxide is an oxidized form of graphene with different functional groups.

In this topic, students will learn about:

- Current applications of nanoparticles
- Types of nanoparticles
- Cleaning of textile wastewater

Lesson 52 : Electrochemical Sensor

Electrochemical sensors are based on the principle of electrochemistry, in which oxidation and reduction reactions are monitored. These oxidation and reduction reactions involve the transfer of electrons between two electrodes leading to the generation of electric current. A typical electrochemical sensor is composed of seven main components. These are gas-permeable membrane, sensing electrode, reference electrode, a counter electrode, electrolyte, filter and transducer. The gas-permeable membrane is the essential component that is primarily hydrophobic. The working of electrochemical sensors involves physical and chemical processes. The physical process is the diffusion of gas of interest into the sensor. The diffused gas gets oxidized or reduced at the sensing electrode. It produces an electrical signal that is proportional

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to the gas concentration. Factors that affect sensor performance are temperature, pressure, and humidity changes.

Usually, three electrochemical sensors are present: potentiometric, amperometric, and conducto metric. The potentiometric sensor uses ion-
selective electrodes to determine the concentration of ions. Amperometric measure current resulti ng from redox reaction and conductometric measures changes in ionic strength composition resul ting from an enzyme reaction.

An example of an electrochemical sensor is an alcohol sensor, also known as a breath analyzer. I n this ethanol is oxidized to acetic acid in the presence of oxygen.

In this topic, students will learn about:

- Composition of electrochemical sensors
- Types of electrochemical sensors
- Alcohol sensor

Lesson 53 : Mass Sensitive Sensor

Mass is a physical property of matter. Mass sensitive sensors determine the amount of mass at the nanoscale. Basically two types of techniques are used in mass sensitive sensors. These are the piezoelectric technique and surface plasmon resonance technique. Piezoelectric sensors convert the pressure or mechanical stress into an electrical force. This technique measures physical quantities like pressure, force, and stress. Piezoelectric materials are used in Piezoelectric technology. These materials can generate internal electrical charges from applied mechanical stress. Quartz is used for mass sensitive sensors. Quartz crystal microbalance is a mass sensitive piezoelectric device. It consists of a gold electrode, quartz crystal, sensitive coating and voltmeter. The quartz crystal oscillates at a defined frequency by applying an appropriate voltage through metal electrodes.

The surface plasmon resonance technique includes an optical phenomenon generated by light. The technique is based on the plasmonic effect on the surface. The plasmonic effect is the interaction between free electrons in the metal nanoparticles and incident light.

In this topic, students will learn about:

- Techniques used in mass sensitive sensors
- Quartz crystal microbalance

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Lesson 54 : Conductometric Sensors

A conductometric sensor is a type of electrochemical sensor. These sensors depend on the measurement of conductivity. Conductivity is the property of a solution or electrolyte to pass an electric current through it. The conductometry monitor changes in the electrical conductivity of a test sample. This method based on the concentration of ions is of great importance in detecting toxic and harmful gases like ammonia. Conductivity is measured by a conductivity cell. The simplest measuring cell used consists of two similar electrodes. An alternating voltage is applied to one of the conductivity electrodes that causes the ions in the solution to migrate towards the electrodes. Conductometric methods are based on chemical reactions. In a solution, the gas reacts chemically, and the conductivity is measured before and after the reaction by a differential method. Conductometric gas sensors are widely used for environmental monitoring. They have multiple properties, including high sensitivity, short response time, low cost, device feasibility, and portability.

Conductometric sensors can be effectively used in enzyme biosensors. The conductivity can be used to quantify the extent of change in the REDOX Enzyme reaction. If the change is too small to detect, resistivity can be measured using a conductivity bridge. Its most important application is to determine the quality of water.

In this topic, students will learn about:

- Conductometric sensor
- Applications of Conductometric sensors
- Working of conductometric sensor

Lesson 55 : Calorimetric Sensor

Calorimetric Sensors involve the measurement of heat that is generated in an enzymatic reaction. These sensors typically utilize thermistors that transform heat generated or lost during a reaction in to an electric signal. The enzyme is immobilized in a packed bad column that is placed in the center of the insulated chamber. The sample is allowed to pass through a coiled column of aluminum that served as a heat exchanger to ensure uniform temperature of the sample entering the reaction chamber. A thermistor is placed at the inlet of the reactor containing the immobilized enzyme and another thermistor is placed at the exit of the reactor to measure the temperature after the conversion of the substrate into the product. A thermistor record temperature changes by altering the resistance. Higher temperature will cause reduction

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in the resistance of thermistor. The difference in resistance between the thermistor located at the entry in the exit point to the reactor is recorded as a measure of the temperature change produce.

There are certain limitations of calorimetric sensor. The necessity of maintaining the insulated jacket to prevent loss of heat, make the system bulky. The sensitivity depends upon accuracy of the thermistor employed generally the sensitivity of the calorimetric sensor is very low. Large sample volumes are required for detectable change.

the quality of water.

In this topic, students will learn about:

- Calorimetric sensor
- Working of Calorimetric sensor

Lesson 56 : Amperometric Sensor

It involves measuring the current generated by electrochemical oxidation or reduction of electronegative species, and constant applied potential. First measurement sensitivity (ability to sense even 10-9 M concentration) and ability to perform measurements on turbid/opaque solutions are its advantages over conventional optical sensors. However, pH-sensing mechanisms require weakly buffered or non-buffered solutions and this is a drawback for this category of sensors.

Amperometry is one of an electrochemical method. The potential applied to a sensing electrode is controlled instrumentally; As a result, current is generated due to REDOX reactions. The current occurring at electrode surface is recorded as analytical signal. A common example of Amperometric biosensor is the Glucose biosensor which has paved the path for biosensors in medical diagnostics.

In this topic, students will learn about:

- Amperometric sensor
- Working principle of Amperometric sensor

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Lesson 57 : Potentiometric Sensor

Potentiometric electrodes measure the potential difference of an electrode in the presence of a low current. The signal is measured as the potential difference between the working and reference electrodes. Different electrodes are used in potentiometric sensors, such as chemically modified electrodes, solid-state devices, membrane electrodes, screen printed electrodes as in glucometers, and ion selected filed effect transistors. A salt bridge is used for maintaining neutrality. It prevents the cell from rapidly processing its reaction to attain equilibrium. It usually uses potassium or sodium chloride solutions. The anions and cations in these solutions have similar conductivity and migratory speed. There are two types of salt bridges: paper salt bridges and Glass bridges. The paper salt bridge consists of filter paper soaked in an inert electrolyte that provides a solid medium for conduction. Glass Bridge consists of a U-shaped tube that connects the two cells. A potentiometer determines the potential difference between anode and cathode. Calomel electrode is an example of potentiometric sensors in which Hg or Hg salt is used. This mixture is packed in a glass tube situated in a second tube consisting KCl solution. Holes are created in an inner tube and a porous wick act as a salt bridge. These sensors are used in wearable wire sensors that are present in watches. These sensors are ion-selective electrodes. They have many applications in health, agriculture and the environment.

In this topic, students will learn about:

- Working and construction of potentiometric sensors
- Role of salt bridge
- Types of salt bridge
- Applications of potentiometric sensors

Lesson 58 : Piezoelectric biosensors

Piezoelectric sensors are piezoelectric materials that have the property of Piezoelectricity. These materials can convert the mechanical push into electricity. Piezoelectricity is a reversible property that can change the dimensions of objects. They can be natural or synthetic. Natural materials have two types. The first type includes DNA, collagen, bones and sugars, etc. The second type includes minerals, quartz, formalin, etc., synthetic material, including lead, ceramics, etc. Rochelle salt is sodium and potassium tartrate that has a piezoelectric effect. Piezoelectric materials also have Curie temperature. It is the point at which magnetic materials undergo a sharp change in their properties. Piezoelectric technology is based on polarity. There are two types of crystals; monocrystals and polycrystals. There is one polar axis in monocrystals,

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and in polycrystals, there is a random polar axis. When voltage is applied to polycrystals, the dipole is aligned, and the piezoelectric effect is observed.

Quartz is a complicated crystal. These crystals are cut along the X, y, and Z-axis. The angle at which a crystal is cut can affect its performance. Piezoelectric sensors use the piezoelectric effect to measure pressure, temperature, and force changes. These sensors are used in daily life. For example, quartz is used under the push button of cigarette lighters. It helps in generating the spark. They are also used in microphones and speakers etc.

In this topic, students will learn about:

- Piezoelectric sensors
- Types of Piezoelectric sensors
- Types of crystals

Lesson 59 : Surface Plasmon Resonance Sensor

Capsule plasma sensors are optical sensors in which plasma plays an important role. Plasmon is the vibration of an electronic cloud in metal that oscillates at a particular frequency. Surface Plasmon oscillates only on the surface. Surface Plasmon resonance is an optical technique that involves total internal reflection. This technique mainly depends upon metal properties, the wavelength of incident light and reflective index, which is temperature-sensitive. Metals have conduction bands with high energy orbitals and a valance band where valance electrons are present. Metals on the sensor must have conduct electrons capable of resonating with the incident light. They should be free of oxides and must not react with other molecules. If gold is used in sensors, it must be of 50 nm thickness with constant incident light and temperature. To use SPR in biosensors, we can immobilize the recognition element on a gold surface. It shows variation in optical properties that change the refractive index when detected. The sensors monitor physiological effects in cells for culturing layers of living cells on gold surfaces. Gold is an ideal metal in these sensors.

In this topic, students will learn about:

- Capsule plasma sensors
- Surface Plasmon resonance

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Lesson 60 : Optical Sensor

The optical sensors use a monochromatic light source to measure a physical quantity. These optical sensors are based on optic law. They have enormous applications. Generally, two light sources are used in optical sensors: natural and artificial sources. The natural source includes all electromagnetic radiations such as infra-red, gamma rays, visible, UV, and others, while artificial light sources include LASER and light-emitting Diodes.

Optical sensors are used for contact-less detection, counting or positioning of parts. They can measure rotation, pressure, strain, volume etc. Light may change its optical properties during sensing, including intensity, phase, polarization, wavelength, or spectral distribution. Optical sensors can be intrinsic or extrinsic. Intrinsic optical sensors are placed internally to sensors, and they measure the internal change. For example: an endoscope that detects internal defects of the body. It has a thin long flexible tube, a lens, an eyepiece and a light-transmitting system. Capsule endoscopy uses a small capsule to provide images of the gastrointestinal tract.

Extrinsic optical sensors measure different physical parameters such as temperature, pressure, liquid level, rotation, force radiation, pH, humidity, strain, electric field etc.

An interesting example is Bank Note Counter. It uses visible light, infrared light and UV light. Visible light detects the fake note. Infra-red light sees the banknote image with low-frequency light, and UV detects the banknote image with high-frequency light.

In this topic, students will learn about:

- Working and uses of optical sensors
- Types of optical sensors
- Capsule endoscopy
- Bank Note Counter

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Lesson 61 : Light sources for optical sensors

Optical sensors use different types of light according to the requirement. Generally, monochromatic light is used consists of an extremely narrow wavelength. The two most commonly used optical sensors are Light-Emitting Diodes (LED) and LASER diodes.

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Light-Emitting Diodes (LED) are transistors that use semi-conductor material with P-type and N-type materials. The colour of the emitted light depends on the material used for doping. In LEDs, current flows only in one direction. LEDs are classified based on colours. These may be Gallium Arsenide infra-red, Aluminum Gallium Phosphide that detects green, and Gallium Indium Nitride is used for ultraviolet, blue and bluish green. For white light, a blue LED is quoted with phosphorous.

For LASER rays production, a material is necessary named LASER gain Medium. It could be an atom, molecule, liquids, dielectric solids or semi-conductors. A LASER gain Medium is required for the creation of a LASER beam. This LASER gain Medium is excited by providing energy to it. Electrons move to a high energy state. When these electrons return to the ground state, they release heat energy and enter in Lasing state. In between the Lasing and ground state, a LASER light is emitted. LASERS are of multiple types, including gas LASERS, solid LASERS, molecular LASERS, free-electron LASERS etc.

In this topic, students will learn about:

- Light sources of optical sensors
- Working of Light-Emitting Diodes (LED) and LASER diodes
- Types of LASERS

Lesson 62 : Types of Optical Sensor

Optical sensors can be classified based on location (extrinsic and intrinsic), output (resistance, current, voltage), and applications. Based on the output of optical sensors, these are classified as photovoltaic cells, photodiodes, and photoconductive devices. Photovoltaic cells are solar cells that convert an amount of incident light into an output voltage. This cell comprises positive and negative junctions of Boron doping (P-type) and phosphorous doping (n-type). The common single-junction silicon solar cell can produce a maximum open-circuit voltage of approximately 0.5 to 0.6 volts.

The photodiodes sensors convert an amount of incident light into an output current. It may contain optical filters or lenses. These may be composed of silicon, germanium, indium, gallium, mercury, etc.

The photoconductive devices measure the resistance by converting a change of incident light into a resistance change. Photoconductivity is an optical and electrical phenomenon. The material becomes more electrically conductive due to the absorption of electromagnetic radiation. The photoconductive materials increase the flow of electrons and reduce the resistance. There are three types of materials used in photoconductive sensors; polyvinyl carbazole, lead sulphide, and selenium. Polyvinyl carbazole is temperature-resistant, while selenium is a metal.

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In this topic, students will learn about:

- Classification of optical sensors
- Materials used in the photoconductive sensors

Lesson 63 : Applications of Optical Sensor

Optical sensors have many applications including health, diagnostics, defense, research, surveillance, environment and process control. They can address almost all aspects of detection technology. The optical sensors range from micro-probes to large devices. They are used in biometric verifications.

Biometric verification is any means by which a person can be uniquely identified by evaluating one or more distinguishing biological traits.

Ambient light sensor is that light which is already present in a scene, or lights. It is a photo detector that is used to sense the amount of ambient light present.

An important example of optical sensor is LASER guidance. LASER guidance directs a robotic system to a target position by means of a laser beam. Laser guided technology is used by military LASER guided precision targeting.

The next application is based on liquid level that is Liquid Level Indicator. The Optical Sensor based Liquid Level Indicator consists of three components: Infrared light emitting diodes coupled with a light transistor and a transparent prism tip.

Pulse Oximeter is another application that determines the level of oxygen saturation in blood in an individual without taking the blood sample.

In this topic, students will learn about:

- Applications of optical biosensors
- Biometric verification
- Ambient light sensor
- Liquid Level Indicator

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Lesson 64 : Temperature Sensor Working and Applications

Temperature sensor is a device that measures temperature through an electric signal. There are four types of scales commonly used for measurement of temperature. These are: Celsius, Kelvin, Fahrenheit, and Rankine. Temperature Sensors are classified into two types: Contact temperature sensors and non-contact temperature sensors. Contact temperature sensors require being in physical contact with the object being sensed. They use conduction to monitor change in temperature. Non-contact Temperature sensors use convection and radiations to monitor changes in temperature. They are used to detect liquid and gases that emit radiant energy.

A thermocouple is a sensor that measures the temperature. It consists of two different types of metals, joined together at one end. When the junction of the two metals is heated or cooled, a voltage is created that can be correlated back to temperature.

Thermostat is an important instrument that senses the temperature of a physical system and performs actions so that the system's temperature is maintained near a desired setpoint. It is made up of a bi-metallic strip. When it is cooled, acts as a bridge to the electric circuit. When it is hot, iron expands less than brass and circuit breaks. Liquid crystals have ability to map out the thermal regions of different temperatures. They change colour with change of temperature.

Thermal cameras detect temperature by recognizing and capturing different level of infrared light. Thermal imaging is a technique that is used for diagnostic purpose using thermal cameras.

In this topic, students will learn about:

- Temperature sensors and classification
- Construction of Thermocouple
- Working of thermostat
- Thermal cameras and thermal imaging

Lesson 65 : L65. Pressure Sensor

A pressure sensor is a device that measures the pressure of gases or liquids and piezoelectric materials. They can also measure other variables such as fluid flow, speed, water level and altitude. Pressure is sensed by different mechanical elements such as plates, shells, tubes, and diaphragms. These elements are designed and constructed to deflect when pressure is applied. The basic mechanism in the pressure sensor is to convert the pressure to physical movements. Pressure can be measured in three ways: absolute pressure, gauge pressure, and differential pressure. Absolute pressure measurement is measured relative to a vacuum. Gauge pressure is

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measured relative to ambient atmospheric pressure, and differential pressure measurements are taken concerning a specific reference to pressure.

Blood pressure is commonly measured by a sphygmomanometer containing an inflatable cuff and a mercury manometer. Recently a wearable wrist biosensor is introduced. It has two types of sensors: photo-plethysmography and impedance-plethysmography. A pressure biosensor is designed to measure blood glucose levels using oxygen levels. An implantable sensor is designed to provide convenient, accurate, and on-demand Intra Ocular pressure monitoring in the home environment. The sensor is excited by infra-red light from a tungsten bulb.

In this topic, students will learn about:

- Pressure sensors
- Measurement of Blood pressure
- Applications of pressure sensors

Lesson 66 : Physiological Sensor

Physiological sensors measure physiological parameters, which are very important for our health. There are four to six parameters related to a person's condition. These parameters reflect the general as well as diseased state of a person. These are also called physiological monitors. Electrodes determine the heart rate; Temperature sensors determine temperature; blood pressure is determined by pressure sensors and glucose changes by electrochemical sensors etc. These sensors measure more than one parameter: heart rate, blood pressure, and pulse rate. They typically consist of a central station that receives generated information; a bedside monitor that displays all the information such as heart rate, oxygen saturation, etc. The telemetry transmitter and receiver receive all the information. These sensors can be classified based on functions and type of sensor. More than one sensor can be used in it. Smart sensors and wearable sensors are types of physiological sensors. Wearable sensors can be used in elderly care systems to detect accurate positions and physical activities etc.

In this topic, students will learn about:

- Physiological sensors and their applications
- Structure of physiological sensors

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Lesson 67 : Types of Biosensor

A biosensor is a small device composed of a biological and a physical unit. The first biosensor was introduced in 1987. These can be classified into different fields. The first is based on the generation of biosensors. The second is based on the biorecognition element, and the third is based on the type of sensor element. The other is based on the n-type of application. The most convenient biosensors are based on bio-receptors. Different molecules are used in biosensors, such as enzyme biosensors, immune biosensors, DNA biosensors and biomimetic biosensors, etc. Biomimetic material is based on natural molecules which are artificially produced. If we want to produce a biomimetic of the enzyme, we will look at the enzyme's active site, which is involved in the enzyme's catalytic activity. Commercial biosensors are easy to use. There are four categories: clinical biosensors, food biosensors, biothreat, and environment. Another classification is based on the transducer of the sensor. There are three types of transducer: optical, mass-based, and Electrometric transducers, including amperometric and potentiometric sensors.

In this topic, students will learn about:

- Types of biosensors
- Types of transducers

Lesson 68 : Latest Research and Current Trends

A sensor is an input device that can detect a change after receiving a signal such as pressure, weight etc. Sensors are becoming the biggest growing market because they have applications in almost every field of life, such as smart sensors, the internet of things and robotic devices. Smart sensors are multitasking devices that detect signals saving time and energy. It contains a digital motion processor. It has three components a sensor, microprocessor and communication device. They have T smart shirt applications that use optical fibres to detect wounds and monitor heart rate. The internet of things includes computing devices, mechanical machines, digital machines, people, animals etc. They have a unique identity to access data and the ability to access data without requiring human-human interaction. It has applications in device connection, data sensing, communication, analysis, and human value. A robot is a programmed machine carrying out a series of actions automatically. Robotic sensors are used to estimate a robot's condition for further actions. These sensors are based on the functions of human sensory organs. Wearable sensing technology is one of the fastest-growing fields, at around 60 billion USD.

In this topic, students will learn about:

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- Application of sensors
- Working of Smart sensors
- Robotic sensors

Lesson 69 : An Introduction to Enzyme Biosensor

The biosensors in which enzymes are used as bio-receptors are called enzyme biosensors. Enzymes are biological catalysts produced by living organisms. The first biosensor introduced was the enzyme biosensor, and the most successful one is the Glucometer used to estimate blood glucose levels. DNS is a method used for glucose estimation. This method requires a large volume of samples, pre-treatment of samples, severe conditions, high temperature, proper lab setup, and trained staff. It is a time-consuming method. In biosensor methods, we need a very small amount of sample. For example, Glucometer uses a small amount of blood. It does not require pre-treatment or any special condition for working. Enzymes act only on their specific substrate and produce their respective products. The sensors in biosensors are dependent on the type of signal being detected. Optical sensors are used when change occurs in optical properties. The pressure sensor is used when pressure is changed. Amperometric sensors are used to detect the flow of current. Enzymes work only under optimum conditions. Their catalytic activity is dependent on pH, temperature and pressure.

In this topic, students will learn about:

- Enzyme biosensors
- DNS method

Lesson 70 : Enzyme Stabilization and Immobilization Methods

The stabilization and immobilization of enzymes, biosensors involve, wiring that is some sort of attachment of an enzyme to the electrode. It involves the co-immobilizing the enzyme and mediator directly onto the electrode surface. The most common methods of enzyme immobilization are

1. physical adsorption,

2. covalent tethering,

3. cross-linking

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4. entrapment

Physical adsorption results from hydrophobic, hydrophilic, or ionic interactions between the protein and a solid surface. There is evidence that physical adsorption of enzymes may stabilize their activity. However, this type of immobilization often leads to enzyme leakage, especially with changes in pH and temperature. Adsorption is frequently used to build layer-by-layer architectures in combination with other immobilization techniques such as cross-linking and/or entrapment behind a membrane. Sol-gels have been used to immobilize enzymes. These techniques provide an environment similar to that of the enzyme in solution.

Entrapment can also be achieved by chemical, electrochemical, or photo- polymerization in a monomer-enzyme solution. Amperometric biosensors entrapping GOX in electrochemically generated polymers have been developed.

Covalent attachment of enzymes to solid supports has been done by activation of enzyme carboxylic side chains with a carbodiimide followed by condensation with support surface amino groups.

In this topic, students will learn about:

- Stabilization of Enzyme on Biosensor
- Different immobilization methods for Enzymes

Lesson 71 : Elimination of Enzyme Leakage

In the fabrication of biosensors, the process of immobilization is very important. Different biological recognition elements are immobilized on a surface matrix. Mainly two methods are used. In the chemical method Immobilized enzyme is eliminated as it depends upon the completion of a chemical reaction and specific chemical bonding. Agarose gel is used for immobilization. It is a linear polymer of D-galactose and 3,6 anhydro-L-galactopyranose. It is used with different chemical reagents, and different resins are produced that can bind with different biological molecules, such as CNBR agarose, epoxy activated agarose, etc. CNBR immobilization method works at mild pH and temperature. Biological molecules retain biological activity, and it offers multiple points for attachment. To prepare CNBR agarose, agarose is treated with sodium hydroxide, and then cyanogen bromide is added, and we get more reactive cyanate ester and amino carbonate. There are two populations of covalently attached enzymes: liable to leakage and resistant to cleavage. These two vary with the extent of cross-linking during CNBR activation. The cyanate ester and amine group bonds are unstable and show enzyme leakage at pH 5. Epoxide-activated agarose is used to avoid leakage. It is prepared by treatment with oxiranes which helps in immobilization.

In this topic, students will learn about:

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- Chemical method of immobilization
- CNBR immobilization method
- Preparation of CNBR

Lesson 72 : Enzyme stability by Pressure

Enzymes are globular proteins produced by living organisms. It has been reported since the 1950s that trypsin and Convo-trypsin are stable under high pressure. Enzymes are composed of different amino acids. The effect of high pressure is heterogeneous due to changes in weak forces depending on the type of enzyme, pressure level, the effect pH, and temperature. The amide group forms hydrogen bonds with water or carbonyls. These bonds are shortened by pressure (0-0.11A). There are two examples for stabilizing and activating enzyme activity: glucose oxidase and polyphenol oxidase. High pressure promotes biological activity, stability, kinetic parameters, and structure changes.

Phenol peroxidase is a plant enzyme produced by several plants and is responsible for the browning of roots. Once pressure is applied to it, there occurs an improvement in biological activity and stability. When its three-dimensional structure was analyzed, it was found that there was a loss of alpha-helix content, change in beta-sheet, Trp, and Tyr residues exposed to a less hydrophobic environment, and SH group content was increased. Glucose oxidase is a common enzyme that acts on glucose. It is composed of two polypeptide chains. These chains are bound with disulfide bridges. Increased pressure leads to a stable molecule as it changes the 3-dimensional structure. The position of tryptophan is changed, and the helical content is also changed.

In this topic, students will learn about:

- Stabilization of enzyme by pressure
- Glucose oxidase
- Phenol peroxidase

Lesson 73 : Nanoparticles

Nanoparticles are used in biosensors due to their unique properties. Two types of nanoparticles are used to fabricate enzyme biosensors: organic nanoparticles and inorganic nanoparticles. It is required to modify the nanoparticles' surface for increasing the applications of nanoparticles. These modifications can be done by physical method and chemical method. In enzyme

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biosensors, these nanoparticles can immobilize biomolecules, catalyze the electrochemical reaction, enhance the electron transfer, and do labeling of biomolecules. Polar surface modifications are required for the attachment of biological molecules to increase solubility and prevention from aggregation. As per requirement, nanoparticles could be modified by electrostatic adsorption, which is commonly used for a linkage approach for carbons: amorphous activated carbon and carbon black. In strong electrostatic adsorption, metals are absorbed over support surfaces by controlling the pH of the solution. Many compounds are used for hydrophilic ligands, such as mercaptopropionic acid, mercapto acetic acid, mercaptosuccinic acid, bis-sulphonated triphenylphosphine, polyethylene glycol SH, short peptide CALNN, and hydrophobic ligand modification. Surface modification of nanostructures with long alkyl tail and hydrophilic ligand head enables multi-functionality on nanoparticles in one step. It also provides a higher yield. Different reactive agents and biomolecules could be attached by this technique, such as proteins, lipids, lipid derivatives, PEG derivatives, etc. So this is one of the most successful approaches as we can attach several nanoparticles with several different molecules.

In this topic, students will learn about:

- Types of nanoparticles
- Functions and modifications of nanoparticles

Lesson 74 : Hexokinase Biosensor

Hexokinase is a metabolic enzyme that converts ATP into ADP. It acts on all hexoses so that it could be glucose, galactose, or fructose. There are four types of hexoses in mammals named A, B, C, and D. Hexokinase catalyzes hexose phosphate's phosphorylation and produces hexose phosphate. The liver, pancreas, and small intestine produce glucose kinase. Its deficiency is a genetic autosomal recessive disease that causes chronic hemolytic anemia. In biosensors, hexokinase is immobilized on an electrode as a biorecognition element, but it has a weak glucose affinity. We use it to identify ATP and ADP. Creatine kinase is an important enzyme of muscle cells as it is released in circulation during injury. In this biosensor, we indirectly determine the amount of creatine kinase. Firstly we determine the amount of ATP. Hexokinase acts on glucose to produce glucose-6-phosphate and ADP. This ADP Will combine with creatine phosphate and produce ATP. GO is used with HK to detect the CK enzyme activity in samples. ATP is consumed during metabolic processes by several enzymes. ATP biosensors were designed for detection in the ileum and colon.

Similarly, an ATP biosensor determines tyrosine phosphate in osteoblastic cells. The fluorescent method is the simplest one. In this, a fluorophore compound diethylamino-coumarin and genetic engineering techniques are used. Biosensors show high fluorescent binding with ADP.

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- Function and working of hexokinase
- Creatine kinase

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Lesson 75 : Cholesterol Biosensors

Cholesterol is a lipid synthesized by animals in 37 steps. Serum cholesterol level is very important and required to diagnose and treat heart diseases, diabetes, and liver diseases. The cholesterol oxidase is employed in biosensors to estimate the serum level of cholesterol. Several strategies are used to fabricate efficient biosensors to estimate cholesterol levels. The LiebermanVurchard test is used to detect cholesterol in the presence of sulphuric acid under drastic conditions. Certain enzymes can act on cholesterol. Cholesterol esterase acts on cholesterol and produces a hydrolytic product. Cholesterol oxidase is a FAD-dependent enzyme and is an oxidoreductase. It can catalyze the cholesterol to cholestenone. Mostly cholesterol oxidase is used for the production of cholesterol biosensors. Fabrication of amperometric cholesterol biosensors is a big example. It contains smart micro-chips based on cholesterol oxidase. It is co-immobilized thioglycolic acid that produces a self-assembled monolayer. As a result, enzymes' activity and stability increase. In another type of cholesterol biosensor, cholesterol oxidase is used, and the substrate used for this is tetramethyl-benzedene. This substrate produces colour when it is oxidized. In the initial stage, it shows a pink colour that turns blue. In this biosensor, graphene quantum dots and platinum dots are used. This biosensor is very effective to detect cholesterol.

In this topic, students will learn about:

- Cholesterol oxidase biosensors
- Fabrication of amperometric cholesterol biosensors
- Production of Cholesterol oxidase biosensors

Lesson 76 : Glucose Oxidase biosensors

Glucose oxidase biosensors are very important. The first Glucose oxidase biosensor was composed of a platinum electrode, an oxygen-permeable membrane, a thin layer of GOx enzyme and outer dialysis membrane. In 1975 first generation of glucose biosensors were introduced

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with several changes. Instead of a dialysis membrane, polyacrylamide gel was used, and glucose oxidase was immobilized. The electrode was dipped in the biological sample. This biosensor was not very successful because interference was observed by ascorbic acid, uric acid, etc., as ascorbic acid gets oxidized at the electrode and gives false readings.

The second problem was due to oxygen gas solubility. It could be affected by certain factors leading to fluctuations in oxygen tension. Second generation biosensors are called mediated glucose biosensors. In these, a reduced mediator is formed by the enzyme action, which reoxidizes at electrodes. Signals are detected for a change in the chemistry of the mediator. Ferrocene and ferricyanide are iron-based complexes, and methylene blue is an organic dye used as a redox indicator. Ferrocene is a good mediator as it is non-reactive to oxygen and pH and shows a rapid reaction with GOx. In the third generation, electrons are transferred directly from enzyme to electrode without any intermediate. These are reagent and mediator less. Two latest devices are; a continuous monitoring glucose system in which needle-type enzyme is used for subcutaneous implantation and a non-Invasive glucose monitoring system in which optical approaches are used.

In this topic, students will learn about:

- Glucose oxidase biosensors
- Problems of old Glucose oxidase biosensors
- Latest technologies used in Glucose oxidase biosensors

Lesson 77 : Lactate Biosensors

In these biosensors, lactate oxidase enzyme is used. This enzyme catalyzes the oxidation of L-lactate to pyruvate in the presence of dissolved oxygen. It forms hydrogen peroxide, which is chemically active and can be oxidized or reduced to give a current proportional to L-lactate concentration. Lactate oxidase is a globular flavoprotein obtained from various bacterial sources such as pediococcus, enterococcus, mycobacterium, etc. Lactate detection plays a vital role in health care, especially in haemorrhage, sepsis, respiratory failure etc. Lactate level is widely used in sports medicine, shock, trauma, and the food industry clinical diagnoses. The baseline lactate level is 0.5 to 1.5mmol at rest. In clinical practice, it is estimated by the chemical method.

Lactate oxidase catalyzes the oxidation of lactic acid to pyruvate and hydrogen peroxide. It catalyzes the reaction with a hydrogen donor in the presence of 4- aminophenazone. Many biosensors use lactate oxidase. Different types of sol-gels are used for the production of lactate oxidase biosensors. Membrane-based biosensors use cellulose acetate, screen printed electrodes use carbon nanomaterial etc. Lactate oxidase biosensors are also produced using different polymers. One is conducting, and the other is a non-conducting polymer in which various organic compounds are used. A commercial lactate biosensor is also available. It is used to

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monitor lactate production during exercise. It gives the result in just 10 seconds. It is a pocket-sized, simple, accurate lactate analyzer to monitor and support performance.

In this topic, students will learn about:

- Lactate oxidase biosensors
- Applications of lactate oxidase biosensors

Lesson 78 : Alcohol Oxidase Biosensors

Alcohol oxidase biosensors are those in which alcohol oxidase is used as a biorecognition element. Alcohol or ethanol are produced in different metabolic processes. Alcohol oxidase is responsible for converting low molecular weight alcohol to corresponding aldehyde or ketone. It uses molecular oxygen as an electron acceptor. It is a FAD-dependent enzyme. Alcohol oxidase enzyme could be divided into four types depending on substrate specificity; short-chain alcohol oxidase, long-chain alcohol oxidase, aromatic alcohol oxidase, and secondary alcohol oxidase. Alcohol oxidase is the key enzyme of methanol metabolism. Its molecular weight is 600Da. It consists of eight identical subunits. Each subunit carries one non-covalently bound FAD molecule as a prosthetic group. It is required for the biological activity of the molecule. It has a 3-D structure with alpha-helices and beta-strands.

Ethanol biosensors use alcohol oxidase. Different industries rely on ethanol production, including health, diagnostics, food industry, medical science, agriculture, and the environment. Ethanol biosensors based on alcohol oxidase have been reported to be stable and practical. An automatic biosensor determines alcohol produced during different industrial processes. The sensor is based on the enzyme alcohol oxidase. Horse reddish peroxidase is another enzyme used to facilitate the detection. Ethanol biosensors are used to check brain activities. It determines hydrogen peroxide and the interfering agents present in the brain. In this biosensor, AO is used. This enzyme hydrolyzes alcohol to produce hydrogen peroxide. This biosensor is very successful in determining the activity in the brain of rats.

In this topic, students will learn about:

- Alcohol oxidase biosensors
- Ethanol biosensors and their industrial uses
- Horse reddish peroxidase

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Lesson 79 : Xanthine Biosensors

Xanthine oxidase (XO) biosensors act on different substrates like xanthine and hypoxanthine. XO requires two flavin adenine dinucleotide molecules for action. It has two sites for binding with molybdenum ions that take part in the reactions. It can bind with eight FeS clusters. XO has a crystallographic structure with two positions for iron and sulfur clusters. Xanthene can be produced by the action of Guanine deaminase, which is further catalyzed by xanthene oxidase to produce uric acid. This enzyme can produce superoxides. It is found in serum and lungs, and its activity is increased during influenza. During severe liver damage, it is released into the blood. Different typed of biosensors have been designed using xanthene oxide. Its main property is that it increases inflammation and delays healing in wounds. If we design a biosensor to control xanthine production, we can improve wound healing as inhibition of xanthene improves healing. For example, a biosensor is fabricated to determine the amount of xanthine oxidase. In this biosensor, flexible material is used that contains xanthine oxidase, which converts xanthine into uric acid. The resultant oxygen reacts with hydrogen peroxide and is used on multi-volt carbon nanogold particles to detect the signals. Xanthene oxidase biosensors are also used to check the fish and meat freshness.

In this topic, students will learn about:

- How Xanthine oxidase functions
- Biosensors based on Xanthine oxidase

Lesson 80 : Cardiac Troponin Biosensors

The Troponin test is one the most important test in determining heart diseases. Troponin is a complex of three regulatory proteins: Troponin C, Troponin I, and Troponin T. It is integral to muscle contractions. I and cTt are used as diagnostic and prognostic indicators in the management of AMI. Troponin C binds to calcium ions to produce a conformational change in TnI. Troponin T binds to tropomyosin, interlocking them to form a troponin-tropomyosin complex, and Troponin I binds to actin in thin myofilaments to hold the Troponin-tropomyosin complex in place. Troponin c binds with calcium ions in the activated state, stabilizing the activated state.

Different tests are used to diagnose acute myocardial infections. In these tests, two enzymes' Lactate dehydrogenase and Aspartate aminotransferase, are used, but these enzymes are not very specific to these infections. ECG is used to detect myocardial problems, but it can diagnose only 57% of the patients, while 25% do not show any signals. However, troponin gives 95% of the results. Human troponin is composed of 209 amino acids residues with a molecular weight of 22.5kDa that can inhibit ATPase. Its functionality can be enhanced in the presence of

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tropomyosin. The isoform of cTi in myocardial tissue differs from the sequence of those in skeletal muscles. Its concentration can go up to 50ng/ml within 3-6 hours and finally to a level around 550ng/ml. This level will return to its standard status in 6-8 days. Several immune biosensors are fabricated and reported to determine troponin. These biosensors use antibodies with two techniques: surface plasmon resonance and electrochemical. Other techniques are also used, but no biosensor is as successful as the myocardial biosensor.

In this topic, students will learn about:

- Structure of troponin
- Human troponin and its functions
- Detection of heart diseases

Lesson 81 : Immunosensors

These are the biosensors in which immunogenic molecule is used as a bio receptor element. The basis of such detection is the complex formation between antibody and antigen. Sometimes, the whole antigen molecule is not required; therefore, AB fragments are used. A typical immuno biosensor is composed of two main components; Biological component and physical component. The biological component is an immunogenic molecule responsible for detecting the target molecule. The physical component detects the biorecognition signals. There are five types of antibodies. IgM is the first antibody to fight against an attack. It is a pentamer unit with ten antigen-binding sites, but only five are functional. IgG reacts later but helps in the permanent eradication of antigen. Antibody affinity is the epitope's strength to bind to an individual paratope on the antibody. High-affinity antibodies have multiple properties. They bind quickly with the antigen, permit greater sensitivity in assays, and maintain this bond more readily under drastic conditions.

The Ab-Ag complex's formation depends on weak physical forces, including Hydrophobic forces, hydrogen bonding, salt bridge, van der wall forces, and Electrostatic interactions. These forces can be changed by changing pH, salt concentrations, etc. The strength of binding of antigen-antibody complex could be characterized by its affinity constant, that is, K. There are different types of sensors used in immuno-sensors. For example, electrochemical sensors involve electric current, voltage difference and conductivity change. The second type is optical immuno-sensors, including absorption, reflection, dispersion, emission and fluorescence. The third type is piezoelectric immuno-sensors which includes mass change.

In this topic, students will learn about:

- Components of immune-biosensors
- Sensors used in immune-biosensors
- Factors affecting the formation of Ab-Ag complex

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Lesson 82 : ELISA

ELISA is a common technique used in Immuno biosensors. In ELISA, E stands for an enzyme, L stands for linked, I stands for Immuno, S stands for sorbent and A stands for assay. It involves two types of biological molecules; the antibody and enzyme. Horse reddish peroxidase enzyme belongs to the ferroprotoporphyrin group of peroxidases. Its optimum pH is 6-6.5. It is a single-chain polypeptide containing four disulfide bridges with six lysine residues suitable for conjugation with a protein. HRP can be conjugated to antibodies glutaraldehyde through disulfide bonds and amino and thiol directed cross-linkers. HRP is smaller than the enzyme labels beta-galactosidase and alkaline phosphatase, and its glycosylation leads to lower nonspecific binding.

Polystyrene is a long carbon chain to which surface flavouring is introduced to change the binding properties of the matrix. An antigen is immobilized on a solid surface that forms a complex linked to an enzyme. The enzyme is incubated with a substrate to produce a measurable colored product. There are five types of ELISA; Direct ELISA assay, Indirect ELISA assay, Sandwich ELISA assay, Competitive ELISA assay and Multiplex ELISA assay. In a direct assay, an antigen is bound to the surface, and an antibody is attached to the antigen; as a result, the biorecognition process occurs. A substrate is added, and colored product can be detected. In an indirect assay, two antibodies are used. Antigen binds to the polystyrene, the primary antibody attaches to the antigen, and the secondary antibody attaches to the antibody through which the colouring agent can be determined. Sandwich ELISA is a sensitive method that measures antigen concentration. The antigen interest is quantified between two layers of antibody capture and a detection antibody.

In this topic, students will learn about:

- Horse reddish peroxidase enzyme
- Types of ELISA
- Sandwich ELISA
- Polystyrene

Lesson 83 : Monoclonal Antibody

A monoclonal antibody is a type of antibody that is produced by a single colony of cells. These antibodies are important reagents used in biomedical research, disease diagnosis, and treatment of infections and cancer. Each monoclonal antibody is produced against a single epitope. An epitope is an antigenic determinant that is recognized by the immune system. Monoclonal antibodies are produced by two procedures: the Hybridoma technique and the Ascites technique.

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The hybridoma technique uses both in vivo and in vitro methods, while the Ascites technique uses only the in vivo method. It means these colonies are produced only in mice or rats. The animal is injected with the antigen to produce an antibody in this process. The cells from the spleen are separated and are mixed with myeloma cells in the presence of ethylene glycol. As a result, hybridoma cells are produced and cultured in specific conditions where monoclonal antibodies are cultivated. In the ascitic method, the monoclonal antibodies are produced by growing hybridoma cells within the peritoneal cavity of the animal. These cells are injected into the mouse and multiple there to produce antibodies. The monoclonal antibodies have applications in diagnostics. They are used in RIA, ELISA and laboratories. A pregnancy test is used to detect pregnancy using a urine sample. It is also used in cancer diagnostics, hormonal disorders, infectious diseases etc. Therapeutic applications of monoclonal antibodies include treatment of cancer, transplantation of bone marrow, transplantation of organs, autoimmune diseases, cardiovascular diseases and infectious diseases.

In this topic, students will learn about:

- Hybridoma technique
- Applications of monoclonal antibodies
- Ascetic technique

Lesson 84 : Antibody Fragment

Antibody fragments can be generated by biological, chemical or genetic methods. Chemical fragmentation utilizes reducing agents to break the disulphide bonds within the hinge region and digestion of the antibody with proteases including pepsin, papain, and ficin. Chemical and Biological fragmentation utilizes reducing agents to break the disulfide bonds within the hinge region.

In Antibody fragmentation process, antibodies may be enzymatically digested to produce either a Fab or an F(ab′) 2 fragment of the antibody. To produce an F (ab′) 2 fragment IgG is digested with pepsin, which cleaves the heavy chains near the hinge region.

One or more of the disulphide bonds that join the heavy chains in the hinge region are preserved, so the two Fab regions of the antibody remain joined together, yielding a divalent molecule (containing two antibody binding sites), hence the designation F (ab′) 2. The light chains remain intact and attached to the heavy chain. The Fc fragment is digested into small peptides. Fab fragments are generated by cleavage of IgG with papain instead of pepsin. Papain cleaves IgG above the hinge region containing the disulphide bonds that join the heavy chains, but below the site of the disulphide bond between the light chain and heavy chain. This generates two separate monovalent (containing a single antibody binding site) Fab fragments and an intact Fc fragment. The fragments can be purified by gel filtration, ion exchange, or affinity chromatography. IgG

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antibodies are immunoglobulin homodimers of heterodimers; they contain 2 heavy chains (50 kDa) each paired with a light chain (25 kDa). Heavy and light chains, as well as their dimerization, are stabilized through covalent disulphide bridges. Enzymatic cleavage of specific disulphide bonds results in fragments of differing composition.

In this topic, students will learn about:

- Antibody fragmentation
- Different methods of fragmentation
- Fragmentation process

Lesson 85 : Lateral Flow Assay

Lateral flow assay is a very common and simplest testing method that is widely used in different industries. It is based on immunological techniques to detect the molecules and is also known as the Lateral flow test (LFT), Lateral flow device (LFD), Latera flow immunoassay (LFIA), and Dipstick. Basically, this method is a qualitative method that indicates the absence or presence of the analyte. This test bypasses many chemical tests and minimal training is required to operate it. This test shows the results that can be checked visually. Lateral flow assays are based on a simple strip that further has four components. The first component is the sample pad. It is part of the strip where we apply our sample. The second part is the conjugate or reagent pad which contains antibodies specific to the target analyte and these antibodies are conjugated with different colored particles. The third component is the reaction membrane which typically consists of a nitrocellulose or cellulose acetate membrane onto which anti-target analyte antibodies are immobilized. The last and most important component is the backing material on which the whole system is set up. Lateral flow assay has many applications in industries, diagnostics, the environment, and other fields of life. Its design is based on a dipstick. One of the most common examples of lateral flow assay tests is the pregnancy test kit.

In this topic, students will learn about:

- Components of Lateral flow assays
- Applications of Lateral flow assays

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Lesson 86 : Optical Immunosensor

Optical immunosensors are sensors in which we use a light source to detect biorecognition elements. In these biosensors, there is a reaction between antigen and antibody leading to the formation of an antigen-antibody complex that is detected by the waves. During its detection, there is a change in the optical properties of that wave. In an optical immuno-sensor, it is important to have a proper geometry and design of the optical platform. Different geometrical layouts include strips, waveguide fibers, planar optical waveguide systems, capillary sensors, and arrays. In the designing and fabrication of immunosensors, the key point is the position of a light source to detect the biorecognition process. In optical biosensors, there are different types of signals which are detected by different methods. Normally we use a single mediator molecule in optical biosensors which is very helpful to detect the completion of the reaction. Hydroquinone is used as a mediator in optical biosensors that includes two hydroxyl groups and a benzene ring. During threat conditions, hydrogen peroxide is catalyzed into the water and the released electrons react with hydroquinone to convert it into benzoquinone. The reason for using a mediator is that hydrogen peroxide and its product are transparent liquids so it is very difficult to detect the reaction. The mediators show different colors in their oxidized and reduced forms. Refractometric sensors measure the refractive index which is the speed of light in a vacuum. For this purpose, a special type of refractometer is used which has y-shaped platters.

In this topic, students will learn about:

- Working of optical immunosensors
- Role of mediator in detection
- Fabrication of immunosensors

Lesson 87 : SPR Immuno Sensor

Surface Plasmon resonance is a spectroscopic method that recognizes the formation of an antigen-antobody complex. A thin metal film is used in these sensors on which the biorecognition element is immobilized and a wave is used to detect the change in refractive index. During the biorecognition process, some physiological conditions are used in which pH is 7.4 with a strength of 10mM and Buffer solution which has two types. One of the types is PBS which contains Phosphate, sodium, and chloride ions and they do not interfere with antigen-antibody complex formation. The second is the HEPES system. Metals and proteins interact with each other. Four types of amino acids are responsible for metal and protein interaction; aspartic acid and glutamic acid, cysteine and methionine, histidine, and the fourth are serine, threonine, and tyrosine. For biosensors, different techniques are employed for the immobilization process. One of them is the use of bacterial proteins produced by a pathogenic bacterial strain

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'staphylococcus aureus' and it produces protein A (42kDa), protein G(65 kDa), and protein G(58kDa). They all have a specific binding affinity with the Fc and Fab region of the immunoglobins. Another important thing in the surface plasmon resonance is the kinetic parameters for the antigen-antibody complex. There are two types of kinetic parameters; one of them is the Association rate constant which indicates the rate of formation of a complex between an antigen and antibody per second. The second type is the Dissociation rate constant which tells about the rate of Dissociation of an antigen-antibody complex. Another technique in SPR is the use of fluorophores to increase sensitivity. In this technique, a fluorophore molecule is attached along with the antibodies. It is a fluorescent chemical compound that can re-emit light upon light excitation.

In this topic, students will learn about:

- Surface Plasmon resonance immuno sensors
- Physiological conditions used in Surface Plasmon resonance immuno sensors
- Types of buffer solutions
- Types of amino acids

Lesson 88 : Electrochemical Immno Sensor

Impedance immune sensor is a type of electrochemical biosensor that determin the opposition measured in ohms and the nature of these sensors is all electrical. These are much simpler because they lack optical and acoustic components; therefore, it offers significant advantages for portable applications. In impedance immunosensors, the most important thing is the immobilization of the antigen or antibody on the surface platform. Different techniques and strategies include gold Nanoparticles which are coated on copper electrodes, gold and silver layers coated on the metal electrodes, magnetic particles, hydroxyapatite powder, ion exchange membranes, and polymeric films. When a biorecognition element is immobilized on the sensing plate it reacts with its complementary antigen and antibody leading to the formation of an antigen-antibody complex. This leads to the change in electrical properties including dielectric constant and resistance. A transducer is used to determine these changes. In these sensors, there is no need for any label molecule or any mediator molecule. Field-effect transistors are another type of impedance immunosensors that have three important parts; gate, drain, and source. We use N-type transistors having an ample amount of electrons. The antibody is immobilized on the gate electrons flow just after detection of the target molecule. Some merits of these sensors include small size, low cost, speedy operations, and suitability for use in an ambulance or during a clinic visit. On the other hand, demerits include the use of small analytes, complexity in detection, nonspecific adsorption, and problems in the stability of biomolecules.

In this topic, students will learn about:

- Specifications and working of Impedance immune sensor

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- Merits and demerits of Impedance immune sensor

Lesson 89 : Impedance Immno Sensor

Electrochemical immune biosensors are those in which the biorecognition process is marked by the generation of electrochemical signals. There are three types of electrochemical immunosensors; amperometric immunosensors, potentiometric immunosensors, and conductometric immunosensors. The immobilization of the biorecognition element on the electrode surface of the immunosensors is the most important thing. GNP is the use of gold Nanoparticles as gold particles have a special affinity for thiol groups. These thiol groups form a mono-assembled layer that acts as a linker between antibody and electrode. Magnetic beads are also used in place of gold particles. These beads are oxides of iron that are immobilized on the surface. Once these beads are immobilized, the antibody and immunogenic molecules are attached to these nanoparticles. In amperometric sensors, the detection of antigens and antibodies is done by using a redox enzyme which is a horse reddish peroxidase that also uses a mediator. These mediators are normally organic molecules and occur in different colors during their oxidized or reduced form. The change in their color is marked by the completion or detection of the target molecule. In potentiometric sensors, different materials are used to create a potential difference. For example; polyaniline matrix. Another substance OPD is also used in these sensors which causes the fabrication of these biosensors. OPD is used in the mediator and hydrogen peroxidase catalyzes the hydrogen peroxide. This leads to the generation of electrons consumed by OPD and the OPD substrate is converted into brown diaminobenzidine. Due to its color, we can detect if the reaction is progressing or not.

In this topic, students will learn about:

- Types of electrochemical immunosensors
- Use of GNP and magnetic beads
- Use of OPD

Lesson 90 : Piezo Electric Immuno Sensor

Piezoelectric immunosensors are immunosensors in which an antigenic molecule and a piezoelectric material are used. It is a unique type of material that can produce an electric current when stress is applied to it. Normally quartz crystals are used as a piezoelectric material that detects the formation of antigen and antibody complexes. Immobilization is an important step and the antigen or antibodies can be immobilized on a piezoelectric sensing platform surface.

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This surface is needed to be sensitized or functionalized so that antigens or antibodies can be immobilized on it. There is a variety of organic and inorganic materials that are used for coating the surface platform where BRE is immobilized. These coating materials are used for covalent attachment to the PZ crystal surfaces. The materials include silane reagents, modified silica, chitosan, ZnO micromechanical membrane, and protein in a based film. There are certain requirements for the attachment of these membranes including the membranes to be thin, coating to be homogeneous, able to incorporate the amino acid functional groups, can contribute to high sensitivity, must have minimum interference, and should generate reproducible results. Two types of technologies are used in these immunosensors; the QCM technique which is a label-free and extremely sensitive device. It measures the mass variation per unit area by measuring the resonance which is distributed by the addition or removal of a small mass. The second technology is the use of microcantilevers which is a rigid structure attached at one end to support and with another end free and these are commercially available in different materials.

In this topic, students will learn about:

- Working of Piezoelectric immunosensors
- Requirements for attachment of materials in piezoelectric immunosensors
- Technologies used in piezoelectric immunosensors

Lesson 91 : Introduction to Cell Biosensor

Cell biosensors have a biological cell as a biorecognition element. In these biosensors, both viable and non-viable biological cells are used depending upon the requirements. There are four types of cells used in cell biosensors. The first type is a microbial cell in which a fungal, yeast, bacterial or viral cell is used. The second type is an organelle, the third includes tissue cells and the fourth includes stem cells used for research purposes. In the development of biosensors, the most important step is the immobilization of the whole cell on a matrix. Four processes are used for immobilization. First is entrapment which bypasses the use of drastic chemical conditions, another method is self-aggregation in which cell is mildly stressed so they aggregate, the third method is adsorption on a solid support and the last is mechanical confinement. Different strategies are used for the immobilization of biological cells. One of them is the use of biofilms formed by bacterial aggregates held together by a mucous-like matrix of carbohydrates adhering to the surface. A biofilm has a nutrient-rich environment with a complex 3-dimensional structure that creates an internal protective environment for bacterial cells. Another method used is the immobilization of scaffolds which is a protective structure in which cells reside. The most important method is the 3-D printing of HeLa cells which are immortal cancer cells. 3 D models consist of a scaffold made up of three fibrous proteins corresponding to the extracellular matrix of a tumor and the matrix is coated with Hela cervical cancer cells.

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- Types of cells used in cell biosensors
- Immobilization processes used in cell biosensors
- 3-D printing of HeLa cells

Lesson 92 : Bacterial Cell Biosensor

Pathogen microbes cause diseases such as E Coli, salmonella, etc. Generally, there are two types of cell biosensors used for the detection of pathogens; the impedimetric biosensor and the optical biosensors. The reason for the creation of these biosensors is that most bacterial infections are often misdiagnosed as they are treated without identification of the pathogen or if they are identified there is a delay of at least 2 to 3 days for the detection of bacterial pathogens. Biosensors offer a rapid and cost-effective method of bacterial detection. The methods currently being used for bacterial detection are cell culturing, microscopic analysis, and biochemical assay but these procedures are time-consuming, cost-producing, and require special equipment with trained users. Biosensors can detect pathogens in a very short time. Bacteria are classified as gram-positive bacteria which retain the violet gram stain due to their thick peptidoglycan layer outside the cell membrane where lipids, proteins, and glycoproteins are present. The other type is gram-negative bacteria that do not give stain as their thin peptidoglycan layer does not expose. Porins are proteins that are present in the membrane of bacteria and it allows the entrance of certain molecules. We can use proteins, glycoproteins, lipopolysaccharides, and peptidoglycan to act as targets for biorecognition as they generate antibodies against the molecule. Polyclonal antibodies raised against specific bacterial strains are most commonly used for the whole bacterial cell.

In this topic, students will learn about:

- Effectiveness of biosensors against pathogens
- Types of bacteria
- Porins

Lesson 93 : Cell Biosensor with Reporter Gene

A reporter gene is a gene attached to a regulatory sequence of another gene of interest in bacteria, cell culture, animals, or plants. The function of the reporter gene is to help in the detection of the target molecule in a biosensor. These are the marker genes to measure the amount of expression. The most commonly used report genes include the LacZ Reporter gene which records for beta-galactosidase that is an enzyme that uses a colored substrate. Luc

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Reporter gene is a luciferase gene that is present in a firefly. GFP Reporter gene produces a fluorescent protein, Egfp gene, and CAT is an antibiotic-resistant gene. There are some types of reporter genes; the first one is Gfp which is coded by a single protein and is composed of 238 amino acid residues with a molecular weight of 26.9kDa. The second type is the crtA gene which is responsible for carotenoid synthesis. Carotenoids are of different colors so when a sample containing this gene is subjected to a test, the whole-cell biosensor can change its color. The change in the color of the media is a signal for biorecognition, therefore, it is also considered a good choice for rapid detection. Other types of reporter genes are the Luc gene which requires luciferin as a substrate and the LacZ gene requires beta galactoside to generate optical signals. These two are very sensitive genes that can detect 2fg and for these genes, chemiluminescent and electrochemical substrates are also present.

In this topic, students will learn about:

- Functions of reporter genes
- Types of reporter genes

Lesson 94 : Cell Biosensor with Regulatory Protein

Regulatory proteins regulate the expression of certain proteins. These proteins could be repressor proteins that lower the expression of a protein to act at the RNA level and activator proteins that activate the level of expression and act at the DNA level. One of the best examples of regulatory proteins is Metallo-regulatory proteins that are produced in all bacterial cells. Metallo-regulatory proteins regulate two protein expressions; the first is membrane transporters and the second is metal trafficking promoters. The function of these proteins is to increase or decrease the expression of proteins by binding to the promotor structure where they produce some changes in that particular structure. To know about the changes, consider an example, copper ions are considered toxic ions in many bacteria. When they enter a bacterial cell, a mechanism is initiated by the regulatory proteins that prevent the bacteria from the toxicity of copper. Once this copper enters a bacterial cell, the protein binds with copper and this protein binds with DNA so there is a change in structure. This initiates the expression of iron protecting protein which removes the copper. Another example includes Salmonella which has a particular gene for the detection of gold salts and is known as gold sensors. Its expression is controlled by a regulatory protein called GolS which is highly specific for Au ions. If we introduce a mutation in the GolS protein then it can bind with other elements such as mercury, lead، cadmium, etc.

In this topic, students will learn about:

- Mechanism of regulatory proteins
- Metallo-regulatory proteins
- Salmonella as gold sensor

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Lesson 95 : Selection of Cell Biosensor

The selection of a host cell for a cell biosensor is very important because it impacts the performance of a cell biosensor and also affects the specificity, sensitivity, reproducibility, reliability, capacity, time response, and background noise. Two strains of bacteria are used; one is Alconivorax borkumensis which shows lower tolerance towards pollutants, and higher tolerance towards salinity. The second is pseudomonas putida which has the highest solvent tolerance. Acinetobacter is a natural adhesive to oil-water interphase so where there is a leakage of oil in the water there occur a huge population of these bacteria because it has got a natural system for the assimilation of crude oil and mineral oil. When this strain was compared with *E. coli*, the poor accessibility was shown by *E. coli* toward oil droplets while acinetobacter completely emulsified the oil droplets. We can also use those bacteria which lack the machinery to detect the target by virtue of genetic engineering techniques. We can engineer the host cell and use that cell as a receptors element and these genes can be used for cloning and expression but they may not provide the same results all the time as it all depends on the experimental conditions.

In this topic, students will learn about:

- Specifications of Acinetobacter
- Types of bacterial strains used for cell biosensors

Lesson 96 : Tissue Biosensor

Animal tissue-based biosensors are those in which animal tissue is used as a biorecognition element. It could be from any tissue such as the liver, kidney, brain, etc as they offer a number of advantages such as easier immobilization, higher activity, stability, low price, and presence of cofactors. Commonly two types of animal tissues are used; the first is genetically modified cells and the other is direct genetic modification. However, in both these types, the objective is to determine the type of analyte in the sample. These animal tissues show the physiological processes determined by different techniques. One of them is the Bio-photonics technique which provides the most versatile basis for tissue-based biosensors. Here are two types of light outputs that can be generated by cells; fluorescence and the other is bioluminescence light which provides the advantages of not requiring an input source of light and they also provide background signals in cells. Another technique is discovered under bioluminescence which is

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known as BRET which is a natural phenomenon and it contains four main conditions involving resonance energy, enzymatic donor, fluorescent protein acceptor, and range (1-10nm). Mostly 30 to 50 % G protein-coupled receptors are commercially available.

In this topic, students will learn about:

- Advantages of Animal tissue-based biosensors
- Techniques used in Animal tissue-based biosensors
- Lights generated by Animal tissue-based biosensors

Lesson 97 : Detection of Bacterial Pathogen

Microbes such as bacteria, viruses, fungi, and yeast are known as pathogenic microbes and these can be detected by using cell biosensors. Different strategies are used to detect pathogens from a disease. One of them is the use of the virus as biological recognition element as the virus is the simplest microbe composed of a protein core and nucleic acid. These viruses are basically parasites so they just live in their host and inject their DNA into the host cell. They use the machinery of the host to replicate inside the host cell and cause diseases. The *E. coli* can be detected by using the bacteriophage T4 virus that only affects *E. coli* and once it affects the *E. coli* cells it enters into the cell and produces a huge population of viruses there. It uses a lytic cycle which leads to the destruction of itself and once a bacterial cell is infected a clear zone is seen instead of a colony. The second approach is quorum sensing which is basically an intra-cellular communication process and is dependent on the density of the population. These sensors are used in two ways first is in which we tagged a reported gene with a quorum sensor. This induces an expression of a reporter gene in *E. coli* a fluorescent light is seen. The other thing is that if the amount of an induce-able molecule is very little we can use a lactonase which destroys that molecule.

In this topic, students will learn about:

- Detection of pathogens by cell biosensors
- Structure and functioning of viruses
- Quorum sensing

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Lesson 98 : Cell Membrane Protein based Biosensor

These biosensors use membrane proteins. Two types of proteins are present in the membranes; peripheral or extrinsic proteins and intrinsic proteins. The peripheral proteins are present in the amino acids with hydrophilic side chains. They can interact with the surrounding water and with the polar surface of the lipid bilayer. The second type is intrinsic proteins which contain hydrophilic regions that interact with the polar heads of the lipid molecules. There are two types of membrane-based biosensors. In the first type, the lipid bilayer proteins are used and only membranes are used as a biorecognition element. The protein of interest is embedded in these membranes. In the second type, cell-based platforms are used in which we take membrane proteins out, express them in another cell, cloned them in a particular cell, and then use it as a source of membrane protein. There are two types of membrane receptor protein-based biosensors; the first type contains intrinsic proteins. These membranes are directly attached to the electrode and once there is a detection of the target molecule the action is transferred to the transducer. In the second type, a piece of DNA is cloned into the cell which introduces a gene into it. Once it detects, target molecule signals are generated. These are being used in research for many years because they show reliable growth, are used for foreign gene expression, and have high transformation efficiency for foreign cloned growth.

In this topic, students will learn about:

- Membrane receptor protein based biosensors
- Proteins present in Membrane receptor protein based biosensors
- Types of membrane receptor protein-based biosensors
- Advantages of membrane receptor protein-based biosensors

Lesson 99 : Biosensor for Plant Pathogen Detection

When plants are attacked by pathogen, they become under stress, as discussed in previous modules. Plants secrete a number of volatile compounds, as an indicator of s Plant microbial pathogens attack crops. These microbes include virus, bacteria and fungi. A rough estimate shows that 20–30% of the field crops are lost annually. An estimate shows that in Georgia USA cost the economic loss estimates to up to 821.85 million dollars in 2013. tress. In lecture a number of examples are described to make the concept clear. Traditional Detection Microbial Pathogens involves interpreting visual symptoms of disease. It is followed by pathogen diagnosis using microscopy techniques to confirm the data. This approach cannot be used until the symptoms start appearing. It is not suitable for on-site diagnosis of plant pathogens. A biosensor is designed to detect methyl salicylate. It is a volatile organic compound released by pathogen-infected plants via systemic response. Alcohol oxidase and HRP enzymes were immobilized on

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to a CNT matrix. Amperometric method was used in the development of sensor. This biosensor works by the following process.

1. Hydrolysis of methyl salicylate to form salicylate and methanol.
2. AOD converts methanol into formaldehyde and simultaneous reduction of Oxygen and to hydrogen peroxide.
3. Reduction of hydrogen peroxide by HRP to water.
4. Generation of amperometric signals.

In this topic, students will learn about:

- Biosensor for the detection of plant pathogens
- Different examples of Biosensors for plant pathogens detection

Lesson 100 : Plant Tissue Biosensor

These biosensors are used to determine the stress level of plants for better yield. There are different reasons for plant stress such as drought, and deficiencies of different nutrients. The first type of stress is a water shortage or drought in plants. The plant shows different signs under drought conditions like slow growth, reduce photosynthesis, tissue damage, or wilting. A biosensor designed used for the detection of water shortage is designed by fabricating a CNTs-based electrode and stomata cells. This biosensor allows determining the physiological response to drought conditions for further studies. The sensor in these biosensors uses the opening and closing of stomata to detect the water condition of plants.

Another type of biosensor is used for determining the nitrogen content of a plant. Plants assimilate the atmospheric nitrogen into amino acids including glutamine (Gln) under rhizobia. Gln is exported to the shoots as a major fraction. During the export, the Gln is released into the atmosphere. The release of less amount of Gln can be used as an indicator of N deficiency.

In this topic, students will learn about:

- Biosensor determining the water level of plant
- Biosensors determining the nitrogen content of plant

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Lesson 101 : Biosensor

DNA is used as a bio recognition element in nucleic acid biosensors. Nucleic acid has two types: RNA and DNA and both are prone to degradation by nucleases but DNA is comparatively more stable than RNA. In these biosensors, DNA as a biorecognition element specifically binds with its target molecule. The target molecule creates a signal by immobilizing the electrode on the electrode surface and the target molecule on the surface. This interaction between DNA and target molecule generates a signal that is detected by a suitable transducer that can be an optical, electrochemical or piezoelectric transducer. Usually, DNA's single-stranded molecule is used as a biorecognition element. This single-stranded molecule with its complementary sequence can bind with another DNA or RNA molecule which is called hybridization. This can also interact with proteins, vitamins, and other molecules. Single-stranded DNA flexibly interacts with proteins to adjust their position for binding. Two types of forces are involved in this interaction; hydrogen bonding and electrostatic interactions. This can be altered by certain factors such as pH and temperature. In the binding of genomic DNA with histone proteins, different forces are used. For Example; polar interactions between the helix and phosphate groups of DNA, non-polar interaction between ribose sugar and histones, hydrogen bonds between DNA and amide group, the salt bridge between amino acid and a phosphate group, and non-specific interactions between the N-terminal tail of histone and minor groove of DNA.

In this topic, students will learn about:

- Functioning of DNA in DNA biosensors
- Forces involved in the DNA interaction

Lesson 102 : DNA as Bio-receptor molecule

PNA is a derivative of DNA and has the same four bases but it contains a peptide bond instead of a phosphate group. In DNA biosensors small size fragments of DNA ranging from 10-to 100 nucleotides are used as probes and are also known as geno sensors. Different approaches are used to increase the functionality and efficiency of biosensors. The use of dendrimers is one of them which are tree-like structures and contain several similar branched DNA units connected through DNA ligase. The dendrimers often adopt a spherical three-dimensional morphology and are systematic around the core. PNA as a receptor molecule is also used to increase the efficiency and functionality of biosensors because PNA has an affinity to bind with DNA receptors. PNA can also bind with the complementary DNA or RNA target molecule but the duplex of DNA -PNA is much more stable than the DNA -DNA duplex. This is mainly due to the absence of phosphatase groups. The rate of hybridization of PNA -DNA duplex increases up to 50000 times as compared to the DNA -DNA duplex due to the absence of repulsion inside the

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duplex. Moreover, the stability of the DNA-PNA hybrid increases with the increasing number of bases such as an increase of one base pair increases thermal stability up to 1°C.

In this topic, students will learn about:

- Characteristics of PNA
- Structure and use of dendrimers

Lesson 103 : DNA Immobilization

Immobilization is the first step in the fabrication of biosensors. Generally, two methods are used for the fabrication of DNA biosensors. One is a physical method in which a complex is formed by using avidin and biotin which is comparatively a smaller molecule and binds to four binding sites of streptavidin which is a large molecule to form a highly stable complex. It is then immobilized on an electrode surface. This complex is stable to different pH values and high salt concentrations. The other type is a covalent method or a chemical method in which the amino-terminated DNA probes are used for covalent bonding onto the modified surface of the electrode. For this purpose, activation of the surface is done by using imide reagents and then is modified with carboxyl, aldehyde, sulphonic, isothiocyanate, and epoxy to attach with DNA. Another type of covalent binding is the chemisorption technique which is based on gold and gold metal is needed in the form of gold nanoparticles where DNA could be attached. These groups could be attached at the 5' or 3' of the surface. This technique is used for the covalent immobilization of DNA probes between thiol-modified DNA probes and the Au surface. It forms a SAM of DNA probes on the gold electrode surface and thus there is a strong affinity between the thiol group and the gold surface forming a covalent bonding of gold-sulfur.

In this topic, students will learn about:

- Methods used for immobilization of DNA biosensor
- Chemisorption technique

Lesson 104 : Electrochmical Behavior of DNA

It has been reported that DNA can transport charge in solution and on the surface where it is immobilized. Three possible mechanisms including superexchange, hopping and domain hopping are given for the transfer of charge by DNA and they all contain a donor, acceptor, and a bridge. In superexchange, the charge flows from donor to acceptor through the bridge while in hopping charge occupies the bridge in traveling from donor to acceptor, and in domain hopping charge occupies the bridge by delocalizing over several bases. Since DNA can transport charge

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so it could be damaged by oxidizing agents, alkylating agents, and high energy electromagnetic radiation such as UV light and x-rays. Moreover, the bases present in DNA also show electrochemical behavior and these could be oxidized by producing their oxidation signals at highly positive potentials on carbon electrodes. For example, the oxidation of Guanine is pH-dependent so two degraded products of Guanine are produced when human DNA is exposed to heavy particles. The convenient oxidation of Guanine is used as a tool in biosensors for the fabrication of reagent-less biosensors. Due to this property guanine can also be used for the preparation of biosensors as guanine can easily be adsorbed on a carbon electrode and is oxidized by using a transition metal complex. In the end, there is a difference in the number of Guanine available before and after the detection of a target molecule.

In this topic, students will learn about:

- Specification of Guanine
- Mechanisms involved in the transfer of charge by DNA
- Damaging of DNA

Lesson 105 : Sensing Techniques of Biosensor

In DNA biosensors the process of biorecognition relies on the hybridization/interaction of receptor and target molecule. So hybridization is the main event for the detection of a target molecule for which two different techniques can be used; direct technique and indirect technique. In direct technique, suitable methods including optical methods are used in which DNA molecule is labeled. It could be labeled at DNA 3' or 5' by using a reporter molecule such as biotin, fluorescent, etc. Generally, bacteriophage T4 polynucleotide kinase is used for labeling. A piezoelectric method is another technique in which piezoelectric crystals are used in the fabrication of highly sensitive mass sensors. They can detect even a change in a single molecule because it causes a change in the resonance. So after the detection of a target molecule, the DNA molecule hybridizes with the target molecule. This hybridization to immobilized oligonucleotides leads to molecular crowding which creates surface stress resulting in cantilever bending. A reference cantilever can also be used to detect a differential deflection. As a result of this crowding, resonance changes and it can be detected easily. Electrochemical methods, acoustic methods, and mass sensitive methods are also a part of direct methods and can be used in the bio recognition process.

In this topic, students will learn about:

- Sensing techniques in DNA biosensors
- Techniques used for hybridization
- Piezoelectric method

Lesson 106 : Label Free Detection in DNA sensor

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The Label-Free Electrochemical detection relies on the process of oxidation and reduction. The method is based on the electrochemical properties of DNA and all four bases of DNA. There are some drawbacks to this method. It requires high background signals, high potential to detect Guanine oxidation, and non-specific adsorption of DNA targets containing Guanine bases. In Label-Free Electrochemical DNA sensors, modified electrodes are used to amplify the signals produced by hybridization. Glassy carbon is an allotropic form of carbon. It is highly pure, nonporous, biocompatible, stable, and inert against erosion and gases. In these biosensors, glassy carbon can be used as an electrode. A modified electrode glassy carbon electrode is used with 3% Cu or cesium oxide to study all four bases of DNA. The oxidation of DNA bases is carried out at pH 7.4 using phosphate buffer solution. To detect HIV target DNA, graphene is used as an electrode as it helps to detect the hybridization at a very low noise level. A single-stranded PNA molecule is non-covalently linked to the graphene surface with a pyrene linker group. The complementary DNA strand binds with the HIV target DNA. The Label-Free Electrochemical DNA sensors are simple, rapid, highly sensitive, and long-term stable.

In this topic, students will learn about:

- Basis of Label-Free Electrochemical detection
- Electrodes used in Label-Free Electrochemical DNA sensor
- Graphene-based biosensors
- Advantages of in Label-Free Electrochemical DNA sensor

Lesson 107 : Physical Labeling of DNA

DNA probe is physically labeled with different molecules such as organic dyes, metal complexes, and cancer drugs which are used to improve the performance of sensors. These organic molecules can be used as indicators as they change their color on different pH values. Metal complexes also form highly colored complexes. Cancer drugs have a high binding effect on DNA. As these drugs are smaller in sizes so they are used as reporter molecules that can bind with DNA molecules using different physical mechanisms. Electrostatic interactions, intercalation binding, binding with minor groove, binding with major groove, and high affinity to a specific DNA sequence are the interactions that occur between a reporter and DNA molecule. There occur two grooves within the structure of DNA; a minor groove occurs where backbones are close together and a major groove occurs where backbones are far apart and interacts with DNA bases. The minor groove binders are crescent-shaped molecules that easily bind with the minor groove of DNA. This usually takes place by directed hydrogen bonding to the base pair at the edges. Binding by intercalation is another type of binding. It is the insertion of a molecule between the planar bases of DNA by van der wall forces such as ethidium bromide intercalated between two adenine-thymine base pairs because ethidium is capable of forming close van der wall forces with base pairs and so it binds to the hydrophobic interior of DNA molecule.

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In this topic, students will learn about:

- Molecules used for DNA labeling
- Grooves within the DNA structure
- Binding by intercalation

Lesson 108 : Molecular Beacons

Molecular Beacons are specially designed molecules for the detection of hybridization. These are used in molecular biology for years. Molecular Beacons are sensitive fluorescent probes hybridizing selectively to designated DNA and RNA targets. These are ssDNA hybridization probes that form a stem-and-loop structure. Molecular Beacons consist of two parts; stem and loop. The loop contains a sequence that is complementary to a target sequence. The stem is formed by the annealing of complementary arm sequences located on either side of the probe sequence. There are two arms of the molecular beacon; one arm is attached with a quencher and the other is attached with a fluorophore. Quenching is a process that decreases the fluorescence intensity of a substance. When the fluorophore and quencher are nearby, the quencher absorbs the energy emitted from an excited fluorophore and suppresses its emission. Molecular beacons do not fluoresce in free form. The probe remains dark in the absence of targets. The quencher and fluorophore transiently share electrons thus eliminating the ability of the fluorophore to fluoresce.

When the probe encounters the target molecule, it forms a stable probe-target hybrid. After hybridization, the quencher moves away from the fluorophore and the fluorescence of the fluorophore is restored.

In this topic, students will learn about:

- Use of Molecular Beacons
- Parts and arms of Molecular Beacons
- Role of quencher and fluorophore
- Principle of designing a molecular beacon

Lesson 109 : Enzyme Labelled DNA Biosensor

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Enzymes are used in DNA-based biosensors to increase their sensitivity. They work on the electrochemical principles so the enzymes used in these sensors also work on this principle. Generally, the enzymes are used in electrochemical-based DNA biosensors and the electrochemical signals of these DNA biosensors are amplified by the chromogenic substrate to detect the enzyme activity. Usually, three types of enzymes are used in DNA biosensors that include Horse reddish peroxide, alkaline phosphatase, and glucose oxidase. Different strategies are used to use these enzymes for effective amplification of the signals that are detected at the electrode. For example, a hairpin DNA probe is used for horse reddish peroxidase. It has two important components, a stem, and a loop. Similarly, a molecule is attached at one end while at the other end an enzyme is attached. This DNA probe does not bind to the enzyme probe until it hybridizes with the target molecule and this binding leads to the generation of signals. In working one end of HDP is attached to the thiol group and the other is linked with a fluorescent probe. As long as the hairpin is closed there is no reaction between enzyme and substrate. As hybridization occurs the hairpin opens to form an HRP complex with a probe. On the other hand, alkaline phosphatase is used as a reporter enzyme and TNB is used as a substrate which is affected by the presence of the phosphate group.

In this topic, students will learn about:

- Use of enzymes in DNA-based biosensors
- Types of enzymes used in DNA-based biosensors
- Strategies applied to use enzymes in DNA-based biosensors
- Working of hairpin DNA probe for horse reddish peroxidase

Lesson 110 : Aptamer based DNA sensor

Aptamers are short, single-stranded DNA or RNA molecules as well as non-natural bases, backbones and small molecules. They bind to specific targets by proper fitting with their target molecule such as proteins, peptides vitamins and living cells. They can form helices and single-stranded loops. Aptamer binding is determined by its tertiary structure. The target recognition and binding involve different interactions. These interactions are shape-dependent interactions, hydrophobic interactions, base-stacking and intercalation.

The shape-dependent attractive and repulsive forces guide the Aptamer to create physical bonds with the target molecules. Aptamer with the affinity for the desired target are selected from a large oligonucleotide library through a process called SELEX (Sequential Evolution of Ligands by Exponential Enrichment). The chromatography and capillary electrophoresis techniques are used to find the best binding Aptamers. High affinity can be created by the ligand-induced structural changes in the Aptamers. It results in the formation of unique secondary structure responsible for high-affinity ligand binding. Aptamers can undergo multiple freeze-thaw cycles and retain their activity.

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In this topic, students will learn about:

- Unique characteristics of aptamers
- SELEX

Lesson 111 : Introduction to Lab on Chip

Lab on chip is the latest technology in which the whole lab function can be transferred on a small chip of less than 1-inch diameter. The development of this technology is related to the human genome sequencing project where fast running technologies were established to sequence the whole human genome. It consists of a small chip made up of silicone, glass, or ceramics on which different patterns are formed called microchannels. Here a reagent cartridge is present that contains the required reagents for the test, micro components for sample introduction such as sample injection or ejection and mixing, and electronic circuits. The lab on-chip has three main features; the safe platforms that use very minute amount of samples and reagents so there is no need for any proper training and no specific safety precautions are required. Another feature is lower fabrication cost that allows mass production for cost-effective disposable chips and the third feature is quality control that is embedded on the LOC so that they can verify the results. For LOC technology, some components must be studied such as materials for chips, lithography, capillary electrophoresis, microfluidics, introduction to sampling, mixing of reagents, and electronic circuits. Advantages of using the LOC technique include less sample requirement for diagnostics, low fluid volume consumption, fewer wastes, lower reagents cost, fast response, fast analysis, small heat capacity, and fast heating.

In this topic, students will learn about:

- Main features of Lab on chip
- Structure and functioning of Lab on chip
- LOC technique
- Advantages of LOC technique

Lesson 112 : Microfludics

The study behavior of fluids at the micro level from 100 nanoliters to 500 microliters is known as Microfluidics. The factors that play important role in Microfluidics are; surface tension, energy dissipation, and fluidic resistance. A microfluidic chip is used to study Microfluidics. It is a pattern of molded or engraved microchannel on-chip. The internal forces must be understood properly in Microfluidics. It requires injection of sample, flow, and mixing with the reagents.

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The elements of the microfluidic chip are required to accurately manage the fluid inside. These elements can be embedded inside or outside the chip.

Reynolds number is the most important parameter for fluid behavior. It is a dimensionless quantity that tells the fluid behavior at the micro level. It compares the momentum and viscosity effect of fluid. The Reynolds number is the ratio of internal forces to viscous forces. It can predict if a flow condition will be laminar or turbulent. In laminar flow, the fluid moves with uniform viscosity while in turbulent flow the speed and magnitude of fluid change. Reynolds number is the ratio of inertial force to viscous force. There are two basic electro-kinetic phenomena: electrophoresis motion and electrosmosis flow. The electro-kinetic motion is the motion of charged particles relative to their surroundings. In electrosmosis flow, the fluid moves under the influence of electric potential applied.

In this topic, students will learn about:

- Microfluidics
- Reynolds number
- Types of fluid flow
- Physical forces of a fluid

Lesson 113 : Capillary Electrophoresis

Capillary electrophoresis is an analytical technique in which biomolecules are separated based on charge. This technique is used in the separation of ions based on their electrophoretic mobility with the use of an applied voltage. This mobility is dependent upon the charge of molecule, viscosity, and mass. The applied electric field is directly proportional to the rate at which particles are moving. In capillary electrophoresis, capillaries with an internal diameter of about 25-75 micrometers are used and an electric potential is applied from the negative to the positive electrode. A small volume of about 10nL is injected at the positive end of the capillary and the separated components are detected at the negative end. This automated technique is also used in DNA sequencing. In the automation of DNA, firstly the DNA samples are oxidized by dideoxynucleotides. Each of these contains different fluorescent lights such as adenine have a green color, cytosine blue color, guanine orange, and thymine is pink color. Once the reaction is completed the samples are loaded on the capillaries and then detected by monitors. Four main components are needed for the automation process. First is a sample injector in which the samples are introduced into the capillary through gravity, electric current, or pressure. Secondly, we need a capillary, the third is a power supply and the last component is a detector. After the completion of electrophoresis, high sensitive detectors are used to detect the separated products either by using UV detectors or fluorescence detectors.

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- How Capillary electrophoresis works
- Applications of Capillary electrophoresis
- Components required for the automation process

Lesson 114 : Photolithography

Photolithography is an important component of LOC technology through which we transfer different geometrical patterns on LOC. In photolithography, a light source of shorter wavelengths is used such as UV light that is produced from mercury discharged lamps to transfer the patterns on the silicon chip. This technique is commonly used for printing circuit boards and microprocessors and it involves several steps such as wafer cleaning, barrier layer formation, photoresist application, soft baking, mask alignment, exposure and development and the last is hard-baking. For cleaning purposes, different types of chemicals are used such as hydrogen peroxide, tri-chloromethane, acetone, or methanol and after cleaning the wafer with these chemicals it is heated to remove any moisture and stored for future use. After that, an adhesive promoter such as hexamethyl-disililazane is applied to form a high water repellent layer. Photoresistors are also used in technology that is a resistors whose resistance decreases when the intensity of light increases. It has two types, the positive photo resister that dissolves on the exposure of light and negative resistors that do not dissolve in the exposure to light such as epoxy-based polymers. But there also occur the unwanted photoresistors and their removal process is known as etching. Etching is of two types; wet etching in which hydrofluoric acid is used to bathe the treated silicon wafers while in plasma etching standard conditions such as pressure under vacuum are created for etching.

In this topic, students will learn about:

- Use and steps of Photolithography
- Etching and its types

Lesson 115 : Soft Lithography

Soft lithography is a collection of lithographic techniques for the fabrication of Lab on Chips when biological molecules are used with the use of a soft material such as poludimethylsilioxane (PMDC). It is a group of silicon-based compounds and is unique in rheological properties. It is optically clear, inert, non-toxic, non-flammable, and is used in contact lenses and medical devices. The most important property of PMDC is that it can make very strong bonds with other PMDC layers of glass by simple plasma treatment. Moreover, it is also gas permeable and this

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material is used for culturing different cells as it controls the amount of gas through PMDC. The method for the preparation of this polymer includes mold preparation by treating it with silicon and mixing it with a curing agent, then pouring it into the mold, baking it, and cutting it according to the required size. The advantage of using this technology is that it is well suited for applications involving large or non-polar surfaces. we can produce different channels of about 30-100nm in a smaller laboratory setting and the main advantage is the use of this technology to make the human cells such as human alveolar epithelial and pulmonary microvascular endothelial cells cultivated in PMDC chip to mimic lung functions.

In this topic, students will learn about:

- Properties of PMDC
- Advantages of PMDC
- Soft lithography

Lesson 116 : Optical Fiber in LOC

Optical fibers are thin, flexible filaments of glass or plastic with a diameter of micrometers. They are commonly composed of a central core made up of glass and cladding. These are used for the transmission of signals through an optical wave with little loss by using the phenomenon of total internal reflection. LOC devices show poor sensitivity due to reasons such as loss of detectable signals, presence of ambient signals, and use of small sample volume and detection limit in the lowest concentration of analyte determined by LOC. So, the optical fibers are used in LOC to overcome these shortcomings. The use o,f optical fibers in LOC has two reasons; the first is the delivery of light source and the other is the detection of signals at a very precise location. Advantages of optical fibers in LOC include enhancement of signals and reduction of ambient signals and its disadvantage includes its complicated designing and fabrication process. In LOC we can use optical fibers in two ways; the first is by incorporating them into the LOC with physical contact with the microchannel structures. These settings offer the best performance because there is a very little loss of signals and reduce ambient noise. The second method includes the use of optical fiber in proximity but they are not in direct contact with microchannels. The advantage of this system is that they are convenient to design and fabricate but its disadvantage is that it can produce noise.

In this topic, students will learn about:

- Reasons to use optical fibers in LOC
- Advantages and disadvantage of optical fibers
- Ways to use optical fibers

Lesson 117 : Diagnostic Target in LOC

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Lab on chip (LOC) is an emerging technology and is becoming an essential part of diagnostic laboratories. It ranges from relatively simple immune chromatographic strips to highly complex systems employing advanced technology and it requires proper training for handling. LOC indicates the early stages of diseases and they are used for the detection of different analytes such as protein or their derivatives, metabolites, nucleic acids, cells, and pathogens. Current LOC devices utilize immunoassay technology including antigen-antibody binding and with the help of this technology, we can detect troponin I, troponin T, prostate-specific antigen for prostate cancer, and a pregnancy test to detect HCG. The best example of LOC technology is the pregnancy test kit that is based upon the antigen-antibody complex in which the pregnancy hormone reacts with antigen to produce a complex and shows either a plus or minus sign within minutes. LOC technology is also used to detect different protein variants; one of them is glycated hemoglobin linked to sugar. It measures the three-month average blood sugar level and is used as a diagnostic test for diabetes mellitus. LOC technique is also used to detect different types of cells such as infectious diseases, cancers, inflammatory responses, hematological parameters, metabolic products, the concentration of ions, and nucleic acids. PCR on-chip is one of the most challenging assays to develop due to additional steps required for the sample It is used to assay different particular genomic details of a subject. It requires cell sorting, isolation, lysis, DNA extraction, signal amplification, and detection of the end project.

In this topic, students will learn about:

- Techniques used in LOC
- Applications of LOC

Lesson 118 : Detection Principles in Lab on Chip

The detection principles are classified into several types. They could be optical techniques, electrochemical techniques, piezoelectric methods, and magnetic methods. The objective is to two key objectives; sensitivity and selectivity to minimize the numbers of false negatives and false positives.

In magnetic detection, the magnetic field is applied to detect the magnetic material and this phenomenon is called ferromagnetism. Ferromagnetism is a property of the material in which all the magnetic ions of the material contribute positively to net magnetization. They exhibit a strong attraction to magnetic fields and can retain magnetic properties after the external field has been removed. Examples include nickel, cobalt, iron, etc.

The magnetic field is used in two ways in LOC for magnetic detection. The first way is to concentrate and the other is to localize the analyte. Magnetic particles can be used as a labeling

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technology for detection without the requirements of fluorescent dye. This technology is used for different operations in LOC.

Superparamagnetisim is a magnetic property that is produced in Nanoparticles and the size of Nanoparticles ranges from 10nm to 150 nm in diameter. The Nanoparticle's magnetization can randomly flip the direction under the influence of temperature. An example is superparamagnetic iron-oxide nanoparticles (SPION).

In this topic, students will learn about:

- Types of detection principles in LOC
- Ferromagnetism
- Magnetic detection in LOC

Lesson 119 : Dry Reagent Storage

Storage of chemicals is one of the important elements of the Lab on Chip. Some particular reagents and chemicals are required for performing the test. It is essential to store these chemicals and reagents in a form that is immediately available for performing the test. These are stored either in dry or liquid form. In dry storage, chemicals and reagents are stored in dried form on the absorbent paper pads such as urine test strips which with the pads allow several determinations simultaneously and the results can be qualitative or quantitative. In urine test strips colored reagents are used that change their color under different conditions and can detect the results even in the common man. For example, the presence of bilirubin, proteins, ions, ascorbic acid, glucose, or pH of urine is checked through these strips. In the detection of pH, the range is about 5-9 so the strips show colors that change from orange at pH 5 to yellow, green and dark blue at pH 9. We can also detect blood in urine by using these dip strips which are highly sensitive and can detect as low as five RBCs per microliter which are lysed on the pad to liberate hemoglobin. Iron present in hemoglobin acts as a nano enzyme and catalyzes the conversion of peroxide into the water with the release of electrons and takes part in the detection of the analyte. Another example is the detection of the bilirubin in urine by using test strips as a diazotization reaction in which the bilirubin combines with a diazonium salt to produce azo dyes.

In this topic, students will learn about:

- Storage of chemicals in dry form
- Urine test

Lesson 120 : Dry Storage of Biological Molecules

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Biological molecules like enzyme and antibody are used as a diagnostic tool in several modern hand held devices in clinical biochemistry. It requires the use of dried and stable biological molecules that retain their biological activity after mixing in aqueous environment. Biological molecules are heat liable. Trehalose acts as a good stabilizer. Freeze drying or lyophilization involves the removal of water molecules from the sample by freezing the sample and then lowering the pressure by creating a vacuum. The water molecules are removed from the sample and protein of sample is converted into powdered form. The protein powder acts as an enzyme by retaining its shape after reconstituting it in water. Trehalose is a natural sugar and has been found as an effective stabilizer of liable proteins during lyophilization and exposure to high temperature in solutions. Trehalose is a disaccharide composed of two glucose molecules joined by an alpha-alpha (1, 1) glycosidic bond. It is a white crystalline substance that occurs naturally in small amount in mushrooms, honey and yeast. It is less hygroscopic. Trypsin enzyme activity can be stabilized in the presence of trehalsoe.

In this topic, students will learn about:

- Dry storage of Biological molecules
- Trehalose as a universal stabilizer

Lesson 121 : Wet Reagent Storage of Biological Molecules

Lab on Chip is a complete lab for which stored reagents are needed. Two strategies are used for storing the reagents; dry storage and storage of liquids. Storage of liquids is very important for LOC as the reagents could be used immediately and the device that is used for this purpose is called a micro-dispenser. Microdispenser mainly performs two functions; one is the storing of liquid reagents and the other is that it dispenses the reagents across different microfluidics platforms. These micro-dispensers are manufactured with several materials based on aluminum and glass polymer. They all must be inert and should not react with other materials. Commonly two methods are used to produce these micro-dispensers; the cold-forming method in which no heat is applied and aluminum is used as a substrate. The advantage of this is that aluminum is the complete barrier for water and oxygen and it allows an extended product expiry date. The second method is the thermoforming method in which plastic polymers are used and heated to the extent to which they become pliable. Then this warm plastic is treated under high pressure to form a cavity into a negative mold. The mold is then cooled in such a way that the plastic becomes rigid again and maintains its shape. Moreover, several types of packaging materials are used for different storage devices for the liquid reagents such as reagents in blisters, glass ampoules, stick packs, thermally actuated paraffin film, hydrophobic chambers, and hydrogel microvalve.

In this topic, students will learn about:

- Strategies used for storing the reagents on LOC
- Uses and manufacturing of Microdispenser

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Coldforming and thermoforming method

122 Sample Introduction in LOC

There are four steps in the functioning of Lab on Chip; sample introduction, sample preparation, sample testing, and detection of signals. The introduction of the sample is very important. Sample can be introduced by two methods; passive method and active method. In the passive method, the gravitational force is the main force. It allows the non-absorption and fast mobility of the aqueous sample. In the active method, different external forces are used through microdevices. Two types of micro devices are used; micro valves and pressure pumps. The forces used by these devices are; electric current, ionic strength, change in mass and change in pressure, etc. the microfluidic analytical systems require micro-pumps and microvalves for the active sample application. Microdevices are required for the sample application, buffer addition, reagent flow, mixing, and delivery. Active sample introduction can be improved by superhydrophobic coating. A hydrophobic coating is a thin surfaces layer that repels water. It minimizes the surface adhesion and increases the blood clotting time. The integrated operation of microfluidic pumps and valves can be controlled by finger actuation. Polydimethylsiloxane (PDMS) is used for controlling the pneumatic valves, flow direction, and pumps.

In this topic, students will learn about:

- Methods for introduction of sample in LOC
- Microdevices used in sample introduction
- Superhydrophobic coating

123 Mixing in LOC

Mixing a sample with its reagent is important in LOC technology because it is necessary to make the sample with the reagent to get either a negative or a positive result. In LOC, as the samples are confined to microchannels, the mixing becomes difficult due to the low Reynolds number with laminar flow. Two strategies are used to increase the control and fast mixing in LOC. One is passive mixing in which the designing of the microchannels facilitates the mixing of reagents with its sample and it involves two approaches; the introduction of some holes in the microchannels that will increase the speed of the fluid and the other is the designing of channels to increase the contact area or time. For example, T shaped mixer is used that consists of two inlets where samples flow perpendicularly to each other and one outlet. Mixing efficiency can be increased by adding obstacles in channels. The second type is active mixing in which mixing

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efficiency is increased by external forces applied to the samples and these forces include acoustic waves, pressure perturbations, magnetic field, thermal methods, propagation of ultrasound waves, and fluctuation of the electric field. This strategy uses active components such as micro-pumps for dosing and microvalves that determine the flow direction of liquids.

In this topic, students will learn about:

- Strategies used for sample mixing in LOC
- T shaped mixer

124 MNG in LOC

Magnetic Nanoparticles are used in LOC to facilitate the number of operations. The magnetic nanoparticles are smaller than one micrometer in diameter. Magnetic nanoparticles consist of two components: a magnetic material such as iron, nickel, and cobalt and a chemical component that has functionality. The functional properties of magnetic nanoparticles largely depend on synthesis method, size, and chemical structure. An important property of Magnetic Nanoparticles (1-100nm) is super Paramagnetism. The iron-based Nanoparticles provide different chemical properties. A zero-valent iron in the core mainly provides the reducing power for the reactions and the oxide shells provide sites for the sorption and adsorption. The surface modification of the magnetic material is essential for most chemical and biological applications. It induces biostability, biocompatibility, catalytic properties, and vehicle for the targeted drug delivery. Magnetic Nanoparticles are the carrier. The core and shells are the potential carrier. The active agents attached to the shell can be adsorbed into the magnetic core. The active agents can interact with the shell through hydrophobic and hydrophilic interaction. Different capping agents including salicylic acid, glutamic acid, and trichloroacetic acid are used. Salicylic acid allows the formation of strong hydrogen bonds.

In this topic, students will learn about:

- Magnetic Nanoparticles and their functionality
- Iron-based Nanoparticles
- Capping agents

125 Sample Processing in LOC

Sample processing and sample pre-treatment are the essential components of Lab on Chip operations. Sample processing and pre-treatments can be of different types depending on the nature of the sample. An analyte is present within an extremely complex matrix. The isolation

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and ‘clean-up’ of a particular analyte are required for the processing of the sample. The processes may include sample filtration, centrifugation, dilution, concentration, target amplification, and extraction. In blood sampling and processing, the use of a superhydrophobic membrane and coating is the first step. It minimizes the surface bio-adhesion of the blood and increases the blood clotting time. It also prevents the loss of target molecules. Another available advanced technique is Multiple Slanted Spiral microchannels to pre-treat the blood. When a blood sample is applied, the forces in the system drive different particles in different directions. The smaller particles migrate near the outer wall of the channel.

In this topic, students will learn about:

- Sample processing in LOC
- Multiple Slanted Spiral micro channels
- Processes in LOC

126 Platform on LOC

LOC or Lab on-chip platform provides an infrastructure for lab on a chip. A microfluidic platform is a board for the setting up of components for multiple operations. LOC platform is suitable for architectural design for a proper combination of functional units by employing a miniaturized fabrication technology. It provides integration, automation, and biochemical processes in parallel. LOC platforms are of different types such as lateral flow platforms, linear actuated platforms, Microfluidic large-scale integration platforms, centrifuged microfluidics platforms, and electrowetting platforms. A lateral flow platform is easily available in the market for pregnancy tests. A Linear actuated platform is a very simple platform in which the flow of fluid is controlled by mechanical devices such as a polymer. The reagents and reaction buffers are pre-stored in pouches.

The second type of platform is the microfluidic Large Scale Integration (mLSI) platform. It has microfluidic networks to handle a large number of samples. It uses microdevices as building blocks. It is an effective, complex, and highly integrated platform. The microdevices used in this platform are microvalves, micro pumps, mixers, multiplexers, etc.

In a Centrifugal microfluidic platform, the microchannels and chambers on a substrate are in the format of a disk. The forces involved in the transport of liquids are centrifugal force, Euler force, and capillary force. An example of this platform is Spiral-Slits chips for centrifugal microfluidics.

The Electrowetting platform is the most important type. Electrowetting is the modification of the wetting properties of a surface with an applied electric field. The operation includes the creation of a droplet, transportation, mixing, and splitting of droplets.

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In this topic, students will learn about:

- LOC platform and its functions
- Different types of LOC platform

127 Retector in LOC

Lab on a chip consists of nine basic parts: injector, transporter, preparatory, mixer, reactor, separator, detector, controller, and power supply. The most important part is the reactor, where almost all the reactions occur under certain conditions. The reactor also consists of several micro-devices, such as a heater that controls the temperature, sensors that detect temperatures, pH, pressure, etc., and actuators. These micro-reactors are useful tools for optimizing and studying chemical reactions. They reduce the chemical amounts, enable higher reaction velocities, and collect information about the reactions. The use of micro-reactors has several advantages, such as the cost and risk are reduced as only a minimal volume of sample and reagent is required for the reaction process. Moreover, it improves heat and mass transfer properties in micro-fabricated structures and produces a high yield. These micro-reactors have three types based on the reaction carried out in the reactor. The first is a gas-phase reactor in which at least one component is in the gaseous form, either in reactants or products; the second type is a liquid-phase reactor for liquid samples and reagents in which both reactants and products are in liquid form. In this reactor, reagents and samples are continuously fed into the reactors. Pumps are used for continuous supply at a constant speed, and the products are immediately separated. The third type is a packed bed reactor in which reaction takes place on the surface of a catalyst such as NPS that has been extensively synthesized using microfluidics.

In this topic, students will learn about:

- Nine basic parts of Lab on a chip
- Advantages of micro-reactors
- Types of micro-reactors

128 Detectors in LOC

Detectors are used to detect the bio-recognition event, and they perform two main functions; the recognition of a bio-recognition event, and its quantification. Detectors are transducers that detect the signals and then convert them into electrical signals by using detection methods such as optical methods, electrochemical, mass spectrometry, and magnetic methods. Optical methods are commonly used because they are easy to integrate, low cost, and easily available. In optical

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detection methods, we detect different properties such as absorbance, fluorescence, chemiluminescence, surface Plasmon resonance, and interferometer-based techniques. Smartphones are also used as optical detectors, and they are provided with high-resolution cameras and white flashlights, that are used as powerful optical detectors and light sources. For example, cell phone-based fluorescent imaging can detect the presence of E. coli in samples by using micro-fluidics channels in which quantum dots coated with an antibody are used. When a water sample is passed through these channels, the E. coli binds with the antibody and quantum dots can be detected. This system uses a lightweight, compact attachment to the cell phone camera. Another example is a mobile water kit used to rapidly detect E. coli and coliform bacteria. A simple platform is produced in the form of a mobile application that is known as mHealth E. coli. We have to download this mHealth E. coli app on our mobiles to detect the presence of E. coli in the water sample.

In this topic, students will learn about:

- Functions of detectors
- Optical detection
- mHealth E. coli app

129 Controller and power Supplies in LOC

Controller and power supply are two main parts of LOC in which the controller is software consisting of certain modules including micro-controller, communication interface, programmable memory, control circuits, signal preconditioning circuits, and power interface circuits. These modules carry out all the operations in a controlled environment. The controller has two types; the first is the controller system on chip which means that all the necessary components are integrated on chips, and the second type is the controller system off-chip, in which the control panel is not integrated into the chip, and there is the main system to control the functioning of the chip. A power supply is an essential unit in the LOC system as LOC devices utilize batteries as a power source. They energize the on-chip components such as transducers, actuators, and electronic circuits. The power system provides power for the system to operate. For example, self-containable and adjustable batteries are used for hi-tech operations on LOC like electrophoresis and PCR. WHO has introduced proper criteria for low-cost diagnostic devices that are known as ASSURED which stand for, sensitive, specific, user friendly, rapid and robust, equipment-free and easily deliverable to the end users.

In this topic, students will learn about:

- Parts of LOC
- Types of Controller
- ASSURED

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130 RT PCR in LOC

Reverse transcription-polymerase chain reaction (RT-PCR) on LOC is used in molecular biology, genetic diseases diagnosis, and forensics to quantify specific RNA within cells or tissues. In this technique, specific mRNA has to be isolated from a cell or tissue. This extraction of high-quality mRNA is critical for accurate RT-PCR performance. The technique used in RT-PCR to get a good quality RNA is called megnetophoresis, a contactless method for manipulating particles in which particles move in a magnetic field. This technique does not affect the properties of samples such as pH, ionic strength, surface charge, and temperature. The advantages of using this technique include; being easy to operate, low in cost, and simple in design. In LOC, closed systems are used to isolate mRNA, which also protects it from RNA degrading enzymes. The micro-fluidics channels are made with special materials such as iron and nickel, as a result of which the magnetophoresis technique is used to isolate RNA. In the end, magnetic oligo-dT beads are used for the purification of RNA. In RT-PCR on LOC, the chip is composed of PDMS glass. There are two separate units for mRNA; RNA microchannel and DNA microchannel. Once the mRNA is isolated, it is transferred to micro-fluidic channels to synthesize cDNA using the mRNA template and the stored reagents.

In this topic, students will learn about:

- Reverse transcription-polymerase chain reaction on LOC
- Megnetophoresis and its advantages

131 Paper based LOC

It is a technology in which a paper is used as a diagnostic tool. A paper-based LOC comprises cellulosic materials as cellulosic fibers guide liquid from an inlet to the desired outlet by imbibition. Imbibition is a special type of diffusion that occurs when water is absorbed by solid colloids, causing an increase in volume. In the case of paper-based LOC, we use two materials; the first is cellulose, a plant-derived polymer, a natural polysaccharide composed of units of glucose and contains pre hydroxyl groups. The second material type is nitrocellulose, a derivative of cellulose, a highly flammable compound formed by nitrating cellulose through exposure to nitric acid or any other powerful nitrating agent. Paper-based LOC has a proper structure containing inlet, microfluidics, and storage of reagents, transporter, mixer, reactor, detector, electronics, and batteries. The most important thing in LOC is the flow of fluids governed by several physical properties of the fluid, including viscosity, density, contacts angle, temperature, and surface tension. But different physical parameters can affect the flow of liquids, such as permeability, geometry, evaporation effects, and pore size of substrate can affect the flow rate of the fluid. There are certain ways to control the fluidic flow in channels, such as

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changing the channel length and width, altering the wettability of paper, diverting some fluid through a parallel change, and changing the fluid's viscosity paper technology is used to separate molecules by using a moving solvent on filter paper and is known as paper chromatography. This technology has certain advantages, such as filter paper can remove containments from the sample and prevent them from moving down the micro-channel. Moreover, paper microfluidics is extremely low-cost, user-friendly, and portable.

In this topic, students will learn about:

- Materials used in Paper based LOC
- Ways to control the fluidic flow
- Paper chromatography

132 Paper based LOC and Assured Criteria

Paper-based LOC is fully equipped with the elements to be used as diagnostic tools and has gained popularity in the consumer market due to its compactness and portability. Paper-based LOC contains several elements: control system, microfluidics, electronics, sensing element, data processing, readouts, display unit, connectivity, and energy storage. For processing samples, different modules are used, such as lateral flow test strips in which there is a sample pad, conjugate pad, test and control lines, and a wicking pad. Another module is the 2D paper-based diagnostics using hydrophobic fluidic barriers such as wax printing. The third module is 3D paper-based diagnostics using stacking of multiple layers folding or origami paper device, and it consists of two layers. The test solutions are loaded to the injection zone, where the chemical reaction occurs in a folded 3D microfluidic device that is then air-dried. An image of the results is visible after unfolding the paper device. Several control systems are also integrated into the paper, requiring switches, valves, a heating system, and an energy source to detect the target molecule; therefore, different sensors could be placed on chips, such as electrochemical, Temperature, chromogenic, and light sensors. For the proper functioning of LOC, several microelectronic pieces of equipment are required, such as transistors, electrodes, capacitors, and inductors. For diagnostic tests, several external instrumentation solutions can be used with paper.

In this topic, students will learn about:

- Elements of Paper-based LOC
- Modules used for processing samples
- Control systems

133 Conducting Ink Technology

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Conductive ink is an advanced technology used to print different circuits and electrodes on paper chips. Electronics are the main elements of this technology integrated into the paper chip, and ink is infused with a conductive material like graphite and silver, which enables the ink to conduct electricity. Several conductive materials could be used for preparing conductive ink, such as conductive polymers, carbon, organic metallic compounds, metal precursors, and metal NPs. There are several methods for preparing ink. For example, an aqueous solution of silver nitrate is combined with poly-acrylic acid and di-ethanolamine, which serves as a capping agent and a reducing agent. After processing, nanoparticles of about 400nm are formed, and it is mixed with hydroxyl-ethyl cellulose that acts as a binder to make this ink thicker. A conductive ink pen is a liquid polymer delivery system in which the polymer is electrically conductive and is rapid to dry. They are used to create or repair conductive traces on printed circuit boards; for example, silver-infused conductive inks are used in modern transit tickets, and the life of a ticket depends upon the number of circuits used in them. The functionality of conductive ink can be increased by adding different types of nanoparticles. For this purpose, a certain ratio of NPs is inserted into a solvent to get a wanted viscosity, which leads to the formation of a liquid called metal ink with NPs in the suspension.

In this topic, students will learn about:

- Conductive ink
- Conductive ink pen
- Conductive materials

134 Paper Battery

Paper batteries act as an energy source for Lab-on-Chip technology. It is based on a thin paper acting as a flexible energy storing device employing cellulose or cellulose-based compounds as a substrate. These can be used both as a battery and a capacitor because they are long-lasting, non-toxic, and provide steady power. Paper batteries work on the same principle as those conventional batteries in which the battery is dipped in ion-based liquid and the battery starts working. As a result, a chemical reaction occurs between the electrodes and the liquid. Therefore, electricity is generated by the movement of electrons from the cathode to the anode. Lithium is used in paper-based batteries, making the battery more energy-dense than lead and nickel metal-based materials. Due to these characteristics, these batteries are widely used in automobiles, laptops, cameras, and other electronic devices. Lithium is a lightweight alkali earth metal with low density compared to pine wood, and it is the least reactive metal as it has only one electron in the outermost shell. The major components used in paper batteries other than lithium include carbon nanotubes used for cathode terminals. Carbon nanotubes must have low mass density, flexibility, high packing density, lightness, and better electrical conductivity than silicon. Other major materials include lithium metal used for anode terminal, Li used as lithium

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oxide and lithium cobalt oxide, different electrolytes such as blood, urine, sweat, etc., and cellulose that acts as a separator. Cellulose has several properties such as excellent porosity, biodegradability, non-toxic, recyclability, high tensile strength, good adsorption capacity, and low-shear strength.

In this topic, students will learn about:

- Paper batteries and working principle
- Carbon nanotubes
- Properties of cellulose and lithium

135 Printing in Paper Chip

Different techniques are used for integrating physical components on paper to get desired patterns. These desired patterns are the architect of the lab-on-chip, and after printing, the biological and chemical reagents are integrated onto the paper. For developing paper LOC, a hydrophobic physical barrier is needed for passive transport of aqueous solutions created on a hydrophilic paper. Biological and chemical reagents must then be deposited selectively with the device either by dipping or locally spotting. Different devices are used for printing technology, such as wax printing, which is a simple and inexpensive method for the fabrication of microfluidic devices. As wax is a solid material at room temperature, thousands of heating elements on the print head cause the wax to melt on the roller. Wax adheres to specially coated paper, and micro-channels are integrated on the paper surface. Inkjet printing is another inexpensive and high-resolution method that requires a paper in the hydrophobic polymer. It is well suited for low viscosity, soluble materials like organic semi-conductors because nozzle clogging could occur if the materials are of high viscosity. Other methods used are photolithography, flexographic printing, laser treatment, and digital light processing (DLP), which is based on optical micro-electro-mechanical technology and uses a LASER light source and micromirrors. Digital mirror devices are used for printing that employs a photocurable resin for printing.

In this topic, students will learn about:

- Techniques for printing on LOC
- Devices used for printing technology
- Wax printing

136 Arrays in biosensor

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In biosensors, more than one sample can be tested at a time or more than one sample can be used for target molecules, so arrays are very important because all the results of the samples need to be arranged. An array is a data structure that contains a group of elements, and the collection of elements have the same name and same data type as these are arranged in natural numbers(any integer greater than zero), and data is organized in such a way that a related set of values can be easily stored or searched. Arrays in biosensors perform many functions, but one of the most important features is their two-dimensional configuration, allowing controls and standards to be analyzed in parallel with unknowns. These arrays can also test multiple samples simultaneously, rapid testing, on-site screening, and reliable detection. Arrays in biosensors are arranged in different geometrical patterns such as circular, cylindrical, linear, planar, spherical and square. To arrange arrays in biosensors, a mosaic technique is used that uses tiny particles to create a whole image in small square tiles called tesserae and rounded tiles known as pebbles. These arrays can also be used in optical sensors.

In this topic, students will learn about:

- Array introduction and functions
- Arrangement of arrays in biosensors
- Mosaic technique

137 Substrate for paper

Paper is used as a substrate for micro-pads. This paper is produced from three different types of plant tissues; the Hemicellulose, lignin and cellulose. Cellulose is extracted from plant pulp by using different methods such as alkaline solutions through which we get three types of cellulose; alpha cellulose that is a pulp fraction resistant to alkali under certain conditions and it is highly stable with highest degree of polymerization. Second type is beta-cellulose that is the soluble fraction is alkaline solution that re-precipitates on acidification and the third type is Gamma-cellulose which remains soluble in the solution. Depending upon the conditions lignin is also present in paper that is darker in colour as compared to cellulose and can affect the quality of paper. The exposure of sunlight and air of lignin causes paper to turn yellow just like newspapers. In paper some additives are also used to get the required functionalities such as binders, biocides, coating agents, dyes, optical fibres, retention aids, sizing agents and wet strength agents. The paper in micro-pads has its internal network in a structure such as Fibrous with inter and intra chain hydrogen bonds, non-fibrous materials, capillaries and pores. A good paper must have multiple properties: surface area affecting the load of reagents, flow rate representing the speed of a migrating analyte, low cost, thickness, pore size, and porosity such as Filter paper which has all these characteristics.

In this topic, students will learn about:

- Plant tissues used in preparation of paper

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- Types of cellulose
- Properties of a good paper

138 Final Fabrication of paper microfluidics

Paper-based microfluidics is also known as micro PADs. Micro PADs are analytical devices that have hydrophobic barriers created by generating hydrophilic channels. There is no external force required for liquid transport due to wicking. There are two basic fabrication methods; chemical patterning and physical patterning. Chemical patterning involves the creation of barriers by blocking the pores inside the paper. Different chemical methods include analog plotting, inkjet printing, etching, spraying, plasma treatment, flexography printing, screen printing, and laser treatment. In physical methods, different physical forces are used. The physical methods include knife Plotter, craft cutting, and embossing. In a knife Plotter, a computer-controlled knife plotter is used for precise cutting of the paper to obtain the desired pattern. In craft cutting, a digital craft cutter issued to fabricate laminated paper-based analytical devices. Embossing is pressing an image into paper to create a 3D design.

The detection of micro PAD can be carried out visually by comparing the color intensity in each reaction zone. Various devices are used to get this calorimetric information such as cameras, scanners, smartphones, etc. By using a suitable software application, it is possible to get the parameter of any color change from a digital picture of a micro PAD. These parameters can be used to obtain the color intensity vs. concentration regression line.

In this topic, students will learn about:

- Micro PADs
- Methods of fabrication
- Detection on micro PAD

139 Final Paper microfluidics deisign

Paper-based microfluidic devices can potentially serve as a cheap and disposable tool for analytics and diagnostics. The paper has three-dimensional structures: X, Y, and Z-axis. Z-axis is along with the thickness of the paper. The micro pads have multiple features like sample collection, filtration, separation of mixtures, mixing with reagents, analysis, display of results, and multiplex assay. Device design can be simply differentiated by the directional flow of the fluid. When the flow of a fluid is in a single direction, it is called 1D. 2D is the lateral flow of the liquid in multiple directions on the horizontal plane. The 3D design adds a vertical component. In 1D, the wet sample is applied and the dry reagents are dissolved by the fluid sample solutions.

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This allows the reaction to occur and the color product can be detected on the paper. In 2D, the biological reagents are fixed on the paper and passive fluid transport occurs in micro pads due to the porous nature of the paper. In fully enclosed 2D PADs, a protective layer is printed thermally to protect the channels from contamination. The 3D paper devices are developed to increase the number of test samples. In this, several sheets of paper need to be patterned, stacked, and aligned sequentially. The 3D paper devices involve waxing and screen printing techniques and stacking of layers. The holes if placed are filled with cellulose powder to provide a vertical microchannel.

In this topic, students will learn about:

- Structure of a paper
- Features of micro PADs
- Types of flow of fluid
- 1D, 2D, and 3D paper types

140 Fluidic control in paper microfluidics

When a droplet of liquid contacts a paper; it leads to the competition between four main phenomena. These are; droplets wetting the surface, paper absorbing and wicking the droplets by capillary action, evaporation proceeding, and diffusion occurring. All these four steps lead to the equilibrium conditions. The time frame for wetting ranges in seconds, absorption in seconds to minutes, wicking in seconds to minutes, evaporation in minutes, and liquid diffusion also occurs in minutes. Wicking utilizes the void spaces present in the hydrophilic fiber that is in the form of pores in the substrate paper. Wicking depends upon the diameter, orientation, and length of the paper. Different methods are employed to have effective fluid control on paper. These are geometry-based methods, and physical-based methods. The rate of fluid flow through a paper can be altered by physical methods using an adsorption pad. The paper device has an absorbent pad on top. The absorbent is in contact with the main channel. This contact produces a delay in the fluid flow that controls the flow of liquid through the paper. The geometry-based methods achieve fluid control through an alteration in the fluid channel. The change could be in channel length, channel width, and inflow path. In chemical-based methods, different types of chemicals are used. For example, wax is used for surface coating in microfluidics and surfactants that can lower the surface tension.

In this topic, students will learn about:

- Steps in fluid control

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- Wicking
- Physical-based methods
- Geometry-based methods
- Time frame for different phenomena

141 Sensors on paper

Sensors are the main components of biosensors therefore papers are used as a substrate on which different types of sensors can be integrated such as chromogenic sensors, electrochemical sensors, pH sensors, optical sensors, pressure sensors, proximity sensors thermal sensors, and humidity sensors working on the principle of capacitance. In humidity sensors electrodes are directly drawn onto the paper using conductive ink which absorbs all the moisture from the air. These humidity sensors exploit the hygroscopic character of cellulose paper which is the tendency to absorb water from the environment. In optical sensors, a filter paper of white color is used for fabrication because any change in its color during the detection of the analyte can be detected easily. In such cases, the chromogenic substrate for the enzyme acts as a sensor. For example, Diaminobenzidine is used as a colored substrate for HRP. In pH sensors, pH test paper is used that is soaked in several pH indicators that gives the paper ability to determine an approximate wide pH range of acids and bases such as Methyl red (4.4-6.3), Bromothymol blue(6.0-7.6) and phenolphthalein(8.2-10). Another type of sensor is the temperature sensor which is made out of a simple resistive structure using metals such as aluminum foil or silver ink. Its principle is based on the change in resistance of the metal due to the change of temperature that is detected as a signal.

In this topic, students will learn about:

- Humidity sensors on apper
- Optical sensors
- Temperature sensors
- pH sensors

142 Applications of paper microfluidcs

Paper-based microfluidic devices have advanced significantly because of certain reasons as they are affordable, automated flow, portable, and biodegradable diagnostic platforms because they could be degraded for the cellulase enzyme that is present in nature. These paper-based microfluidics have a variety of applications in three main sectors: health, environment, and food quality. This technique has several advantages such as it can analyze simple lateral flow tests, but it can also perform complex analyses requiring multiple-stage reactions. The micro-pad

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devices can be fabricated in 1D, 2D and 3D devices for a variety of applications. Research is being made on paper-based microfluidics but these are not much popular among people and available in the market in lesser amounts. The main reason for this is a gap between research and development. We must look at the developed products under certain conditions such as their practical utility, operational simplicity, stability of devices, and availability of associated equipment.

In this topic, students will learn about:

- Paper-based microfluidic devices
- Applications and advantages of Paper-based microfluidic devices

143 Introduction to nanosensors

Nano-biosensors are a type of biosensors consisting of Nano-materials which are the sensing component of a device to enhance the output of sensors. These nanostructures are functionally integrated into the nano-bio sensors in two ways; as an intermediate layer or as biological events in which the transducer is combined with nano-material. Nanomaterial integration in nano-bio sensors plays a key role in enhancing the performance of the device. Different types of nanomaterials are used in nano-bio sensors to improve their sensing mechanisms such as carbon nanotubes, nanowires, nano-rod, nano-diamond, metal nanoparticles, metal oxides, and quantum dots. Nano biosensors function at the nano-scale despite their sizes and these are the sensory points that are biological, chemical, or physical. These nano-bio sensors have dimensions at the nanometer scale and they can convey information at the macroscopic level by using nano-particles. Nano-biosensors are fabricated by using a technology called NEMS which stands for Nano-electrochemical system and it used four processes including photolithography, chemical bonding, thin-film etching,, and surface etching. Since gold nanoparticles are highly unstable so another technique known as Silk fiber is used that is extracted from silkworms consisting of sericin protein(outer coating) and fibroin protein(central core). Sericin protein is composed of 18 different amino acid residues of which 32% is serine and it contains flavonoids and carotenoids that act as gumming agents. Sericin is an amorphous hydrophilic protein that consists of random hydrophilic and hydrophobic blocks of amino acids while fibroin is made up of amino acids to form beta-pleated sheets. These protein structures consist of a network of inter and intra-chain hydrogen bonds.

In this topic, students will learn about:

- Nanomaterial integration in Nano-bio sensors
- Materials used in Nano-biosensors
- NEMS
- Sericin protein

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144 Nanomatertials act as nanoenzymes

Enzymes are biological catalysts and these are protein-based molecules but recent researches show that other nanoparticles can also perform an enzymatic activity such as they can act as catalase, oxidase, peroxidase, and superoxide dimustase enzymes. These nanoparticles have many advantages over natural enzymes as they are low in cost, highly stable, have robust catalytic performance, and allow smooth surface modification. These nanoparticles have been classified into two classes: scavenging free radicals and generating free radicals. The first class is antioxidants which include superoxide, dismutase mimetic NP, and catalase mimetic NP while the second class is pro-oxidants which include peroxidase mimetic NP and oxidase mimetic NP. Superoxide dimustase is a detoxification enzyme that is the most powerful endogenous antioxidant enzyme in the cell. It acts as a component of the first-line defense system against reactive oxygen species. Depending upon the protein fold and the metal cofactor, superoxide dismutase has three types including Cu/Zn type which binds to both copper and zinc for its functioning, and Fe/Mn type which binds to either iron or manganese and Ni type which binds to nickel for its activity. The cerium oxide nanoparticles exist with two oxidation states that can perform two types of catalytic activities: superoxide mutase activity and catalase activity. Cerium has a variable electronic structure and the energy of 4f electron is nearly the same as that of outer 5d and6s electrons with dual oxidation states of +3 and +4.

In this topic, students will learn about:

- Types of superoxide dismutase
- Advantages of nanoparticles as enzymes
- Classification of nanoparticles

145 CNTs in nanobiosensors

Carbon nanotubes are large molecules of pure carbon that are 1-3 nanometers in diameter but the length could be from hundreds to thousands of nanometers. It is composed of a graphene sheet consisting of hexagon carbon atoms that are rolled in such a way that a carbon nanotube is produced. These carbon nanotubes are very stable because they are a single chain of unbroken covalent C-C bonds entirely composed of sp2 hybridization and they provide a very unique strength. Electrical properties are also introduced into these nanotubes just by the virtue of pi bonds. Carbon nanotubes in Nanobiosensors are the most important biosensors as they have an ultra-high surface area that permits efficient loading of multiple molecules along the length of the nanotube sidewall. But these are non-functional due to certain reasons such as they are carbon-based so they are highly stable and are insoluble in aqueous media. They can be functionalized by using two techniques: the chemical method and the non-chemical method.

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Carbon nanotubes are functionalized either by covalently or non-covalently attaching to another moiety and the three main classes are used for the dispersion of CNTs including organic acids, surfactants, biopolymers (peptides and DNA), and synthetic polymers. They are mainly functionalized to make them bio-compatible, to increase their solubility in physiological solutions, and to increase their binding ability with the target molecule.

In this topic, students will learn about:

- Carbon nanotubes
- Properties of Carbon nanotubes
- Functionalizing techniques of carbon nanotubes

146 Quantum dots in nanosensors

Quantum dots are Nano-sized crystalline structures ranging from 2-10 nm composed of semiconductors. Qds possess superior fluorescent properties with less photo-bleaching (fading) which is a photochemical alteration in a fluorophore molecule. These quantum dots produce different colors depending on the size of the nanoparticles. After the excitation of electrons in an atom, the atom emits light. The color of the light produced by QDs depends upon the energy levels and varies from one atom to another. Two types of colors are mainly produced: deep blue color for which the size of quantum dots should be small and there must be a higher energy difference between the valence band and conductance band. The other type of color is Red color for which small-sized quantum dots are required and the energy difference between the valence band and conductance band must be lower. Quantum dots have a direct relationship with the wavelength as the biggest QDs produce the longest wavelength with the lowest frequencies while the smallest dots make a shorter wavelength. Moreover, big dots produce red light while small dots produce blue light and intermediate-sized dots produce green light. These quantum dots have a wide range of applications such as they are being used in place of organic dyes in biological research, they can be used in nanoscopic light bulbs to light up and they are being tested as sensors for chemical and biological warfare agents such as anthrax.

In this topic, students will learn about:

- Quantum Dots in Nanosensors
- Colors produced by Quantum Dots
- Applications of Quantum Dots

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147 Nanodimonds

Nano-diamonds are pure carbon nanostructures, the architect of which are octahedral, and diameter ranges between 2-8nm. They deliver a wide range of therapeutics successfully such as small molecules, proteins, and nucleic acids because they are spherical, biocompatible, and have low toxicity. Several techniques are also available to tailor the surfaces of ND to facilitate the attachment of various medicinal molecules. The structure of diamond nanoparticles includes three main aspects; first is the shape of ND that is either spherical or elliptical; secondly, their core resembles the structure of a diamond and third is that the surface of diamond nanoparticles is the surface of graphite. These nanoparticles have a wide range of properties including biocompatibility, chemical resistance, induction of doping, hardest of the known metals, electronic properties, highest atomic density, stability in both acid/basic environments, and magneto-optical properties. Advantages of using these nanoparticles include a high absorbing capacity and a specific surface area of 5nm ND is equivalent to 50nm particles. The inter-particle distance between NDs is uniform and they are capable of penetrating the blood-brain barrier (a highly selective semi-permeable border). Sometimes a nitrogen atom has substituted for one of the carbon atoms known as nitrogen-vacancy and this improves the optical and magnetic properties as well as the performance of NDs.

In this topic, students will learn about:

- Nano-diamonds
- Structure of Nano-diamonds
- Advantages of Nano-diamonds

148 Nanowires in nanosensors

A nanowire is a nanostructure. A nanowire is made from a variety of conducting and semiconducting materials such as metals, metal oxide, silicon, and carbon nanotubes. Nanowires are hollow from the inside. Its ratio for length to width is greater than 100. The current flow through nanowires in any one direction and it is extremely sensitive to minor perturbation. The tailoring of nanowires involves three methods: doping, the introduction of heterostructures, and the use of higher bandgap material. The bandgap is the energy difference between the top of the valence bond and the bottom of the conduction band. A type of nanowire is known as a core or shell nanowire based on two types of materials: core and shell. Core and shell are made up of different types of materials and elements. They have a large surface area and the core improves the efficiency of nanowires. The combinations of core nanowires involve semiconductors with semiconductors, a semiconductor with metal, metal oxide with metal oxide, and metal with metal heterostructure. These nanowires are capable of exhibiting catalytic activity. Photocatalysis is a

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process in which a light source interacts with the surface of semiconductor material. Photocatalysis is used for the removal of pollution. The CDs-ZnO core-shell nanowires were designed to enhance hydrogen production in water spitting.

In this topic, students will learn about:

- Structure of nanowires
- Tailoring of nanowires
- Core or shell nanowires
- Photocatalysis

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149 DNA nanosensor materials

DNA-based Nanobiosensors are an important class of Nanobiosensors. This unique class of biosensors uses DNA as a bio-recognition element and specialized Nano-materials. This type of Nanobiosensors is used in the diagnostics and cell biology process. In DNA-based Nanosensors, it is required to attach the nanostructures to the nucleic molecules. The structure of DNA depends on two bases: structural forms of nucleic acids and modes of assembly of nucleic acids. At the structural level, there are three forms: linear duplex, spherical duplex, and circular duplex of DNA. The modes of assembly include naturally programmed tRNA, natural scaffold, synthetically programmed, and synthetic scaffold. They make use of Nano-materials as transducers such as semiconductor nanoparticles, quantum dots, silver, perfluorocarbon, organic polymers, and fluorescent lanthanide nanoparticles.

Perfluorocarbon is a type of compound in which only carbon and fluorine are present. It is highly used in biological studies because it is biocompatible, colorless, hydrophobic, inflammable, inert, non-toxic, and highly dense. When they are used in medical procedures, they are rapidly excreted from the body.

Luminescent lanthanide nanoparticles are lanthanide-based materials. They are highly luminescent ranging in size from 10-300 nm. They can be coated with organic silica towards functionalizing with organic groups. The colors produced by lanthanides are fader than transition metal colors. These are: highly stable, highly fluorescent, can conjugate with nucleic acid, and are stable in a wide range of pH values and the biological environment. The lanthanides at millimolar concentration can serve as nuclease mimics to cleavage both RNA and DNA.

In this topic, students will learn about:

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- DNA-based Nanobiosensors
- Structure of DNA
- Perfluorocarbon
- Luminescent lanthanide nanoparticles

150 Optical nanosensors

Optical Nanosensors are a highly specialized class of Nanosensors. Optical nanosensors are also known as PEBBLES. They are used in cells to determine their physiological functions. The sensors can be inserted into the living cells with a minimum of physical perturbation. PEBBLES are spherical optical nanosensors with sizes ranging from 20-200 nm in diameter. A PEBBLE is an abbreviation of Probes Encapsulated By Biological Localized Embedding. They are biocompatible with negligible biological effects. The sensor response time is less than 1ms. These sensors are fabricated in microemulsion. A PEBBLE consists of Fluorescent indicator dye, polyacrylamide matrix, and reference dye in the matrix. There are five methods for the delivery of optical sensors in the cell. These are endocytosis, picoinjection, liposomal delivery, membrane penetration peptide, and gene gum.

Fluorescence lifetime imaging microscopy (FLIM) is an imaging technique. The technique is dependent on the lifetime of the fluorophore signal. It produces an image based on the difference in the exponential decay rate of the fluorescence form of a fluorescent sample.

In this topic, students will learn about:

- Optical Nanosensors or PEBBLE
- Methods used for Optical Nanosensors
- Fluorescence lifetime imaging microscopy

151 Electrochemical Biosensor

Electrochemical sensors are categorized as most widely accepted sensor in Nano biosensors. These sensors work on the principle of electrochemistry in which electrochemical signals are measured with the help of electrochemical methods. The electrochemical nanosensors depend on reactions between two things the Nanomaterial Biological element and Target analyte. As the process produces or consumes electrons. The electrons are measured as Voltage, Current and Impedance. These sensors are characterized as Highly sensitive, BioCompatible and require minimal Power. The electrochemical biosensor is classified in to three categories Amperometric Nanobiosensors, Voltammetric Nanobiosensors and Potentiometric Naobiosensors.The best example of electrochemical biosensor is e-nose or e-tongue. A e-nose is basically mixture of gas

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sensors or chemical sensor which copy human nose. In general term, Gas sensor arrays are defined as 'electronic nose (e-nose),' while chemical sensor arrays are defined as 'electronic tongue (e-tongue)'. These sensors are used for Rapid sensing although the price is lower than standard used equipment. In case of e nose we use sample in form of vapors and in e-tongue the sample is liquid

Students will be able to learn

1. Example of Electrochemical nanobiosensors
2. Classification of electrochemical nanobiosensors.

152 Electronic Nose

An electronic nose is defined as a device used to sense odors in a volatile sample. This device mimics the mammalian olfactory system. Basically an electronic device consists of two elements one is array of electronic chemical sensors and second is pattern recognition system. Electronic Nose is describing as a Gas sensor. It employs a number of technologies to work as an olfactory device-nose. The odor is basically measured in real time. The electronic nose utilizes sensors as for function of receptor. When a specific sensor obtains molecules, it transfers the signal to a brain for processing. Bioelectronic nose uses olfactory receptors proteins, as bioreceptor element. It works similar to human nose. Bioelectronic nose detect the smell of cadaverine, an odorant molecule produced by rotting of food. Cadaverine has a distinctive unpleasant repulsive smell. It is a colorless liquid diamine. The Detection of cadaverine smell is dependent on the binding of cadaverine with membrane bound olfactory receptors.

Students will be able to learn

1. Definition of electronic nose.
2. How cadaverine is produced in a rotten meat sample.
3. Fabrication of electronic nose for the detection of cadevarine.

153 Electronic tongue

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Electronic tongues basically used to analyze foods and beverages and are well known sensing technologies that greatly addition to quality management. Electronic tongue is instrument that estimates and compares the tastes. The principle of Electronic tongue has array of chemical Sensors for the detection of chemical compounds. The combination of all sensors consequences a unique fingerprint on which appropriate methods used for pattern recognition of generated data. An electronic tongue has Multi-sensor systems based on array of chemical sensor. The E-tongue uses an array of many chemical sensors for the detection of different compounds present in solution including acids, astringent, Bitter, Salt, Sugars and Umami. In case of electronic tongue, the data is evaluated by Chemometric method**.** Chemometric models can be used to classify the samples and detect Concentrations of target species quantitatively. There are many advantages of electronic tongue mostly in situation where human expert panels cannot be applied due to valid reasons. The example of detection is Caffeine C8H10N4O2. In E-Tongue Caffeine can be detected on the basis of it chemistry and presence of amide and amine functional groups. Electronic tongues have many applications in different industrial areas e.g. pharmaceutical industry, food and beverage industry.

Students will be able to learn

1. Description of e-tongue.
2. The working principle of e-tongue.
3. What is meant by Chemometric studies.

154 Biosensors in Food

In organisms, the energy is assimilated by ingestion of different food items to ensure and maintain life. Food provides nutritional support for an organism. Food contains lipid, proteins, Carbohydrates, Vitamins, and minerals. Basically food is present in three forms raw, processed and cooked. A huge amount of pesticides and fertilizers are used in the crops to enhance the crop production and control pest. The remains of pesticides and fertilizers contaminate the crops and are hazardous to human and other organisms. The food is processed after harvesting and after processing it must be in packed form or canned. Canning is a technique for preservation of food in which first is to process the food contents and then properly packed in container. Canning gives edible long shelf life after the processing time. During the processing of food, a number of trials happen, one of the primary challenges is to select the quick approach for detection of allergenic components and pathogens present in the food. Biosensors give a way for the rapid detection of allergens, pathogens as well as pesticide residues in food. The main areas of any biosensor application includes detection of contaminants, freshness, confirmation of product contents and checking of raw materials. Biosensors act as logical devices using a biological material as a recognition molecules coupled within transducer or transducing microsystems. These biosensors sense Taste**,** Freshness**,** Color**,** Flavor, Texture and Consistency of foodstuffs

Students will be able to know about

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1. Different sources for contamination of crops.
2. Benefits of using biosensors in food industry.

155 Detection of ethylene

Ethylene is defined as plant hormone popular due to its important role in fruit ripening. Ethylene has gaseous nature, and have small size. ethylene represents as growth regulator tangled in every phase of the plant's life, from seed germination to tissue senescence. The basic properties of ethylene include it's a colorless combustible gas with a faint "sweet and musky" odor when pure. Ethylene is a hormone that affects the ripening and flowering of many plants. It regulates the ripening and aging of plants. As ethylene is produced in minor quantities by many fruits and vegetables. The ethylene amount is detected by electrochemical sensor with lower limits of 10 ppb. The most collective used biosensor for ethylene detection is a chemo resistive sensor made simply by involvement copper complex with nanotubes of single-walled carbon. The resulting devices show sensitivity in form of sub-ppm and high specificity to ethylene.

Students will be able to learn

1. The role of ethylene in plant physiology.
2. The physical properties of Ethylene gas.
3. Detection of ethylene

156 Detection of benzene

Fruit juices are present on numerous shapes, including a nectar type, a cloudy product, a fully clear juice, a concentrate product and fruit drinks. There are three steps in the treating of any fruit juice sequentially consist of washing, extraction, clarification and last one is preservation. Benzoic acid is usually added in form of food preservative as different salts e.g. Sodium benzoate, Potassium benzoate, Ascorbic acid (E300), Calcium benzoate and Erythorbic acid (E315). The presence of these preservative results in the production of benzene Which of dangerous effect on health for its carcinogenic nature. In 1993, it is said that benzene formation is due to decarboxylation reaction of benzoic acid and vitamin Benzene is a chemical that is a colorless in nature or light yellow liquid at room temperature. It is highly flammable. The production of benzene results in the development of acute and chronic lymphocytic leukemia and Non-Hodgkin lymphoma. For the degradation of benzene, Pseudomonas putida is very effective in term of microbial biosensor. It is Gram-negative bacterium that normally lives in water. Microbial biosensor able to degrade many aromatic compounds.

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1. Different preservatives that are used in fruit juices.
2. The production of benzene.
3. microbe that can produce benzene degrading enzyme.

157 Detection of Antibiotics in milk

An antibiotic is a kind of antimicrobial substance that is active against bacteria. They may either kill or inhibit the growth of bacteria. The widespread use of antibiotics in farm animals has contributed to the Control of diseases Antibiotics are also used to increase the life of the raw milk by some farmers. The presence of antibiotics in milk constitutes an adulteration. Cows having mastitis infections involved in antibiotic treatment in their milk, therefore, milk extracted from cows under antibiotic treatment is either discarded or collected into a separate tank for other purposes. Milk having antibiotic is not preferring for use in human consumption. Antibiotic remains can affect people's health. And can develop antimicrobial resistance. Penicillin in very minor concentrations found in milk may cause reactions in highly sensitive individuals. The presence of antibiotic is defined as in adequate concentrations to stop dairy starter microorganisms. It causes economic losses to main fermented milk industries. P. aeruginosa was used as a bioreceptor element for the developed of a biosensor. It can detect selectively bind with cephalosporin group of antibiotics. The hydrolysis of cephalosporin takes place on the microbial layer placed on the electrode. There is a change in the electric potential. The variance between working electrode and reference electrode is related to amount of drug present

Students will be able to learn

1. Use antibiotics to increase the life
2. Beta lactam antibiotic.
3. The intrinsic antibiotic mechanism of *Pseudomonas auregenosa*

158 Biosensor for Vitamin B12

Vitamin B12, also recognized as cobalamin, is one of the important water-soluble vitamin. Vitamin B12 is present in animal products such as Fish, poultary and eggs naturally. The defined deficiency of Vitamin B12 causes anemia. It is caused by common factor. One factor is called as Lack of intrinsic factor. Intrinsic factor is defined as protein synthesize in the stomach. which is required to absorb vitamin B12. This deficiency of this type of B12 anemia is called pernicious anemia. Surgery is used to removes or bypasses the end of the small intestine. This is the actual part where intestine is where vitamin B12 is absorbed. The symptoms of vitamin B12 deficiency anemia include Weak muscles, Numb, Trouble walking, Nausea etc. The structure of B12 is based on a corrin ring. There are two types of methods available for the estimation of B12. The

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Traditional quantification method are expensive, long extents of time. This makes their use impractical for a quick diagnosis that would allow adequate treatment in case of deficiency. There are at least three types of biosensor methods are available, which are quick and efficient. Three approaches are used for the fabrication of a Biosensor for the detection and estimation of Vitamin B12 in the samples.

Students will be able to learn

1. Structure of B12 vitamin and give its importance.
2. Design an immunobiosensor for the estimation of vitamin

159 Urea fertilizer in food

A fertilizer is man-made origin material that is useful to soil or to plant tissues to supply one or more plant nutrients vital to the growth of plants. There are several types of fertilizers including Inorganic fertilizer, Nitrogen fertilizer and Phosphorus fertilizer. Whereas urea is the Organic fertilizers. The important macronutrient in fertilizer includes Nitrogen, Phosphorus (P) and Potassium (K) The three numbers on fertilizer signifies value of present three macro-nutrients primarily used by plants. A series of three numbers like 10-10-10, 20-20-20, The higher the number, the more concentrated available nutrient is in the fertilizer. A fertilizer having one macro-nutrient will have marked as "0". For example, if a fertilizer is named as 10-0-0, then it means only contains nitrogen. Urea has highest Nitrogen content (about 46 percent). In case of Urea fertilizer, it can deliver only nitrogen which is primarily involved for bloom growth. Several procedures are describing to measure urea determination in Biosensors. including Spectrophotometry, Fluorimetry, Potentiometry Amperometry. The biosensors are enzyme based. The conventional analytical methods used are not time effective and typically laboratory bound whereas biosensors have ability to use easily, convenience and the ability to supply real time signals. Urea is degraded by urease enzyme They are nickel-containing metalloenzyme of high molecular weight. So urease based biosensors are used for its detection.

Students will be able to learn

1. Composition of Fertilizer
2. Mechanism of Action of Urease enzyme in the catalysis of urea.

160 Pestcide detection in food

In agriculture high pesticide use can main source to environmental pollution. Pesticide residues moved in different media, including water and food products, which may serve as straight ways

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for human exposure These compounds mimic the nature of hormones and in certain cases lead to death. Pesticides can be classified giving to the types of pests which they kill e.g. Insecticides as name suggest kills insects. Herbicides kill plants and Rodenticides means to kill rodents (rats and mice). There are mainly three pesticides that cause a serious problem such as Organophosphates, Organochlorines and last Carbamates. Organophosphates are important chemical substances that are made during the process of esterification between phosphoric acid and alcohol. These three chemicals are the key components of herbicides, insecticides and pesticides. Organophosphorus pesticides e.g. methyl parathion are extensively usage in agriculture field for insect pest control. Organophosphate pesticides (OPs) irreversibly inhibit the activity of acetyl cholinesterase present in human body and animals. It leads to intervention with function of important vital organs and in severe cases causes respiratory paralysis and death. For detection of pesticide There are two approaches that are widely used for the fabrication of biosensor to detect pesticides, based on organophosphates. Organo Phosphorus hydrolyze enzymes They can hydrolyzes methyl parathion into p-nitro phenol (PNP).

Students will be able to learn about

1. Different types of pesticides that are used.
2. Different soil microbes that can biodegrade pesticide.
3. Chemistry of Paranitrophenol.

161 Detection of pathogens in food

Detection of contamination in food is very important. food can be contaminated during different stages of food processing by Food borne pathogens which can cause serious diseases. The occurrence of food-borne pathogens and toxins must be quickly determined at various stages of production, processing and during distribution. As the meat industry is related with the health hazards e.g. deadly pathogens, toxins and heavy metals. So there is need to hold the dreadful situation and deliver safer product to customer. Biosensors are categorized as latest tool of detection in the fast rising industries comprising food industry. Biosensor detection is not only used for detection of usual pathogens including Escherchia coli, Staphylococcus, Salmonella, Clostridium perfringens, and Campylobacter. These microbes were detected by using special bioreceptor. For this purpose, DNA oligonucleotides sequences for the selected genes from eleven target microbes were used as a bioreceptor element. These ssDNA probes work together with targets present in the solution, the microcantilevers start twisting toward the gold side is recognized as bending up (positively), and bending to the opposite side denotes to bending down (negatively). The microcantilever array sensor detection, multiple bacteria can be quickly and precisely detected at levels 1–9 cfu/ml with excellent specificity. Bacterial Bacteriophages are also used for bacterial detection.

Students will be able to learn

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1. The importance of pathogen detection in food.
2. Detection of more the one pathogenic sample in a food sample.

162 Heavy metal ions in food

Metal is a solid material which is normally having properties of hard, malleable, fusible, and ductile nature, with property of good electrical as well as thermal conductivity. Metals are divided into three categories as per periodic table such as alkali metals, alkaline earth metals and last is transition metals. The division of metal is also made on the basis of density as light and heavy metal. Metals cause high damage by following activities including breaking in DNA strands, molecular cross linkage, Oxidative stress and damage by ROS. Heavy metal spread due to Untreated industrial wastewater which is released in open drains and used in the irrigation fields. This water contained high concentration of heavy metals. The metals are accumulated in soil. Plant uptakes water from soil, the metals also absorbed by roots of plants and accumulated in plants. Heavy metals are moved from soil pores to plants in mostly ionic forms, which can differ by metal. Pb, Cd, Cu, Zn, and As bio accumulated in 22 vegetables grown subsequent decreasing order: leaves > stalk/root/tubers > legume/melon vegetables. Generally whole cell biosensor or cell Biosensors are used as Baroreceptor element. There are two types of CB Turned off Cell Biosensor and Turn on cell Biosensor. In 90% tannery industry worldwide, chromium (Cr) (III) salts are castoff as tanning agents. During this process a substantial excess of a tanning agent is applied. Chromium is a basis of causing environmental pollution and also in groundwater contamination All these cases are due to wrongful discharges, accidental spills of industrial wastewaters from unsuitable landfills. The whole cell biosensor can settle plants, inoculation of soil near to their roots should permit the assessment of Cr (VI) bioavailability in rhizosphere and plant tissues.

Students will be able to learn

1. Definition of heavy metals.
2. Explain how heavy metal ions get into the crops.
3. Different approaches for Biosensor design to detect heavy metal ions.

163 Cyanide in food

Cyanide is a rapidly temporary substance that is usually known as a poison. It is originating in two forms Cyanide can be defined a gas, hydrogen cyanide, a salt, potassium cyanide. It is present in combustion products of synthetic materials such as plastics. Cyanide is used to variety paper, textiles and photographic chemicals. With reference to biological cell, there are two classes of sources of Cyanide. Endogenous Sources and Exogenous Sources. Large numbers of

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people are daily visible to low concentrations of cyanogenic compounds. The exposure cause Sedative Relaxant effect on heart & muscle, Vertigo, Vomiting and Collapse. Cyanide is also present in food it is said that if it contains HCN at 200 ppm in the form of cyanogenic glycosides. It involves calculating the inhibition of respiration. Cyanide inhibits with the proper functioning of cytochrome C in the respiratory pathway. As a result, less oxygen is spent and less carbon dioxide is produced by bacteria in the presence of cyanide. Other compounds such as pesticides and herbicides also reduction amount of bacterial respiration There are three types of approaches are used. The microbial cell is used as bioreceptor element. A microbial cyanide sensor was organized, consisting of immobilized *Saccharomyces cerevisiae* and an oxygen electrode.

Students will be able to learn

1. Cyanide is known as a poison.
2. Different plants containing cyanide ion.
3. Inhibitory mechanism of Cyanide on respiratory enzyme

164 Anti oxidants in food

Antioxidants are compounds that inhibit oxidation. Antioxidants are famous for their valuable health effects by giving protection against extreme free radical accumulation. Oxidation is defined as chemical reaction that yields free radicals, thereby helping them to chain reaction leading to chain reactions. Free radicals that may harm the cells of organisms. In the food industry, the total antioxidant capacity (TAC) of a food is determined through chemical analyses that include chromatographic, Electrophoretic and Spectrophotometric techniques. Quantification of different anti-oxidants is crucial for the quality of the product. Recently, a great deal of care has been dedicated in developing biosensors for food analysis and procedure control to achieve high specificity, rapid response and reliability. A wide range of biosensor types have been developed e.g. Enzyme-based, Tissue-based immunosensors, DNA biosensors and Thermal Biosensors. In Reaction mechanism of CeO2, NPs with antioxidants for the colorimetric detection of antioxidants spends caffeic acid as an example. In presence of antioxidants, color of the NPs deviates from white/yellowish to dark brown due to surface oxidation and creation of charge transfer complexes.

Students will be able to explain

1. Different foods having anti-oxidant activity.
2. What are anti-oxidants.
3. Name different enzymes that are used in the fabrication of food anti-oxidant Biosensors.

165 Packaging material

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Students will be able to explain

1. Different foods having anti-oxidant activity.
2. What are anti-oxidants.
3. Name different enzymes that are used in the fabrication of food anti-oxidant Biosensors.

166 Intellegent labels

The main objective of packing is to keep the contents clean, fresh, Hygienically Safe and intended to increase the shelf life. Packing materials includes Cellulose, Plastic, Glass, metal such as Al, Tin, Steel. The food encircled in the package may need protection from various factors such as shock and temperature. One of the most general materials suited for packaging in food industry is petroleum plastics (synthetic polymers). Such polymers comprise low and high density polyethylene, Polypropylene (PP), Polyvinyl chloride (PVC), and Polystyrene, Polyethylene terephthalate (PET). Smart Packaging can be broadly categorized into the following types: Passive, Active or Functional packaging. In functional packaging, three types of Bio sensing approach could be used to control the inside environment of the packaging. Active packaging typically revenues having active functions beyond the inert passive containment and protection of the product. Intelligent and smart packaging frequently comprises the ability to sense the inner atmosphere of the package. As Transition metals in foods, can bring oxidative degradation of food components, So, Metal-chelating active packaging materials are commonly made by immobilizing metal attractive active compounds onto the active packaging material.

Students will be able to learn

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- Different packaging materials.
- The functions of packaging.

167 Natural environment

All prepacked food needs food label that shows obligatory information. All foods subject to general food labeling necessities and any labeling delivered must be in accurate form. The information must be present on label according to law. The name of the food must be obviously indicated on the packaging. If the food has been treated, it must be mention in title of packaging for example 'smoked bacon. A processed food is food that has been changed in some way during preparation. These systems can be placed outside or inside the package. Intelligent labels are used as an umbrella term for any labeling or coding that practices technology to add functionality and data beyond a traditional barcode. This includes Barcodes, Radio Frequency Identification Tag (RFID) and Quick Response Code. The Tempering or opening of the food packing would increase the level of oxygen within the packet. Modified Atmosphere Packaging technologies keep fresh food by reducing its exposure to oxygen because Oxygen leads to oxidation, which can reason discoloration, spoilage, off-flavors and textures. The application of a novel nano-indicator, containing titanium oxide, as an oxygen intelligence in. QR code mechanism in the similar way as a barcode at the supermarket. Every QR code consists of a number of black squares and dots which signify certain pieces of information. Smartphone can scan this code; Smartphone can translate that data into something that can be easily recognize by humans. A barcode is called as sentinel system of food which is being established only to detect pathogen in food packages. The barcode membrane is attached to specific-pathogen designed antibody. The Dark black color of barcode shows the presence of contaminating bacteria and as a resultant barcode become unreadable upon scanning. Radio-frequency identification (RFID)-based sensors are used in the smart packaging to detect deviations in the physical stimuli.

Students will be able to explain

1. Different types of intelligent labels.
2. RFID system.
3. Benefits of intelligent labels
4. Food sentinel system.

168 An introduction to environmental pollution

All the physical surroundings on Earth are called the environment. Physical elements found space, landforms, water-bodies, climate and minerals. Biological elements comprise plants, animals, micro-organisms, and man. Cultural elements include economic, social and political

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conditions which are mainly man-made features. It includes everything living and everything nonliving. Through many ecosystem services natural environments play a important role in protecting health. Various populations rest on nature for basic material, such as fresh water, wood, fuel, and nutritious food. Biosphere is defined as comparatively thin life-supporting stratum of Earth's surface, ranging from a few kilometers into the atmosphere to the deep-sea vents of the ocean. The biosphere is defined as a global ecosystem consist of biotic and abiotic factors from which they derive energy and nutrients. Just like atoms interact to make a molecule or species interact in a community, ecosystems and biomes are connected with one another to form the biosphere. A critical example of Earth-wide connection is the water cycle. Living organisms need water, and there is a large number of living organisms on our planet. The water cycle moves the water that living things use from ecosystem to ecosystem across the biosphere.

Students will be able to learn

1. Natural environment and its components
2. Living and nonliving things are related in the Ecosystem.

169 Chemical warfare as Air pollutants

The contamination of physical and biological components of the atmosphere due to which normal environmental affected is called as environment pollution The substances that involve in causing pollution are reflected as pollutants, Environmental pollution are classified on different factors such as chemical structure, mode of action, source or by quantity. A pollutant can be any chemical or geochemical substance, biological organism or physical substance, which has been free into the environment by man and has damaging, unpleasant or inconvenient effect. Depending on nature of a pollutant, pollution is categorized as air, water, soil or land, noise, radioactive and thermal pollution in unassuming terms, pollution is the starter of harmful chemicals into some areas of the environment. Pollution can have Natural causes or be caused by Humans. Persistent organic pollutants known as "forever chemicals" are organic compounds. that are unaffected to environmental degradation through chemical, biological, and photolytic processes. Because of their perseverance, POPs bio accumulates with potential opposing effects on human health and the environment. Generally speaking, there are five types of pollutions. These are air Pollution, water pollution, land Pollution, radioactivity Pollution and noise Pollution.

Student will be able to describe

1. Different types of pollution.
2. Persistent organic Pollutants.

170 WHO guidelines for Air quality

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Chemical warfare agents (CWAs) can results in damaging effect on human body. These chemicals are manufactured chemical substances that are adept of causing a lethal effect on humans, animals and plants. The toxic chemicals that are limited in CWAs are usually adept of prompt incapacitation, Sudden death and Permanent detrimental effects towards health. CWAs were major used in First World War. Mustard gas, or sulfur mustard, is a chemical agent that bases severe burning of the skin, eyes and respiratory tract. Nerve agents, sometimes also called nerve gases, are a class of organic chemicals that disrupt the mechanisms by which nerves transfer messages to organs. The disruption is triggered by the blocking of acetylcholinesterase. Lewisite is a blister agent defines as an oil in its pure form and can seem amber or black in its impure state. It can easily enter clothing and rubber masks. During world war I a chemical having cyanide is used as weapon it has ability to rapid kill the spreading. This chemical is basically as blood agent that easily spread in bloodstream on inhalation or skin contact. At cellular level, it cuts the body's ability to take in oxygen. In case of high doses, effects of these chemicals are quick and catastrophic and lead to coma. The gas is flammable and possibly explosive. It's lighter than air, so if it's dispose if in the sea, it would bubble to the surface.

students will be able to learn

1. Different types of Chemical warfare
2. Effect of these chemicals on health

171 Environmental Biosensors

World health organization has provided instructions for the Air quality. There are severe dangers to health but also from exposure to Ozone (O3), Nitrogen dioxide and Sulfur Dioxide. **Ozone** is present in inorganic molecule (O3). It is a pale blue gas with a typically pungent smell. It is an allotrope of oxygen that is much less stable than the diatomic allotrope Oxygen breaking down in the lower atmosphere to dioxygen. Ozone present at ground level is not mixed with the ozone layer in the upper atmosphere. Ozone is considered as one of elements of photochemical smog. The photochemical smog is formed by the reaction with sunlight of pollutants. Volatile organic compounds (VOCs) emitted by vehicles, Solvents and industry. The highest levels of ozone pollution occur during periods of light weather as an air pollutant, NO2 as nitrate aerosols form some part of Particulate Matter and, in the presence of ultraviolet light of ozone. SO2 is a characterized as colorless gas having sharp odor and produced from main event of Burning of fossil fuels. A SO2 concentration having 500 µg/m3 is not more high over average periods of 10 minutes. When SO2 combines with water, it forms sulfuric acid. It has demoralizing effect on environment. Sulfur dioxide is the main constituent of acid rain which is a cause of deforestation.

Students will be able to learn

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1. WHO guidelines for air pollution
2. Different types of pollutants

172 Air Pollution

Environmental biosensors are used for environmental checking of Organic pollutants, possibly toxic elements, Pathogens and Heavy metals. Different types of means are used for environmental analysis. The most common traditional methods are typically grounded on chromatographic use for detection of molecular pollutants attached with different types of mass spectrometry. Chromatographic tools for sample pretreatment are sensitive but are reliable but also time-consuming. the equipment is expensive and wants qualified personnel to perform analysis. The traditional methods are not suitable in term of real time measurements, for example as in the case of accidental release of pesticides, where rapid Miniaturized equipment is used for environmental monitoring. The use of environmental biosensors in the field has been added to the sustainable development of society. Biosensors can measure pollutants which caused adverse effects on both ecosystems and human health.

Students will be able to explain

1. Traditional techniques used in Environmental sciences.
2. How environmental sensor is different from a diagnostic biosensor.

173 Air born pathigen detection

Environmental Pathogens are microorganisms that usually live in environment. Environment pathogens exist outside of a human host for a large amount of their lifecycle. When they come into interaction with humans they result in disease at measureable frequency. An airborne pathogen can make a person sick simply because they have inhaled the pathogen. The rhinovirus is a contagious virus that causes people to get colds. Streptococcus pneumonia, which causes pneumonia, and the Influenza virus that causes people to get the flu. The main diseases caused by Air borne pathogens include Respiratory infection, heart disease and Lung cancer. The detection of environmental pathogens often requires analysis of matrices, such as air, and objects, such as the surfaces of biomedical devices. Dust particles, which are dispersed and heterogeneous in nature, may interfere with the aerosol collection and sensing methods. There are basically two Strategies to collect the sample. First is Use of different filters to collect the pathogen and second is Use of gravitational collector. Samples must be positioned in such a way to capture the maximum amount of aerosol for sufficiently sensitive sensing. Care is to reduce amount of dust present. The types of aerosol particles describe their settlement. The larger particles ($d > 100\ \mu m$) will settle. Smaller than 100 µm will typically travel farther.

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Students should be able to learn

1. Characteristics of air borne pathogens.
2. Strategies that are required to be opted for the sample collection.
3. Role of dust particles and aerosol size in Biosensor designing

174 Biosensor for air pollution

Environmental Pathogens are microorganisms that usually live in environment. Environment pathogens exist outside of a human host for a large amount of their lifecycle. When they come into interaction with humans they result in disease at measureable frequency. An airborne pathogen can make a person sick simply because they have inhaled the pathogen. The rhinovirus is a contagious virus that causes people to get colds. Streptococcus pneumonia, which causes pneumonia, and the Influenza virus that causes people to get the flu. The main diseases caused by Air borne pathogens include Respiratory infection, heart disease and Lung cancer. The detection of environmental pathogens often requires analysis of matrices, such as air, and objects, such as the surfaces of biomedical devices. Dust particles, which are dispersed and heterogeneous in nature, may interfere with the aerosol collection and sensing methods. There are basically two Strategies to collect the sample. First is Use of different filters to collect the pathogen and second is Use of gravitational collector. Samples must be positioned in such a way to capture the maximum amount of aerosol for sufficiently sensitive sensing. Care is to reduce amount of dust present. The types of aerosol particles describe their settlement. The larger particles ($d > 100\ \mu m$) will settle. Smaller than 100 µm will typically travel farther.

Students should be able to learn

1. Characteristics of air borne pathogens.
2. Strategies that are required to be opted for the sample collection.
3. Role of dust particles and aerosol size in Biosensor designing

175 Lead as a pollutanat

Lead (Pb) naturally present in the environment as well as present in manufactured products. Lead as heavy metal mostly discharges into air as suspended particles. The motor vehicles and industrial sources are major sources of lead air emissions. lead is commonly utilizing in aviation gasoline for piston-engine aircraft. The air emitted lead can be inhaled or ingested, primarily through contact with contaminated soils or other surfaces. Lead is a most common environmental pollutant. The lead in fuels contribute to air pollution, especially in urban areas. Plants exposed to lead can absorb the metal dust through their leaves. Lead is a naturally-occurring heavy metal

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constituting 10–30 $mg \cdot kg^{-1}$ of Earth's crust. Lead poisoning is basic type of metal poisoning caused by lead in the body. Lead poisoning can occur as a consequence of ingesting, inhaling, or via direct contact with lead-containing compounds. The common metalloenzyme having 3 vicinal thiol groups in coordination with Zn(II) is Aminolevulinate dehydratase

Students will be able to learn

1. The environmental sources for the pollution of lead.
2. The role of Pb in the inactivation of enzyme

176 DNA ribozymes

DNAzymes are essentially RNA-cleaving DNA enzymes. Deoxyribozymes are DNA enzymes. These enzymes are called as first Deoxyribozyme discovered by Ronald Breaker. This Deoxyribozyme also called as GR-5. The structural differences present between RNA and DNA also play have a key role in Deoxyribozymes, such as the additional methyl group of the DNA base thymidine compared to the RNA base uracil or the tendency of DNA to adopt the B-form helix while RNA tends to adopt the A-form helix. It is also illustrated that DNA and RNA both form structures but there are some differences in each form which does not affect their catalytic activity. DNAzymes consists of two separate strands. The Catalytic strand (blue/cyan) and second is Substrate strand (black). DNAzymes are dependent of metal ions just like other enzymes such as lead, zinc, cobalt etc Deoxyribozyme are basically ribonuclease, and are also artificially synthesized. GR-5, catalyzes Pb2+-dependent cleavage of a single ribonucleotide phosphoester. They are capable of performing a specific chemical reaction, but not always catalytic. There are many advantages of DNAzymes such as DNAzymes have found practical use in metal biosensors. A DNAzyme based biosensor for Pb ion are used to detect lead ion.

Students will be able to learn

1. Definition of RNA –cleaving DNA enzymes
2. Types of Pb dependent DNAribozymes

177 Biosensor for lead detection

The detection of metal as important pollutant having major routes of human exposure results from lead-based paints, contaminated soils and foodstuffs. A novel DNAzyme biosensor was designed. The sensing component for Pb ions was compose of basically three layer.

- Strand 1 GR-5 DNAzyme.

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- Sensitizer Ru phenyl based compound intercalation
- Strand 2 Complementary substrate

The hybridization of DNAzyme and substrate along with Ru(phenyl) complex is used for emission of high ECL. ECL lead biosensor is the intercalation between Ru 32+ in duplex and GR-5 lead-dependent DNAzyme. The properties of biosensor include high sensitivity as well as high specificity, along with an ultra-low limit of detection of 0.9 pM and a dynamic range from 2 to 1000 pM, which enables analysis of trace amounts of lead in serum samples. An electrochemical lead sensor is based on lead-dependent changes in the conformational state of a surface-immobilized, methylene-blue modified DNAzyme assembly. The designed a biosensor for Pb(II), is based on the Pb(II)-induced release of fluorescence-labeled aptamers from complex with a quencher-labeled short complementary sequence. The biosensor showed a Pb(II) concentration-dependent increase of fluorescence intensity in the low micromolar range and a detecting linear range from 100 to 1000 nM, and great specificity even in the existence of various interfering metal ions. DNA aptamers for Pb(II) were successfully selected to develop light-up biosensors with good sensitivity and specificity for Pb(II) detection. Aptamers with high binding affinity and good specificity were selected.

Students will be able to learn

1. Functioning of lead biosensor
2. A novel DNAzyme biosensor

178 Water pollution and its types

The addition of Chemical, oil, Microorganism and toxin in any water body become the cause of water pollution. Water pollution is divide into two sources. one is Point source water pollution defined as come from a single source. Second is nonpoint source pollution refers to contamination does not come from single source. Basically there are many types of water pollution the most popular is agriculture type pollution, in this type farms release large quantities of organic matter and saline drainage into bodies of water. The one of the most common contaminants found is eutrophication. Eutrophication is defined as high concentration of nutrients, especially phosphates and nitrates into water body. The process of eutrophication promotes excessive growth of algae. Due to high levels of organic matter and presence of decomposing organisms results in depletion of oxygen, and depletion leads to death of organisms, which turns into green having rotten egg odor. The water come from domestic equipment other than toilets is called greywater. Industrial Wastewater is the water used to make different commercial products in all phases of production. Once this process water has been used, it is considered waste and needs to be treated before it is discharged. Drinking water have two main sources of radioactive contamination. Naturally occurring radionuclides are radium, uranium, and radioactive gas radon. Radioactive contamination comes from man-made sources

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develops into thick sludge causing harms for fish. These oils can also contaminate water for drinking and other purposes.

Students will be able to learn

1. Different types of water pollution.
2. Main source of Nitrates in water pollution.

179 Biochemical oxygen demand

The organic pollutants present in water are monitored with the help of Biochemical oxygen demand (BOD) which is an important index for monitoring pollution. BOD is an important characteristic of water quality. BOD is defined as the amount of dissolved oxygen which is Required to oxidize all biodegradable organic compounds. Nitrates and phosphates are two type of chemical responsible for increase or any decrease. Nitrates and phosphates are plant nutrients and can cause plant life and algae to grow quickly. When plants grow quickly, they also die quickly. This contributes to the organic waste in water. Decomposition by bacteria resulted in high BOD level. The BOD test is also widely used at wastewater treatment facilities to assess the biodegradation efficiency in wastewater purification processes. There are many benefits of using traditional means for BOD test and these traditional means are considered as universal for analysis of waste waters. The important point is these methods don't require any expensive equipment. This technique is not good in term of low responsiveness. Yeasts are popular biomaterial for use in almost all types of biosensors. As Yeast show resistant to negative environmental factors and play role in biosensor's recognition element. BIOX-1010 is best on-line analyzer for BOD measurement in waste-water treatment plants.

Students will be able to learn

1. Biochemical Oxygen demand.
2. Basic design for a classical BOD sensor.

Use of BOD sensor in an environmental setting.

180 Heavy metals

Heavy metals are well-known environmental pollutants due to their toxicity, persistence in the environment and Bio accumulative nature. Their natural sources include weathering of metal-bearing rocks and volcanic eruptions. Anthropogenic sources include mining and various industrial and agricultural activities. Heavy metals are present in personal products like Lead,

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Arsenic, Mercury Aluminum, Zinc, Chromium and Iron. These metals are commonly found in a wide variety of personal care Lipstick, whitening toothpaste, Whitening creams, Eyeliner and in nail color. Some metals are intentionally added as ingredients, while others are contaminants. Porus silica-based Biosensor for heavy metals bio sensing platform, based on nanostructured porous Si, thin films. The biosensor is for label-free monitoring of heavy metal ions Ag+ > Pb2+ > Cu2+). Bio sensing principle is based on the enzyme inhibition. Optical transducers are used in the biosensor. The detection limit range of 60–120 ppb.

Students will be able to learn

1. Nature of Heavy metals
2. Porus silica based biosensor

181 Detection methods for water pathogens

The presence of pathogens in drinking water constitute a serious danger for human health. Some biosensors were recently proposed for their environmental monitoring. Government of Pakistan follows the WHO standards for Drinking water quality. In any 100 ml sample of treated water entering the distribution system there should be no detectable E. coli or thermotolerant coliform. Drinking water should be free from known pathogenic microorganisms and indicator bacteria. Pathogenic and thermotolerant microbes are considered as signs of fecal water contamination. Coliform commonly known as" indicator organisms", refers to a wide groups of bacteria that found throughout the environment and used to evaluate the water quality. These organisms have a wide distribution in soil, water surface, vegetation's as well as on the skin or intestinal tract of worm-blooded organisms including humans. Coliform bacteria ferment lactose sugar producing gas (CO2) and acid when incubated at 35°-37°C. The most important Virulence factor in E. coli Shiga-toxin (Stx) 1 and 2. These are toxic proteins. It causes damage to lining of capillaries, eventually leading to anemia, thrombocytopenia and organ dysfunction or failure. It has a major role in causing the infection by binding with the micro villi of the intestine. Intimin is expressed on the bacterial surface where it binds with the receptor. Receptor of Intimin is called translocated Intimin receptor. New sensor on the other hand, detects the E. coli directly in a water sample New techniques uses something called DNA-magnetic particle technology. It essentially seeks out and isolates specific bacteria using lots of tiny, nano-sized magnets.

Students will be able to learn

1. Coliform bacteria.
2. WHO standards for potable/drinking water.

182 Endocrine disrupters

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The endocrine system is comprised of ductless glands that produce hormones that regulate metabolism and growth The best examples of Endocrine Glands include pituitary gland, Thyroid gland, Pineal gland, Adrenal gland and Ovary and Testis. The term Endocrine disruptors are defined as chemicals that act on the endocrine system of humans and animals by mimicking the hormone, blocking and/or Interfering in some manner. Some of the major chemicals that are known as water contaminants are endocrine disrupting chemicals such as disinfection byproducts, fluorinated substance. Fluorinated substances also can be found in drinking water. Binding of EDs to hormone receptors, with resultant alteration in the transcription of messenger RNA, followed by altered gene expression, Other mechanisms involve alteration in Hormone synthesis and metabolism. 4-Nonlyphenol is an endocrine disrupting chemical. It has hormone-like effects in both wildlife and humans. Steroidal hormone as pollutants The steroidal female reproductive hormones, can act as pollutant. In order to design biosensor, Saccharomyces cerevisiae is used as a bioreceptor element in the biosensor to detect estradiol. The yeast cells were immobilized into polymeric matrix to create a sensor material. This will generate a single-use and low-cost field-portable biosensor device. Sensor readout with a conventional smartphone equipped with a filter set.

Students will be able to learn

1. Definition of endocrine disrupters
2. Two important endocrine chemicals

183 Introduction to biosensors for diagnostics

Diagnostic biosensor usually helps to monitor diseases, drug discovery, detection of pollutants, detection of disease causing pathogens. In the fabrication of Biosensors there are five main points that are focused such as Nature of Analyte or target molecule, Sample matrix, Types of Bio recognition element, Transduction process and Immobilization Method The appearance of specific target element related to a particular condition could be motioned as biomarker. Early diagnosis increases the chances for successful treatment of diseases like cancers. A biomarker can be any biological indicator that can be measured. Biomarkers can be DNA, RNA, protein, toxin, cell or metabolites). They are measured from a tissue biopsy or a liquid biopsy using a biological sample (blood, urine, saliva, sweat. Tears, diseased tissue). Physiological or morphological biomarkers are also used and Body temperature. Biomarkers can be either quantitative or qualitative. **Qualitative biomarkers** could be involved in a pathogenic process detection within an yes/no analysis, **Quantitative biomarkers** are involved in pathogenic process detection with a threshold effect. There are five different levels at which the diagnostic biosensors are focused, including, Biomarker, Sample Bioreceptor, Transducer and Immobilizing layer. A biomarker is a signature molecule/element for a pathological or specific condition. An ideal biomarker is specific, informative and cost effective.

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Students will be able to learn

1. What is a Diagnostic Biosensor?
2. Applications of Diagnostic Biosensor

184 Sample matrix in Diagnostic Biosensors

In diagnostic Biosensors a number of biological samples are used, for the detection of single or multiple analytes e.g. Blood, Plasma, Serum, Urine, Cerebrospinal fluid, Tears, Sweat and Tissues Cells. Humans produce 30 gallons of tears per year. protein-rich, antibacterial liquid Tears serve one primary purpose and that is to clean and lubricate the eyes. Tears are secreted from the lacrimal glands located at the back of the top eyelid. All tears contain water, lipids, lysozyme, lipocalin, glucose, and sodium. This protein-rich, antibacterial liquid goes from the outer edge of the eyeball toward the cornea and lubricates the entire eye surface every time we blink. There are three types of tears Basal tears. Emotional Tears, relax tears It is clear from all discussion is that all the biological matrix which are used in diagnostic are complex in nature. IN a biological sample one or few analyses are needed to be detected. In Biosensor technology certain strategies are employed to specifically detect the target molecule with amplified signals. Modification of support layer to specifically binding with the target molecule Amplification of weak signals

Students will be able to learn

1. Biosensing system in Diagnostics
2. Classification of Biosensors in Diagnostics
3. Sample matrix in Biosensor

185 Diagnostic biosnsor transduction

Biosensor is a sensing device that combines a Biosensing element with a transducer. The Biorecognition element determines the degree of activity and specificity of the Biosensor. The ability of Biosensor to detect low concentrations that is Limit of detection is dependent on the transducer, as it transforms the Biorecognition signals into a quantifiable output signal. Different types of sensing techniques are used in Diagnostic Biosensor technology. These include Electrochemical Sensors, Optical Sensors, Acuostic Sensors or Gravimetric sensors, Thermal or Calorimetric sensors, Magnetic sensor, Sensors and receptor element. In diagnostic Biosensing a number of Biosensing elements are used and biosensors are named according to the Biorecognition element.

186 Electrochemical biosensors in diagnostics

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Electrochemical sensors are based on the principle of Electrochemistry. In diagnostics electrochemical sensors are the most used sensors. The electrochemical sensors are basically utilizing as setup of electrode. On the base of measurable parameters, there are three major categories. The electrochemical transducers can detect the signals by using direct and indirect approach There are different properties of the signals on the basis of which the electrochemical transducers are used. Potentiometric Biosensors is used for charge accumulation, Current for Ampermetric biosensors, Conductive properties for condutometric sensors and Resistivity of impedemetric Biosensors. Generally conductive sensors and impedemetric sensors are considered under one type. Amperometric Biosensors are used to detect small molecules, like glucose. Typically, amperometric sensors use enzymes for the detection of small molecules. In a glucose biosensor Peroxidase enzyme oxidize a REDOX reaction. Potentiometric Biosensors allow label-free detection, simply they do not require any enzyme, to get a signal. Sweat is a biological fluid. Sweat analysis is noninvasive. Impedimetric biosensors offer label free detection of target molecules. They can detect small molecules, large molecules and even cell Impedance is a measure of the overall opposition of a circuit to current.

Students will be able to learn

1. Categories of electrochemical transducers
2. Impedimetric biosensors

187 Revised optical transduction in diagnostics

In optical transducers light source is used as a detect for the Biorecognition event. Monochrome and white light sources both are used in optical biosensors, depending upon the requirement, type of senor and Biorecognition event. Optical transducers are one of the popular transducers in diagnostic biosensor technology. Different optical principles are used. Refractometry is define as method of measuring substance refractive index. The ratio of the speed of light in a vacuum to the speed of light in another substance is defined as the index of refraction for the given substance for example, measure the composition. Reflectometry uses the reflection of waves at surfaces and interfaces to detect objects. Ellipsometry is define as an optical technique for measuring the dielectric constant of thin films. Ellipsometry determines the change of polarization upon reflection and compares it to a model. Surface Plasmon Resonance imaging is a label-free optical detection technique used to monitor. It analyzes biomolecular interactions in real time. The imaging capability enables users to visualize the entire working area and work in a multiplex format.

Student will be able to describe

1. Difference between Refractometry and reflectometry

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2. Definition of Ellipsometry

188 Thermal transduction

The principle of Thermal transduction based on calorimetry in which changes in temperature are measured. Thermal biosensors use thermistors, Thesis thermistors are basically temperature dependent resistors, as detector unit. The enzyme electrode used for the first amperometric biosensor, the first thermal bio-sensor was called "enzyme thermistor. Enzyme reactions that are exothermic can be coupled in a single system. The total heat change will be high thereby increasing the sensitivity of the sensor. A series of enzyme-catalyzed exothermic reactions that can be employed to detect the target molecule content in a sample. More recent re-search suggests the use of thermopiles instead of thermistors as this would simplify the ambient temperature control of the system and hence the miniaturization of the device

Student will be able to learn

1. Definition of Thermal transduction.
2. Name and heat output of different enzymes during catalysis.
3. Advantages and disadvantages of thermal transduction.

189 Revised LAMP

Loop-mediated isothermal amplification technique is used for detection of DNA. LAMP is good detection method that amplifies DNAs by using common enzyme DNA polymerase. The technique of LAMP is carried out in isothermal conditions. The LAMP results are highly repeatable, accurate, highly selective for the target DNA sequence. The target DNA sequence can be recognized by using ten DNA sequences in two stages. It uses a single-step test tube at around 60–65 °C for approximately thirty minutes. The reaction time for LAMP takes ten minutes, The total time for the entire LAMP process 50-60 minutes. LAMP need cost effective equipment and consumables. In this way, LAMP is characterized as more cost effective. LAMP is only work to RNA upon use of reverse transcription-coupled LAMP (RT-LAMP). The detection of prostate-specific antigen mRNA in one PSA-expressing cell mixed with 1 million PSA-negative cells. This strategy can identify multiple amplicons with a small DNA melting temperature (Tm) difference (~0.2 °C) it can sensitively detect LAMP amplicons with the initial DNA concentrations ranging from 10 to 108 copies.

Students will be able to explain.

1. What is Loop-Mediated isothermal amplification (LAMP).

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2. Basic principle of LAMP in diagnosis

190 Revised Acuostic diagnostic

Gravimetric transduction uses mechanical acoustic waves which are produced with the help of piezoelectric materials. The mass changes affect signal response change in Biorecognition layer that is why large molecules are used to detect. Acoustic detection is label-free, but the use of nanoparticles as mass-labels to increase mass sensitivity has been reported. The best known acoustic biosensors are QCM, surface acoustic wave and cantilever biosensors. QCM based biosensors were the first mass-sensitive biosensors reported and are still the most commonly used acoustic bio-sensors. The thickness shear mode (TSM) quartz crystal resonator, commonly consisting of a thin AT-cut quartz plate with two metal electrodes on two surfaces. In contrast to that, mechanical waves of SAW biosensors propagate along the surface of the piezoelectric substrates. Surface acoustic wave have higher resonance frequencies as compared to QCM devices. SAW sensors are class of microelectromechanical systems which rely on the modulation of surface acoustic waves to sense a physical phenomenon. In atomic force microscopy, Cantilever biosensors used cantilevered sharp tips. The mechanical waves of SAW biosensors propagate along the surface of the piezoelectric substrates. This allows for higher resonance frequencies, the mass sensitivity increases with increasing frequency. Cantilever biosensors originate from the cantilevered sharp tips used in atomic force microscopy.

Students will be able to learn

1. The importance of Acoustic Transduction principle
2. Types of Acoustic Biosensor in Diagnostic
3. Advantages and Disadvantages of Acoustic Biosensors

191 Revised Magnetic transduction

Magnetic biosensors defined as paramagnetic or super-paramagnetic particles, for detection of biological interactions and estimate changes in Magnetic properties. Paramagnetism is a form of magnetism. Paramagnetic properties are due to the presence of some unpaired electrons. It from the realignment of the electron paths caused by the external magnetic field. Superparamagnetism is Paramagnetism that occurs below the curie temperature. Curie temperature is the temperature above which certain materials lose their permanent magnetic properties, which can be replaced by induced magnetism. Bio-recognition element onto the magnetic particle's surface having diameter in the range of 50nm-10 micrometer with paramagnetic properties This results in a number of additional functionalities, such as transport of bio-molecules to a specific location on the chip; on-chip magnetic immuno-separation of bio-molecules and cells; and, testing or

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accelerating bio-molecular binding events. Magnetoresistance is defined as change in resistance of material when magnetic field is applied. The change is greater in case of alternating layers of both ferromagnetic and non-magnetic metals. This results in quantum interferences. However, GMR biosensors based on magnetic transduction and characterized as less common than biosensors based on electrochemical or acoustic detection. The magnetic principles require more sophisticated readout electronics and magnetic labels are attractive for purification tool as there is no magnetic background in case of bio-logical samples.

Students will be able to explain the

1. Magnetic transduction principle
2. Types of Magnetic transductions in Diagnostics
3. Advantages and disadvantages

192 Revised radioactive transduction

Radioactivity is defined as particles which are emitted from nuclei as a result of nuclear instability. Transduction principles showing radioactivity must be remembered for use in biosensor measurements. However, there are many disadvantage using radioisotopes particularly regarding technical and safety issues. It is the spontaneous emission of radiation in the form of particles or high energy photons resulting from a nuclear reaction. Radioactive decay is a random or stochastic process that occurs at the level of individual atoms. Alpha particles carry a positive charge. An alpha particle is consisting of two protons and two neutrons. Beta particles carry a negative charge and high energy electrons. Gamma rays are neutral. Gamma rays are waves of electromagnetic energy, or photons. A radioactive antigen competes with a non-radioactive antigen for a fixed number of antibody or receptor binding sites

Students will be able to explain

1. To define radioactivity
2. Transduction principle based on radioactivity
3. Reasons for not using radioactive transduction in Biosensors

193 Biorecognition layer

Biorecognition layer or Biosensing layer is platform for the Biosensing event in a biosensor. Biosensing layer holds the Biological component of the Biosensor. It is placed on the surface of a transducer and determines the performance of Biosensor There are three main components of the Biorecognition layer. Biorecognition element could be any biological molecule having strong

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affinity for its analyte. A coating material could be any polymer, gel. There are many Factors effecting the performance of Biorecognition layer **e.g.** Thickness, High surface density of Bioreceptor molecule and Dilution of Sample matrix. Thickness of the Biorecognition layer is very important for the good performance of the Biosensor. If the sensing layer is too thin the process of Biorecognition may not be occur properly. Due to thickness the distance between the bioreceptor element analyte complex and the transducer will be increased, in turn signals will be masked.

Students will be able to explain

1. Biosensing layer.
2. Strategies to inhibit the nonspecific binding.
3. Novel design of sensing layer

194 Lipid Biorecognition layer

Lipids are building block molecules in Biological cells. Biological Lipid membranes show structure biocompatibility for development of new types of biosensors. They provide fast response and high sensitivity. Biological membranes are good candidates for health diagnosis. Biological membranes have continuous double layer of lipid molecules in which protein molecules are embedded. This lipid bilayer is fluid, with individual lipid molecules able to diffuse. The membrane lipid molecules are amphipathic. (Polar and non-polar parts. Phospholipids), for example, have non-polar fatty acid "tails" and polar phosphate "heads." Polarity is an important property of molecules that determines how they will interact with other molecules They have an ability to reproduce some features of biological membranes on the surface of transducer. Lipid bilayer is stable and the sterol and phospholipid molecules can move freely in the environment. The phospholipid rotates on its axis to interact with its immediate neighbors. Whereas in Lateral Movements, the phospholipid moves around in one leaflet. Flip-flop Movement it is possible for phospholipids to move between both leaflets of the bilayer in transverse movement, in a "flip-flop" manner. Phospholipids in the lipid bilayer can either move rotationally, laterally in one bilayer, or undergo transverse movement between bilayers.

Summary

1. Composition of lipid bilayer
2. How Biological membranes are good candidates for health diagnosis

195 Non covalent hydrophilic binding

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Non covalent immobilization is one of the method of choice for attaching the biological molecule on the surface of transducer. There are two types of non-covalent physical methods. Mechanical confinement and Use of physical forces for non-covalent immobilization. Physical methods are characterized by weak interaction forces i.e. Vander wall forces, Hydrogen bonding through no-covalent interactions, Mechanical suppression of Bioreceptor element within the support is Physical methods. The small units of amino acids combine together to form a protein molecule. Different chemical groups include Hydrogen, carboxyl group, amino group and a variable group is attached with the alpha carbon of an amino acid. group. Acids are very stable in water since they are partially charged in their protonated form. Hydrophilic amino Acids are fully charged in their deprotonated form. Chemistry of the protein is dependent on the chemistry of amino acid residues present in the primary structure of the protein. Laccase enzyme protein has an isoelectric point of 3.3. Protein can acquire a charge below or above the isoelectric point. The enzyme protein was bound to the surface of a negatively charged immobilization layer, by the virtue of global positive charge on the molecule. Nucleotides are the building blocks of DNA molecule. These nucleotides consist of three main parts: nucleotide bases, phosphate group and sugar.

Students will be able to explain.

1. Non covalent bindings
2. Attachment of Protein and DNA molecules

196 Non covalent Hydrophobic immobilization

Hydrophobicity is the physical property of a molecule. The composition of amino acid is used for the estimation of protein. The hydrophobic scale measures the contribution of amino acid residue on its surface. Amino acids with hydrophobic nature avoids aqueous environment. Hydrophobic amino acids are those with side-chains that do not like to reside in an aqueous (i.e. water) environment. They have non polar or less polar side chain. In the process of protein folding, the protein interior embedded by the non-polar group of amino acids. The hydrophobic amino acid residues are found in the active site of the enzyme in the hydrophobic pockets. ‘Lipid-like' characteristics of the non-polar pockets can be used for hydrophobic binding. Hydrophobic pockets with volumes in the range 0.4 nm -3nm.The pockets are roughly 1%-33% of the volume of a typical protein. The hydrophobic pocket is very important for the binding. In hydrophobic pockets, the ligands find their binding site more easily. Hydrophobic pockets shown in protein as active site. Immobilization of laccase via supra-molecular interactions between the laccase’s hydrophobic pocket and Laccase immobilized through its hydrophobic pocket with hydrocarbons that are attached to carbon-based support.

Students will be able to explain.

1. Explanation of Hydrophobicity

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2. Structure of hydrophobic pocket

197 Covalent immobilization by Amino resin

i. Amino resins are synthetic thermosetting resins.
ii. They are formed by the copolymerization of amine, amides with aldehydes.
iii. They are used for improve the bonding.
iv. Amino resin is a highly porous solid support material.
v. They can bind with the lysine residues at the side chain of the protein.

198 Covalent immobilization by Epoxy resin

The immobility of the enzyme can be easily performed in epoxy resins. The covalent binding at different point is provided by the epoxy resins. The binding occurs between resin and the enzyme. Epoxy methacrylate resins are epoxy activated resins produced with High degree of crosslinking Control Porosity Covalent bonding forms stable link between the enzyme's functional groups and the support matrix.is a thermosetting epoxide polymer. Bisphenol-A and Epichlorohydrin react to form epoxy resins. When mixed with a catalyzing agent, it polymerizes. Applications coatings, adhesives and composite materials. Epoxy merthiolate resins work under very mild pH and temperature conditions Different protein groups formed different covalent bond. Epoxy merthiolate resins carrier resins are stable during storage. Different nucleophile groups are present on protein surfaces and epoxy activated surface provide an environment to all nucleotides to react chemically.

Students will be able to explain.

1. Epoxy resin

199 Test formats

Biosensors employee the test formats that are standardized in laboratory methods. Generally, those method types are employed that are based on a support material. In biosensor the Biorecognition process occur on a solid support that is a sensing layer therefore those labs based methods are used that are established or a support material. There are two types of methods that are established in laboratories are used in Biosensor technology. The use of special class of

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molecules that can exist in more than one stable states with the change in the physical state. Immunological methods. Are based on antibody and antigen complex formation on a matrix. The shifting occurs between the states. Vitamin A exhibits two reversible stable forms. The basics of ELISA (Enzyme linked immunosorbent Assays) are similar to the format useful in biosensor measurements. ELISA test is performed on 96 well plate on which antigen is immobilized on ELISA plate. A complex is formed when antibody is added in plate (after the immobilization of the antigen).

Students will be able to explain.

1. Non covalent bindings
2. Attachment of Protein and DNA molecules

200 Acidochrome molecular switch

Any molecule that can exists at two or more stable states in response to change in pH is known as acidochrome. Acidochromic molecules are found in nature and produced by a number of plants. Acidochrome are used as pH indicator in scientific studies. Phenolphthalein, Acidochrome molecular switches are found in a wide range of plants like roses, cornflowers, primroses and violets. Acidochrome changes their color in response to change in ph. Acidochrome reversibly change their chemical structure at different pH values and show different colors. In acidic and neutral solutions, it is colorless, but changes to a pink when the pH is greater than 9. It belongs to the class Phenolphthalein changes structure when the Hydrogen ion concentration is between 10-8 and 10-9 M. Cyanidin is a natural organic compound. This pigment is found in number of fruits includes: cranberry, blueberry, grapes, cherry and raspberry. The red cabbage has dark colored leaves (red or purple). The color of the plant depends upon the pH value attain by soil. For example, in acidic medium, the red cabbage either appears red or pink in color.

Students will be able to explain.

1. Non covalent bindings
2. Attachment of Protein and DNA molecules

201 Photochromis molecular switch

Shifting occur between the stable states. The molecular structure also sifted between these two stable states in response of light, the response changes as the light wave properties changes. Photochromic pigments change their "color" that absorption spectra in response to light.

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Fluorescent photochrome was used as an Oxygen sensor. As the UV interacts with imidazole ring, the ring opens. The opened ring structure creates a benzoylimino group in the presence of oxygen. This photochemical reaction used in a number of applications. Beside this, fluorescent probe can be used as an oxygen sensor The lowest detectable ratio is ~0.2%. Cyanide as well as Mercury ions could be detected by Cyanohydrin through metal-assisted elimination.

Students will learn

1. Most common application of photochromic switches
2. Different organic molecules display chromogenic changes

202 DNA switch

DNA-based molecular switches are nano devices. They are highly specific sensors, diagnostic tools. Among DNA nanostructure classification, a molecular switching is one of the basic class that turn off some of the process and turn off depends upon switch binding of substrate. In Biosensor technology the principle of molecular switching is used and are known as molecular beacons. Single stranded DNA molecule is used as a probe. A fluorescent dye A quencher in close proximity coming light result in visible emission due to quenching. . Switching off quenching will detect the target molecule. Hydrodynamic DNA switch detects the target molecule by change in the oscillating movement of DNA probe. It detects the size /shape of analyte. When analytes bind to oscillating DNA nanolevers on the movement is slowed by the additional friction imposed by the analyte. Effects on the oscillating movements reveals size or shape of the target molecule. A series of three-way junction DNA-based molecular switches. It has two domains. Recognition domain which interacts with the target of interest and induces a structural change in the TWJ, Transduction domain, which then becomes accessible and binds to a fluorescent molecule.

Students will learn.

1. Most common application of photochromic switches
2. Different organic molecules display chromogenic changes

203 ELISA test format

Enzyme linked immunosorbent Assays (ELISA) based are one of the formats applied in biosensor measurements. Microplate used in ELISA for the immobilization of immunogenic molecules. In biosensor the immunogenic molecules are immobilized on the surface of active transducer. Enzyme linked immunosorbent Assays (ELISA) assay based on the formats which is

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applied in biosensor measurements. In principle, four categories of test formats are available: Direct Detection, Sandwich assay and Competitive assay. In direct assay, label-free transduction principle is used. It is single step process and permit monitoring of binding events to the biosensor surface in real time manner. The signal response changes as the analyte binds to the Biorecognition element immobilized on the transducer surface. In a sandwich there are two Biorecognition elements. Capture antibody that captures the target molecule. Detection antibody that detects the capture antibody. Response is obtained when the detection antibody binds with the capture antibody. In competitive Assay, there are two types of Analytes. In a competitive assay analyte and a known concentration of labeled analyte (derivative). The competitive assay could of two types

- Surface competitive assay
- Solution competitive Assay

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Final Term Bt505 Biosensor Handout

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